

CFA Level 3 Notes

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1 Asset Allocation

1.1 Capital market expectations I: Framework and macro considerations

CME. Capital market expectations: expectations concerning risk and return. Precision of forecasts not so important. Rather, cross-sectional (across asset classes) and intertemporal consistency more important. Poor cross-sectional consistency leads to poor risk-returns, poor intertemporal distorts connection between portfolio decisions and investment horizon.

Assets. Start with the minimum universe of client assets. The projections should be for these assets. Determine level of detail needed based on assets. Review history. Find information sources. Monitor, implement, defend views. Improve (qualitative and judgmental usually more important, if quantitative errors are reasonably small).

Depending on situation, CME demands may be low (e.g. just local and restricted to a few asset classes over the long term) or high (e.g. global and diverse asset classes and time periods).

Challenges in forecasting. Lags, accuracies, revisions, definitions, biases (survivorship, status quo), appraisal (smoothed) data, stationarity (and regime changes), synchronicity (different timezones, collected at different times of the month), level of detail, normality (and leptokurtic-ness), complexity (too much?). Ex-post data are sometimes not a good guide to ex-ante forecasting (e.g. Covid or Argentine peso problem). Data mining, trends or discoveries only specific to situations and time periods. Misinterpretation of correlations or lack of correlations (e.g. maybe non-linear relationship if no correlation), psychological biases that affect forecasting: anchoring bias, status quo bias, confirmation bias, over-confidence bias, prudence bias.

High-frequency data. Using higher-frequency data gives more precise sample variances and covariances (because capturing the finer fluctuations), but sample mean doesn't get much more precise (the average is the average is the average).

Moreover, sample means precision is defined by confidence interval, which is in turn defined by the variance, and while the variance gets more precise, the variance can grow or shrink!

See Examples 3 and 4 on page 15 showing how to find unconditional expected returns.

Model uncertainty (most serious; mis-specification) + parameter uncertainty (uncertainty in estimation; mitigate through estimation errors) + input uncertainty.

Investment outcomes linked to economy. E.g. technology bubble and GFC (use Grinold-Kroner). Trend + cycles. Trends hard to forecast; they change and only know change afterwards; true exogenous shocks impossible to predict.

Exogenous shocks: policy changes (domestic and trade), new products and technologies, geopolitics, natural disasters, natural resources, financial crises. See how economies develop after a crisis: Type I (lower output level but growth rate recovers), II (output unchanged but lower growth), III (lower output level and growth) economic growth after GFC (which are regime changes).

Economic growth (initial estimate; real) = labor force size growth + labor force participation rate growth + increasing capital inputs + TFP growth.

Faster economic growth leads to higher equity AND government bond yields. In the below of equity value (capital appreciation component only; omits dividends), S is earnings per GDP; if GDP_t is nominal, then V_t is nominal:

$$V_t^e = GDP_t \times S_t^k \times (P/E)_t$$

where the latter two terms can't keep growing. However, GDP can keep growing, so equity value is over the long run linked to GDP. However, the short run changes in S and P/E ratios will affect the forecast baseline. Dividends:

$$dy = \frac{\sum D}{MV} = \frac{DPR \times E}{MV} = \frac{DPR}{MV/E} = \frac{\sum D/E}{MV/E}$$

where MV is market value (=P). From Example 8:

$$\text{Times FV} = e^{\Delta(P/E) \times T}$$

So, if $\Delta(P/E) = 0.90\%$ for 74 years, then

$$\text{Times FV} = e^{0.009 \times 74} = 1.95$$

So, market is 95% overvalued.

Continuously compounded (i.e. logarithmic) growth = real GDP growth +

inflation + S growth + P-E ratio growth + dy. This is from Grinold-Kroner. However, note: higher economic growth **doesn't reliably** lead to higher equity market returns because (1) higher growth rate already built into prices, and (2) if capital stock growing quickly, rate of ROIC could be lower.

Economic forecasting:

- Econometric modelling. Structural models based on theory. Reduced-form models are more compact, less dependent on theory. Data-driven models are the extreme, being very heuristic. Consistent process, disciplined procedure, generates explicit forecasts. Can be mis-specified. Bad at predicting turning points.
- Economic indicators. Leading, current, lagging. Leading economic indicator (LEI) examples: share price, manufacturing metrics, inflation, interest rates, monetary data (6-9 months ahead). Diffusion index: individual LEIs compiled, and see if they net point up or down. Use for nowcasting. Note: historical data revisions lead to look-ahead bias. Captures turning points, easy to track. Only provides binary (yes/no) answers.
- Checklist. Manually monitor big range of data and use for judgment, statistical analysis, etc. Flexible (can adjust list and weights to each item) but subjective. Subjective, inconsistent, subject to cognitive biases. Takes a long time because looking at a long, wide-varying list.

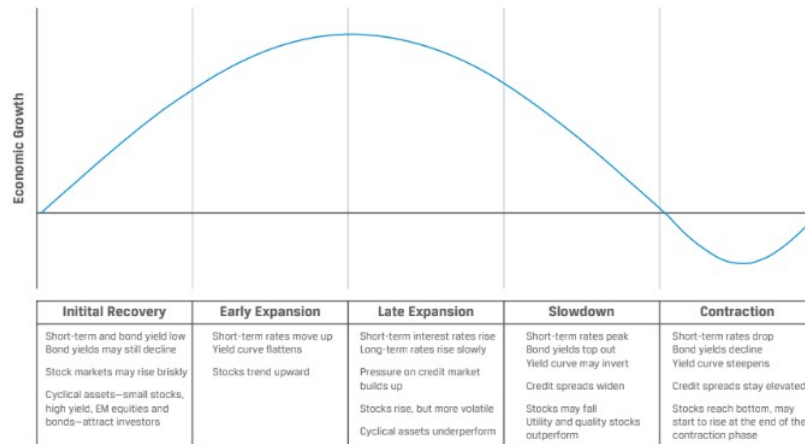
Business cycle analysis. Business cycles arise from

- Decisions made based on imperfect information
- Require lots of resources and time to implement
- Costly to reverse

Business cycle phases (p31):

- Initial recovery. Short-term rates are low, bond yields bottom out, stocks and other assets start doing well.
- Early expansion. Short rates rising, central banks withdraw stimulus, yield curve flattens. Stocks rise.
- Late expansion. Interest rates rise, yield curve continues to flatten, stock markets get jittery over when deceleration will come. Yield curve flattens because short-term rates rise (because of heavy private sector borrowing for capex and business expansion, and because of central bank actions to suppress inflation) while long-term rates don't rise much (they are long-term). A flattening yield curve means bond yields are rising slower than short-term rates.
- Slowdown. Short term rates are high and likely to peak, yield curve may invert, stock market may fall, credit spreads widen.

- Contraction. 12-18 months typically. Short rates drop, yield curve steepens. Stock markets fall. Credit spreads widen.



Highest signal-to-noise ratios over range of likely expansion and contraction phases (1-3 years). Longer than that, turning points become more likely. See Example 10.

Inflation and deflation. Assume aggregate supply curve is upward sloping and AD curve moves; this generates procyclical inflation. If AS curve moves instead, countercyclical inflation. Note: AS curve is flatter with more globalization.

- Inflation is procyclical (high near peak, low in trough) → procyclical bond yields → yield curve slope is countercyclical. Bond yields are high near peak because high short rates + low long-term rates = low or negative (inverted) term premium.
- Deflation. Bad for indebted: nominal debt burden increases; opposite of hyperinflation when nominal debt burden shrinks.
- Cash (short-term interest-bearing securities) values rise with inflation and fall with deflation.
- Bonds: rising inflation induces capital losses. Persistent deflation benefits high-quality bonds (bondholders of them) as cash flows are nominal and can 'buy more' with deflation; impairs creditworthiness of lower-quality bonds.
- Stocks. Normal inflation expectations built in. Change a bit, but more if inflation is very different from expectations or normal. If inflation rises, suffer as central bank will raise rates aggressively.
- Real estate. Normal inflation expectation built into real estate values. Rises with higher inflation, falls with deflation beyond normal levels.

Lessors of short leases will benefit most in inflation. If inflation rises a lot, RE with LT leases and little LT fixed-rate debt will suffer, but RE with lots of LT fixed-rate debt will do well as nominal value of debt falls in relative terms (i.e. real debt value falls).

- Inflation benefits firms able to pass on rising costs, i.e. those with pricing power.
- Deflation bad for firms that are asset-intensive (nominal value falls), commodity-producing, highly leveraged firms (nominal debt burden rises).

Both real and nominal bond yields, as well as equity, have return rates that are linked to the economic growth rate over the long term, though effect can be diluted if (1) already reflected in market prices, (2) capital deepening reduces returns on invested capital.

Monetary and fiscal policies. Fiscal policy has lags (e.g. policy) but has automatic stabilizers (e.g. progressive tax regime, means-based transfer payments), and generally investors analyst look at it for the LT. Monetary policy has bigger ST impact; central banks can move more aggressively than merely 'accommodating' economic growth using Taylor rule:

$$i_{target} = r_{neutral} + \pi_e + 0.5(Y_e - Y_{trend}) + 0.5(\pi_e - \pi_{target})$$

For the Taylor rule, need to know how expectations (growth, inflation) are formed, time horizons, values in formula, how growth and inflation likely to evolve. If short rates are negative, estimate the $r_{neutral}$ instead of using the negative rate for long-run calculations.

Zero or negative interest rates. Prior to GFC, it was thought that negative rates would lead to people carrying more cash and keeping less in the bank, leading to credit contraction and rising interest rates, thwarting central bank attempts at negative rates. However, move to cash didn't occur because modern economies can't just use physical cash. Instead, keeping cash in banks has a convenience value, allowing small negative interest rates that are sustainable.

QE. Increases central bank balance sheets. This should have allowed investors to buy assets (stocks, real estate) but often was just used by businesses to fund dividends and stock buybacks rather than capex. Households also just saved because of leftover problems from the GFC.

Negative interest rates and QE should work similar to low positive interest rates, but greater uncertainty and historical data less reliable. Quantitative models also break down. QE will also distort asset markets.

Fiscal policy. Politicized, e.g. if a government has rising debt, wants central bank to reduce interest rates and inflate away debt.

Exhibit 6. Nominal rates = real interest rate + expected inflation rate. Fiscal policy determines real interest rate (as businesses compete with government over borrowing). Monetary policy determines expected inflation. Tight monetary policy + tight fiscal policy = **low** nominal rates because of low real rates + low expected inflation. loose monetary + fiscal policies = high nominal rates. Other combinations = mid nominal rates.

Government bond maturities and tenors (maturity management). Large purchases or sales at maturities will temporarily change the yield curve but not after such purchases and sales end. If e.g. sales of assets at certain maturity are limited, yields will fall as supply falls and demand is still there - pension funds for example might want to continue to buy a lot of the asset so demand will remain high.

Yield curves.

- Shows where in business cycle we are at. But it is endogenous with the business cycle, so the business cycle might influence the yield curve and vice versa.
- Yield curve model links bond yields to policy rate.

International interactions.

- Large well-diversified economies (e.g. G7) are less impacted by international events than smaller ones (e.g. Chile).
- Current account = trade, net investment income, transfers. Capital account = net investment for FDI, portfolio investment transactions. $X - M = (S - I) + (T - G)$. Current account composition reflects secular trends, while capital account composition reflects short-term adjustments. Current and capital accounts kept in balance by changes in GDP, relative prices, interest rate and asset prices, and FX rates.
- If taxes reduced to stimulate investment, with no change in G , then T falls and I rises. Higher Y might mean increase in S , but probably not enough to offset increase in current account deficit. The deficit is funded by increase in capital account, i.e. borrowing from abroad, selling assets abroad, reduced FDI. This will increase exports from other countries. Higher Y and lower T means equity values rise. Interest rates rise in the IS-LM model, which in turn raises bond yields.

Trilemma. Can't simultaneously have free capital flows, fixed FX rate, and independent monetary policy. For fixed FX rate, interest rates must be the same if unrestricted capital flows; interest rates can be different if capital flows restricted. For fixed rates, always need credibility to fix forever. If (1) unrestricted capital flows (2) FX fixed, then yield curves will be same. But if credibility under threat, yield curves will differ.

Even if non-domestic interest rates remain unchanged, real returns earned by investor will not equal non-domestic real interest rate unless PPP holds. And PPP does not hold. So, real interest rates will not be the same across countries. Real interest rate differentials might hint at FX rates however.

Global S and I will determine real interest rates. They might be different between countries but they will trend together. Notes from q14 and 16

- Deflation may make cash attractive if a 'zero lower bound' is binding on the nominal interest rate. Otherwise, deflation is simply a component of the required short-term real rate. Nominal interest rate = real interest rate + inflation. If inflation is -2%, i.e. deflation, then if nominal interest doesn't fall below 0%, the real interest rate is +2% and holding cash is a gain.
- Falling inflation results in capital gains of bonds as the expected inflation component of bond yields falls. A smaller bond yield = increase in bond price (and so, the yield at maturity falls).
- To equalize risk-adjusted returns across countries, interest rates must generally be higher in the country whose currency is expected to depreciate. This dynamic leads to a situation where the currency overshoots initially, and then the higher interest rate country's currency depreciates.
- q51 of PMP FI active credit strategies. How to calculate bond VaR from yield volatility. How to calculate the bps from bond FV, price, and Mod-Dur.
- Recommendations for Standard VI(A)–Disclosure of Conflicts advises that employee compensation packages based on short-term performance be disclosed to clients, which is currently not allowed by Ravinder's employer. Thus, she should renegotiate her compensation package to either remove the performance aspect of the structure so she can disassociate from the practice or seek to lengthen the performance period assessed to, at minimum, one year. If Ravinder disclosed the compensation package to her clients without her employer's permission, she would be in violation of Standard IV(A)–Loyalty. If she asked her clients to renegotiate their contracts, she may potentially violate Standard III(A)–Loyalty, Prudence, and Care because the new contracts may not be as favorable as the current contracts, thus potentially harming her clients.
- Standard VI(A)–Disclosure of Conflicts requires a member or candidate who becomes a director of a publicly listed company to be isolated from those making investment decisions concerning the publicly listed company at which the employee is a director. One such way of doing this would be to not participate in investment strategy meetings when Naivasha is being discussed. But Ravinder would violate Standard I(D)–Misconduct if she

shared nonmaterial information if the directors of Naivasha considered it to be confidential. She would also violate Standard III(A)–Loyalty, Prudence, and Care if she excluded Naivasha shares from her clients’ portfolios because this approach may negatively affect their investment performance.

- Minerva’s strategy is consistent with a quantitative strategy. Minerva looks at firm characteristics that correlate with economic macro-level factors and constructs factor exposures from these characteristics, which it uses to invest across a large number of firms. The factors are back-tested and statistically validated. Stocks in the portfolio are traded at predetermined levels when certain thresholds are met (systematic rebalancing). These are all characteristics of a quantitative strategy. In addition, a large proportion of return is explained by factor exposures ($88\% = 9.57\% / 10.90\%$) and it has a low alpha ($0.13\% = 10.90\% - 1.2\% - 9.57\%$). Both of these indicate low unexplained return (idiosyncratic risk) in the portfolio.

1.2 Capital market expectations II: Forecasting asset class returns

Forecasting expected returns is most important.

Forecasting methods: (1) formal tools, (2) surveys, (3) judgment.

- Formal tools: statistical, DCF, risk premium. Statistical: sample statistics (means, variances, correlations), shrinkage estimation, time series estimation (these are reduced form).
- Surveys.
- Risk premium: (1) equilibrium models like CAPM, (2) factor model, (3) building blocks.

Forecasting fixed income returns: DCF, building block, and equilibrium models.

DCF. Most precise. Macaulay Duration = modification duration $\times (1 + \text{YTM})$. If investment horizon shorter than MD, capital gain or loss impact will dominate. **Example 1 on p73.**

Building block approach: one-period default-free rate + term premium + credit premium + liquidity premium. Only riskfree rate is observable.

- Short-term rates. Use the rate matched to the maturity of the investment horizon. If doesn’t match, use longer one (which will include the term premium), forecast short-term rates without term premium, short-term instrument futures, Taylor rule, etc.

- Term premiums. Positive and increase with maturity, roughly proportional to duration, and vary over time. Four drivers: (1) inflation uncertainty - higher inflation means higher inflation uncertainty and higher term premium, (2) Ability to hedge recession risk - if aggregate demand driven then nominal bond returns negatively correlated with growth and inflation; if aggregate supply driven then positive correlation; (3) supply and demand of bonds - the relative supply of ST and LT bonds is part of the term premium because this factor shouldn't influence the path of ST interest rates, (4) business cycle. Economic confidence (related to (2) and (4)) lower means higher term premium. The term premium reflects macro-related risks, inflation, reinvestment, etc. risks.
- Term premium is also the difference between expected returns on riskless rate and default-free bonds. Riskless rate = no inflation uncertainty, no interest rate risk. Default-free doesn't have these.
- Credit premium. This is **additional** to compensation for expected losses (so, the buildup approach is additional to the bond's cash flows, principal, default losses, etc.). Both default losses + credit premia in credit spreads. Historical analysis shows credit premiums exist, though they are not uniform and consistent through time. Credit premiums of different bond ratings (e.g. AAA v B) are driven by different factors (e.g. default probability, downgrade bias). Steep yield curves, associated with low point of business cycle, are historically associated with high OAS spreads - so, steep yield curves are associated with both term and credit premia.
- Liquidity premium. Yield spread between highest-quality issuer and lower-tier quality issuer as a starting point, then subtract off credit premium, factors such as how new it is, etc.

Equilibrium model. Include fixed-income asset classes in an equilibrium model. Imposes consistency across asset classes and useful for Black-Litterman. To be discussed later.

Risks in emerging market bonds. Heterogeneous group. As a group, more risks than developed economies. Watch these ratios:

- Fiscal deficit to GDP ratio above 4% is a problem.
- Debt-GDP ratio of 70-80% is a problem.
- GDP growth of less than 4% means only a slow catchup to developed economies.
- Current account deficits of more than 4% of GDP indicate lack of competitiveness.
- FX reserves less than 100% of ST debt (better to have more than 200%).

- Others: foreign debt percentage greater than 50% of GDP or greater than 200% of current account receipts.

Forecasting equity returns. Equity returns across a variety of countries: averages are similar (5-10%) and confidence intervals are wide, so no statistical differences. Two approaches: Grinold-Kroner and Singer-Terhaar.

P-E ratio and E-GDP ratios are volatile, but they tend to cancel out, leaving equity return growth dependent on just GDP (E's cancel out) which is stable.

Grinold-Kroner (supply of returns) approach to equity returns (nominal, absolute return - it's not relative to the riskless rate):

$$E(R_e) = \frac{D}{P} + (\% \Delta E - \% \Delta S) + \% \Delta P/E$$

Values are nominal. $\% \Delta E$ is annual real earnings growth + annual inflation. Middle term is basically EPS. Expected cash flow = expected income annual return = $D/P - \% \Delta S$. Horizon is infinite. Over long run, growth $\% \Delta E$ = nominal GDP growth, $\% \Delta S = 0$ and $\% \Delta P/E = 0$.

Equity premium = ERP. Two definitions. Equity versus bills: ERP = equity expected return minus riskless rate. Equity versus bonds: ERP = expected equity return minus expected return on defaultless bonds.

Risk-premium approach (demand of returns) to equity returns. Equity risk premiums are just as hard to forecast as equity returns themselves. Equilibrium approach: Singer-Terhaar model calculates equity risk premium by assuming global CAPM segregated from individual CAPM where the asset i is not fully integrated because of ϕ :

$$RP_i = \phi RP_i^G + (1 - \phi) RP_i^S$$

ϕ is the degree to which asset is globally integrated (e.g. if country's market is fully integrated with global economy, then $\phi = 1$), which is distinct from the degree of correlation to the global market $\rho_{i,GM}$. Fully-integrated market (i.e. $\phi = 1$) risk premium:

$$RP_i^G = \beta_{i,GM} RP_{GM} = \rho_{i,GM} \sigma_i \left(\frac{RP_{GM}}{\sigma_{GM}} \right)$$

Fully segmented market risk premium:

$$RP_i^S = 1 \times RP_i^S = 1 \times \sigma_i \left(\frac{RP_i^S}{\sigma_i} \right)$$

Can assume global Sharpe ratio = individual Sharpe ratio = compensation for risk is same in every market. Emerging markets are less integrated so country-specific effects are relatively more important than global ones.

The Singer-Terhaar model implies that as a market becomes more globally integrated (ϕ rises), its required return should decline because global investors can now diversify away a larger portion of the market's risk (since $RP_i^S > RP_{GM}$ assuming same Sharpe ratios, RP falls as ϕ rises).

Emerging market equity risks. Specific country effects more important than for developed economies generally because less integrated into global economy and standards. Corporate governance standards (allowing parties to misuse assets), accounting standards, weak disclosure rules, weak property rights, lack of enforcement, weak governments, etc. Declining GDP per capita means potential political strife.

Forecasting real estate returns. Not enough transactional data, so use appraisals. Appraisals don't bias the mean but too smooth. Low-quality properties, properties with short leases, hotels will be impact most by real estate cycle.

Another problem with forecasting real estate returns: each property is different; heterogeneous.

Capitalization rate is used in CRE:

$$caprate = \frac{NOI}{PropertyValue}$$

In the steady state, $g_{NOI} \approx g_{GDP}$, and so

$$E(R_{re}) = caprate + g_{NOI} = caprate + g_{GDP}$$

And in the short-term, caprate fluctuates:

$$E(R_{re}) = caprate + g_{NOI} - \% \Delta caprate$$

g_{NOI} for commercial RE = GDP growth rate. We begin with caprate and NOI growth because that's the 'baseline' return. Then **subtract** $\% \Delta caprate$, because starting with a given caprate if it falls the property value rises and expected return rises. This is analogous to the Grinold-Kroner equities model, except no share buybacks; NOI and caprate are like the dividends and capital appreciation.

Example. If 1,000,000 property rises 1% and NOI of 10,000 rises 10% to 11,000, then expected return = initial caprate + g_{NOI} - % change in caprate = 1% + 10% - x%. Initial caprate = 1%, while new caprate = $\frac{11000}{1010000} = 1.09\%$. So, $x = \frac{1.09-1}{1} = 9\%$. So, expected return = 1% + 10% - 9% = 2%.

The above equation is actually

$$E(R_{re}) = caprate_0 + (1 + g_{NOI})(\% \Delta P)$$

$$\begin{aligned}
&= \frac{NOI_0}{P_0} + \frac{NOI_1}{NOI_0} \left(\frac{P_1 - P_0}{P_1} \right) \\
&= \frac{NOI_0}{P_0} + \frac{NOI_1}{NOI_0} + \frac{NOI_1}{NOI_0} \left(\frac{P_0}{P_1} \right) \\
&= \frac{NOI_0}{P_0} + \frac{NOI_1}{NOI_0} - 1 + \frac{NOI_1}{NOI_0} \left(\frac{P_0}{P_1} \right) + 1 \\
&= \frac{NOI_0}{P_0} + \frac{NOI_1}{NOI_0} - 1 - \left(\frac{NOI_1}{NOI_0} \left(\frac{P_0}{P_1} \right) - 1 \right) \\
&= \frac{NOI_0}{P_0} + \frac{NOI_1 - NOI_0}{NOI_0} - \left(\frac{NOI_1/P_1 - NOI_0/P_0}{NOI_0/P_0} \right) \\
&= caprate_0 + g_{NOI} - \% \Delta caprate
\end{aligned}$$

Caprates are positively correlated to the business cycle because they are related to long-term interest rates (e.g. high caprate when high interest rate at business cycle peak); but mitigated by countercyclicality because they are related to credit spreads (e.g. high caprate when credit spreads during recession are high).

The $\% \Delta caprate$ is calculated **not** by taking differences e.g. $5.7 - 5.5\% = 0.2\%$ but rather by $0.2\%/5.7\%$.

Other features of RE:

- RE has a high duration, so it's sensitive to interest rates.
- Liquidity premium of 2-4% for CRE.
- Equilibrium model (e.g. Singer-Terhaar): remove impact of smooth (otherwise, β is too low) and add liquidity premium (equilibrium models assume liquidity).

Forecasting exchange rates. Difficult. FX changes driven by: goods and services, and the capital account.

Good and services: PPP and current account. PPP is difference between inflation rates (real FX rate: ratio of prices converted to common currency): expected change in real FX rate should be 0 through quasi-arbitrage; works slowly so better over LT. Trade flows: generally don't exert big effect as **net** trade to GDP is usually small. Competitiveness, sustainability of current account surpluses and deficits. Note: Current account surplus: national savings exceeds investment: $(T - G) + (S - I) = (T - G + S) - I > 0$.

Capital account. Consider degree of capital mobility. Capital seeks highest return. Expected changes in FX rate will reflect differences in:

$$E(\% \Delta S_{d/f}) = (r^d - r^f) + (Term^d - Term^f)$$

$$+(Credit^d - Credit^f) + (Equity^d - Equity^f) + (Liquid^d - Liquid^f)$$

Expected returns in the money market (nominal rate r), government bonds (r , term), corporate bonds (r , term, credit), public equities (r , term, credit, equities), private assets (all terms).

- Shorter-term changes to FX happen in the capital account, less in goods and services.
- If $E(\% \Delta S) > 0$ then expect depreciation (actually, the FX strengthened right before this e.g. overshoot according to Dornbusch because prices are sticky and can't move initially, and now will depreciate as in CIP when prices can move slowly).
- UIP relates only to the first term. Contrary to this, carry trades are profitable. So, there must be risk, consistent with the above equation.
- **Hot money flows.** Fast money flows in or out of countries. From changes in interest rate - see Mundell Fleming. Problem for central banks, so they try to sterilize these flows.

Portfolio balance. If country has growing investable assets (or just any assets), then to attract investment, FX rate has to weaken. This happens when, e.g. assets become underpriced and investors don't buy 'enough' of the growing assets. Countries want FDI the most as it grows their economies, though private RE and private equity inflows are good too - no need to repay; but foreign debt can create sustainability issues.

Note: In the questions, a short-term (1-month) government rate rise leads to a strengthening of the currency, not a fall.

Forecasting volatility. Use in Singer-Terhaar models.

- Two types of VCVs: (1) sample VCV, and (2) Multifactor VCV.
- VCVs generally: asset number can't exceed observations (number of observation series common to entire time series). Otherwise, singular matrices.
- Imposing structure with multifactor model leads to much smaller VCV matrices, and improves cross-sectional consistency (i.e. internal model consistency). But, can lead to biased and inconsistent VCVs.
- Shrinkage estimation of VCVs. To get benefits and information of both sample and multifactor VCVs, use shrinkage estimation. Target VCV = built from multifactor VCVs.
- General rules. Need 10-to-1 generally observations to assets for sample VCV. Factor VCV can be used even if asset number exceeds number of observations.

Private equity, private RE often have smoothed returns (e.g. from appraisals, or averaging missing data). To add back volatility:

$$R_t = (1 - \lambda)r_t + \lambda R_{t-1}$$

where r_t is the unobservable true return, and the observed returns are R . Note $0 < \lambda < 1$ and can be re-arranged as

$$Var(r) = \frac{1 + \lambda}{1 - \lambda} Var(R) > Var(R)$$

ARCH models. Address volatility clustering and other conditional heteroskedasticity.

$$\begin{aligned}\sigma_t^2 &= \gamma + \alpha \sigma_{t-1}^2 + \beta \eta_t^2 \\ &= \gamma + (\alpha + \beta) \sigma_{t-1}^2 + \beta (\eta_t^2 - \sigma_{t-1}^2)\end{aligned}$$

where the last term is the shock to the variance. The higher $\alpha + \beta$, the more variance is 'remembered'. **Example effects**

- If economic growth trend rises, equity rises, real interest rates rise. Shift from bonds to equities.
- If greater global integration, then required equity return falls due to Singer-Terhaar. So shift to developing markets.
- At bottom of business cycle, (1) reduce bond durations (because interest rates are about to rise), (2) move away from intermediate maturities (as they will not do anything) and invest in short-term bonds (more liquidity and once mature, can buy higher-interest rate bonds) and long-term bonds (when there is a flight to safety or as economy recovers, rates will fall and bond prices will rise), and (3) get exposure to credit (as economic recovery will reduce credit risks and increase credit prices).
- Monetary and fiscal policies. If central bankers doing what is expected, then no changes (expectations already built into asset prices) unless effect is unexpected. Moreover, if there's a deliberate policy shift (e.g. from interest rate targeting to money growth targeting) then consider.
- Current account balances. Long-term, if deficit then r_e needs to be higher to increase savings over investment, and if surplus then r_e falls.
- Capital account balances.
- Mock4A. v1 q3 calculation. v2 private found flow-through. v3, yield curve is upward sloping, spread risk. v3 q11 calculation. v5 ethics questions. q26 - how to compare PEG - PE v sector average or just straight up PEG? Term premium calculations. q35 upward yield curve. Costing seagull options.

1.3 Overview of asset allocation

Investment governance - high level. Investment management - execution level.

Governance: investment committee, investment staff (e.g. investment managers), third-party resources (external investment managers, investment consultants, actuaries, etc.). Governance is about determining long- and short-term objectives of investment program, allocate decision rights and responsibilities, specify processes, establish reporting frameworks, and doing governance audits.

Investment objectives. Objective statements: simple one-sentence description. Ultimate goal to find best risk-return trade-off. In the objectives, also consider liquidity, risk tolerance. Specify short and long-term objectives.

IPS. Changed slowly over time. More variable aspects in appendix. Should contain some information about how to rebalance.

Investment committee: strategic asset allocation and rebalancing.

Reporting. Where are we relative to goals and objectives, and has management added value? Benchmarks: (1) how well PM relative to purpose, and (2) gap between policy portfolio and portfolio actually implemented. Management reporting: which parts of portfolio performing well and which not, and why. Governance reporting: addresses strengths and weaknesses in execution.

Governance audit. Should be done by third party, to test governance structures effectiveness. Audit to determine whether program can survive unexpected market turmoil, decision-reversal risk, considers staff turnover, key person risk.

Economic balance sheet. Beyond just financial assets and financial liabilities. They include **extended (portfolio) assets and liabilities**, such as human capital, mineral resources, intellectual property values, prospective payouts for foundations. Extended liabilities include goals (non-legal liabilities). Include PV pre-retirement earnings (extended asset) and PV consumption (extended liabilities). Economic balance = economic net worth. Use economic balance sheet, consider risk with relation to economic BS, to arrive at asset allocation decision. Note: policies (e.g. policy to disburse grants) is not an extended liability.

Economic balance sheet: Life-cycle balanced funds (target date funds). Retirement vehicle. Coordinate asset allocation with human capital. Human capital is partly equity-like (varies depending on state of economy, and shrinks as working horizon gets closer) and bond-like (steady wages). With constant risk tolerance through life, more equities when young and more bonds with older.

General note: Assets minus liabilities in economic BS = economic net worth.

When calculating economic BS, don't exclude 'restricted spending' items, e.g. funds restricted to scholarships.

Approaches to asset allocation:

- Asset-only. Typical objective: maximize Sharpe ratio for given level of volatility. Typical risk measures (for some of these, use Monte Carlo simulations): standard deviations of returns, tracking error or risk, semi-variance, maximum drawdown, VaR. Mean-variance optimization MVO. Considers only expected returns, risks, and correlations of asset classes. No consideration of liabilities.
- Liability-relative, liability-driven investing (LDI). Typical objective: fund liabilities and invest excess assets for growth. Typical risk measures: shortfall risks like volatility of contributions. Account for economic BS liabilities. Example: MVO applied to surplus (PV assets minus PV liabilities), where liabilities might be modelled as a short position in a bond matched to the PV and duration of liabilities, and then risky-asset portfolio for rest. Another approach: liability-hedging portfolio + risky-asset portfolio for balance of assets.
- Goals-based, goals-based investing (GBI). Typical objective: achieve goals with probabilities of success. Typical risk measure: max probabilities of not achieving goals. Account for economic BS quasi-liabilities (not strictly legal liabilities but almost like them) and goals. Often specify asset allocations for sub-portfolios, where each sub-PF is about a specific goal. Usually for individual investors and are more specific (e.g. not like pension benefits, for many people). (Note: asset segmentation by some life insurers is similar to goals-based investing; assign assets into sub-PFs for specific lines of business for competitive reasons - to get certain credit ratings on certain business lines).

Modelling asset class risks. 3 'super classes' of assets:

- Capital assets. Source of value, such as interest and dividends. Valued by NPV.
- Consumable or transformable assets. Assets such as commodities which can be transformed into something of value.
- Store of value assets. E.g. currency and art.

Some are multi-class depending on the situation, e.g. precious metals in both consumables and store of value. Some asset classes don't fit into any, e.g. hedge funds which invest in asset classes (HFs are rather 'strategies') and volatility which is an investment characteristic.

To group assets into classes for effective investment, asset classes should be (1) homogeneous, (2) mutually exclusive, (3) diversifying (i.e. should not be

highly correlated with other asset classes; here, don't just use pairwise correlations), (4) when summed together should make up most of investable assets, and (5) should be economical to invest in in a significant proportion (shouldn't be prohibitive due to transaction costs, etc.) by not having excess class granularity.

Often, asset classes are: (1) global public equity, (2) global private equity, (3) global fixed-income, (4) real assets (real estate, commodities, etc.). Often subdivide into developed and developing countries, or other classifications, because of diversification potential (and degree of market integration), information efficiency, institutions.

Strategies. Investment vehicles such as HFs providing good returns relative to risk involving skill are 'strategies' or 'diversifying strategies'. Then, it is assigned a percentage asset allocation, just like other asset classes. Other, regular asset classes = no skill required.

Asset allocation = best done between broad classes like equity versus bonds. More historical data, better investment diversification and other properties. However, for policy implementation, sub-asset classes become more important.

Traditionally, asset allocation used asset classes as units of analysis, e.g. MVO on global public equity, global private equity, global fixed income, real assets based on expected return, return volatility, return correlation. However, factor-based models 'look' at investment from a different angle. Factor-based models don't by default however expand investment outcomes, so don't lead to superior outcomes. Factor-based models take advantage of the fact that asset classes have some common factor exposures (e.g. liquidity) across e.g. equities and bonds. So controlling such common factor exposures is hard with asset classes as units of analysis, ergo factor-based models. But, factors aren't directly observable. Examples of factors: see p160.

Strategic asset allocation. Policy portfolio or strategic asset portfolio: expected to achieve IPS. Maximize utility subject to portfolio weights. Simplest allocation:

$$w^* = \frac{1}{\lambda} \left(\frac{\mu - r_f}{\sigma^2} \right)$$

where LHS is optimal weight of risky asset, λ is investor's degree of risk aversion.

Strategic asset allocation: Asset-only. Typically use MVO to get efficient portfolio frontier of PF choices and then look at Sharpe ratios of these PF choices, but note Sharpe ratios have weaknesses (e.g. doesn't specify absolute amount of returns or risk, doesn't say anything about diversification). The **global market PF** is the most efficient PF for risk-budgets (minimizes diversifiable risk); in selecting a PF, need to justify why deviating from global market PF (e.g. why home bias?).

Strategic asset allocation: liability relative. Example: DB pension, where underfunded potentially means pension sponsor has to make up with extra contributions, which is PV contributions. Going with an asset-only allocation (based on maximum Sharpe ratios, for e.g.) could lead to higher risk and potential for greater PV contributions. So might be better to go for lower risk PF, i.e. liability-hedging PF (matched in interest rate sensitivity to plan disbursements) + return-seeking PF. Generally, use more bonds than equities relative to asset-only allocations (though equities can be used for greater inflation protection and in underfunded plans to boost returns). **Liability glide path**, often used in underfunded plans, technique where plan sponsor specifies proportions of liability-hedging assets and return-seeking assets, duration of liability hedge as funded status changes. Increases funded status by reducing surplus risk over time (as liability cash payments near).

Note: funded ratio = pension assets to liabilities.

Strategic asset allocation: goals based. Take optimized portfolio and assign funds to different goals, may require some adjustments but overall frames it as mental accounting. Once we can show the investor that their goals are going to be taken care of, and then on top of that add risky assets - which would otherwise scare the investor if they were just added to the 'total portfolio' without segregated accounts. In effect, the segregated accounts allow us to add risk assets to the portfolio without scaring the investor.

GBI. Divide goals into scenarios, use different required rates of return on each. Specify time horizons, required probability of achieving.

If presented with an opportunity for an early-stage VC investment in the same industry that the investor works in, this would be risky and illiquid and LT. However, if the investor is still working-age, then it is not diversifying to their human capital (because VC is part of same industry).

GBI. If the required probability of success is high, then required discount rate should be lower.

Implementation choices.

Tactical asset allocation. ST deliberate deviations from strategic asset allocation - there is (1) systematic TAA (relying on rules) and (2) discretionary TAA. This is different from **dynamic asset allocation**, which is a strategy of deliberately deviating from SAA by LT signals or economic views. TAA is about market-timing. TAA can trigger higher capital gains taxes, so do a cost-benefit analysis.

Active-passive distinction. Can be a combination of active (e.g. hold only equity instead of the global market portfolio which is a mix of equity, bonds, other

assets) and passive (e.g. use an all-equity index). Use quantitative measures such as tracking risk and active share to quantify degree of active management.

Even passive investing, e.g. by indexing, incurs some transaction costs from rebalancing portfolios to keep up with the rebalanced index. However, active investing usually leads to greater taxes because of higher turnover.

Risk budgeting. Another way to look at asset allocation. Example: risk-parity approach. There is active and passive risk budgeting.

Factors to consider in passive-active investing choices.

- Whether there are investable indexes.
- Scalability. If the active strategy is a big proportion of the investor's assets, the active management value may begin to decline. If it's too small a proportion, taxes and other transaction costs may be too high.
- Feasibility of investing within IPS constraints. E.g. if investor specifies ESG, then may need active management in addition to passive.
- Beliefs about information efficiency and market efficiency.
- Tradeoffs. Costs like taxes.
- Tax status. More important for individual investors who are taxed.

Generally pensions want passive management as the pension fund executives have lots of other responsibilities. However, might sometimes delegate to an external investment manager for active management.

Rebalancing. Mainly about risk control, not return enhancement. Can be done because of index changes, price changes causing high-return assets to grow higher in proportion in PF leading to higher risk - risk beyond IPS, equity-bond reflecting IPS risk-return changes as asset prices change. Rebalancing is countercyclical (sell best-performers and buy low-performers).

Calendar rebalancing. Target weights on a periodic (e.g. monthly, annual) basis.

Percent-range rebalancing. Rebalancing within ranges (trigger points). Set trigger points using + and - or using cost-benefit analysis.

Rebalancing can be to trigger point, back to original IPS policy weights, or to midway, etc.

Considerations:

- High transaction costs favor wider ranges, e.g. international assets.

- High risk assets = wider ranges because of transaction costs. Theoretically, should have narrower but there are transaction costs.
- Risk-averse investors have tighter rebalancing ranges.
- Less correlated assets have tighter rebalancing ranges - they are more likely to deviate from SAA. (Note: for risk purposes however, we want low correlation between asset classes). If one asset class above target weight, another must be below; if classes are positively correlated, then an upward market move means the overweight class increases a lot in value while the underweight one also increases instead of decreasing, so further target weight changes are less likely.
- Momentum favors wider ranges. Mean reversion = tighter ranges.
- Illiquid investments complicate rebalancing. Usually wider rebalancing ranges as they are hard to sell. Or perhaps rebalance elsewhere in PF to offset this risk. Usually assign much smaller proportions in SAA to illiquid assets than would normally be expected from regular trading.
- Derivatives allow synthetic rebalancing. Often low transaction costs.
- Taxes are costly to discourage rebalancing and encourage asymmetric and wider ranges.
- Higher volatility = narrower range given all else equal. A small divergence under high volatility can easily become a much bigger divergence. However, due to transaction costs, often higher volatility = wider range (as in volatility-based rebalancing and equal probability rebalancing). Usually, higher volatility of illiquid assets = wider (transaction costs are cheaper than expected U losses), and higher volatility of liquid assets = narrower (transaction costs are higher than expected U losses).

Whether to rebalance to range limits or to targets exactly: weigh transaction costs with alignment to target weights. Also, rebalancing now affects rebalancing in future (if more aligned to target now, easier later). Studies show if fixed transaction costs, better to get to target weights and if transaction costs are variable then use range limits.

Factor-based allocation, liability-relative investing, goals-based investing will all give different rebalancing requirements.

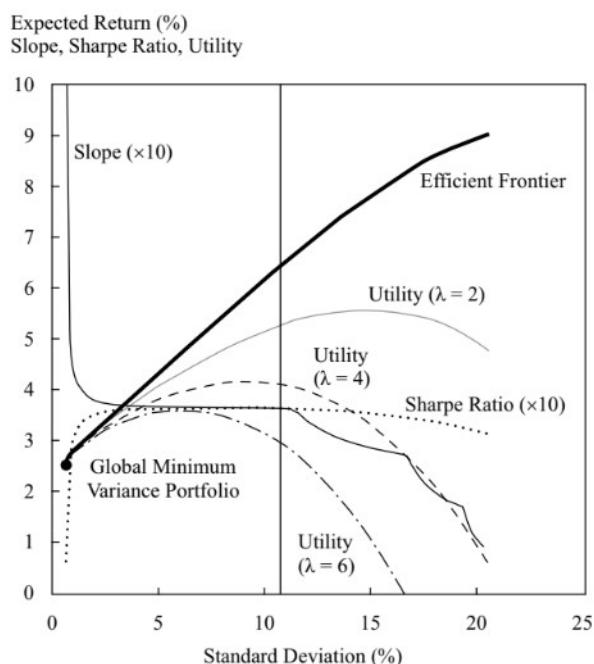
1.4 Principles of asset allocation

SAA is 2 steps: (1) asset allocation, (2) TAA. Can be done in 1 step but usually split into 2 steps because (a) each step is complex, and (b) clients want to review (1) infrequently and (2) more frequently and splitting is more convenient for this.

Mean-variance optimization. A 'single-period' (e.g. week, year, etc.) framework. Most common way of determining SAA. Use utility to determine an investor's risk-adjusted return preference (i.e. preference which takes into consideration both returns and risk):

$$U_m = E(R_m) - 0.005\lambda\sigma_m^2$$

where m is asset mix, R_m is the return of the asset mix, λ (generally from 1-10) is the investor risk version coefficient (higher if more risk-averse; $\lambda = 0$ is risk-neutral), and σ_m^2 is expected variance return for the asset mix. Values in % - e.g. 10% = 10, 10% standard deviation is 100%².



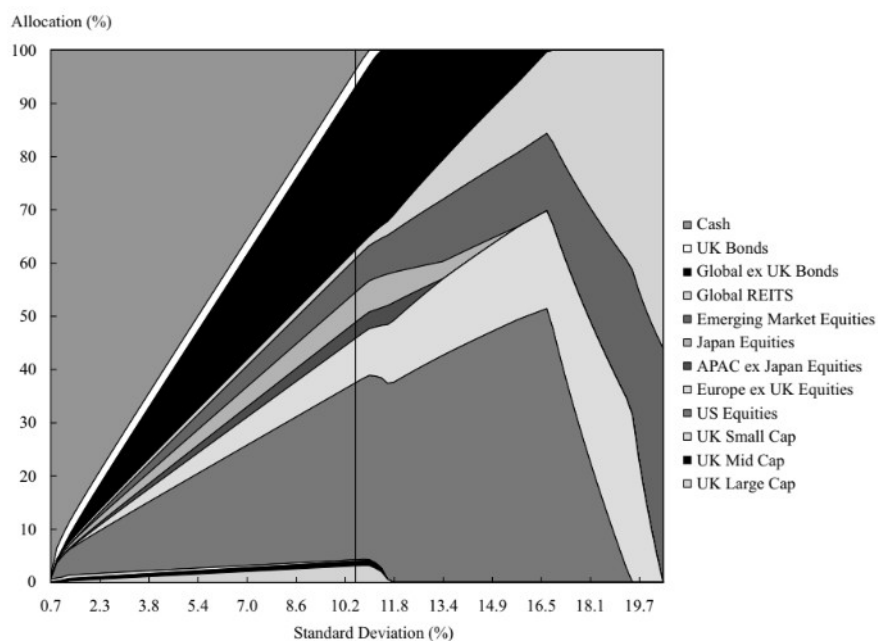
Above chart, with lower risk-aversion = higher inverted utility curves. Use 0.005 if percentages are numbers; otherwise, use 0.5. U_m is a 'certainty-equivalent return', i.e. utility of risky return of asset mix; optimize this without constraints, but generally $\sum w \neq 1$ with large long and short positions. However, usually want $\sum w = 1$ for **fully-invested, non-leveraged** PF (and additionally often $w_i > 0$).

Risk-aversion, λ . Consider both risk preference (willingness) and risk capacity.

Efficient frontier. Between minimum-variance PF and maximum expected return PF. Note max expected return PF is 100% allocation to single asset with highest return (which is not necessarily the asset with the highest level of risk).

Can have efficient frontier (1) with cash as an asset (default in the text) or (2) cash is 'part of' riskfree asset.

Kink in above chart is when asset in efficient frontier leaves or enters mix (corner portfolios).



Efficient frontier can be remapped as above asset allocation graph. **Corner portfolios.** Efficient frontier points where asset class enters or leaves the mix or a constrain becomes binding or ceases to be binding.

Safety ratio. If have a minimum return requirement R_L , e.g. have to withdraw a certain amount in X amount of time without reducing the principal, then look for highest safety ratio:

$$\frac{E(R_p) - R_L}{\sigma_P}$$

Monte Carlo simulation. MVO is single-period + no rebalancing; MC is multi-period + rebalancing (triggering capital gains and losses), and provides probabilities, maximum potential drawdowns (i.e. risk), transaction costs, taxes, evolving financial markets, etc.

Analytical v Monte Carlo. MVO is analytical. Analytical only works if path independent (e.g. cash inflows and outflows independent). Monte Carlo works even if path dependent.

Criticisms of MVO.

- Output is sensitive to inputs. But this is true of any model where forecasts are inputs. Note: MVO most sensitive to expected returns, more than volatilities and correlations.
- Asset allocations highly concentrated in a subset of available asset classes, as it tends to be very sensitive to differences in expected returns - it easily maximizes and prioritizes asset classes with the highest expected returns. MVO uses historical returns and covariances, and risk-aversion coefficient as inputs. Compared to historical data, return optimization (RO) and Black-Litterman would likely result in asset weights equal to and closer to global market portfolio, respectively. RO works backward compared to MVO: it takes the global market portfolio, asset volatilities and correlations, and then works out the implied expected returns by outputting the beta. BL is like RO in that it starts with the global market portfolio, but the investor can add and blend their views, and then implied expected returns are outputs.
- MVO only focusses on mean and variance and not other factors like skewness and kurtosis.
- Sources of risk not diversified despite asset allocation being diversified.
- MVO not connected to liability or consumption series.
- MVO is single-period framework.

Approaches to address MVO criticisms.

Reverse optimization. MVO: expected returns, covariances, risk aversion coefficient are inputs, and optimal weights are outputs. Return optimization: assumed optimal weights, covariances and risk aversion coefficients are inputs and expected ASSET returns (implied or imputed) are outputs. From this, calculate expected PF return. The process of reverse optimization leverages the condition that at the optimal portfolio weights, the **marginal utility** contributed by adding a dollar of any asset must be equal. Mathematically complex. Often, reverse optimize the market portfolio (i.e. find the implied returns that justify the market portfolio), or reverse-optimize the investor's PF: market capitalizations of chosen asset classes as starting optimal weights, policy weights, fundamental weighting schemes, etc. Use the total PF for the investor and then calculate betas per asset class, add to r_f and get expected returns.

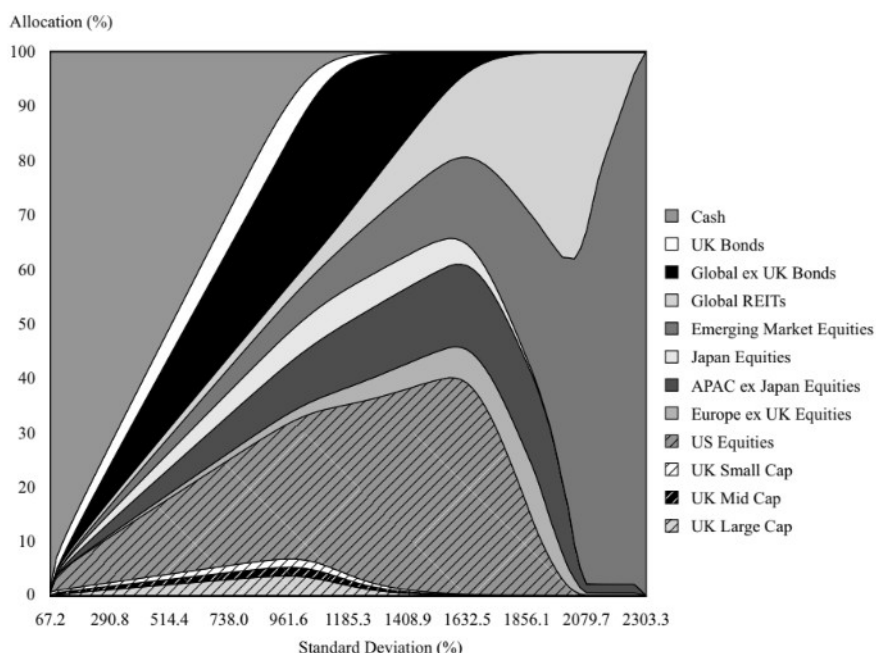
Black-Litterman. Start with excess returns (in excess of r_f) from RO and then alter them to reflect investor's views, and then feed into optimizer to get **well-behaved** optimal weights (generally, not major kinked changes; if changing returns directly and putting them into optimizer, often get huge changes due to the MVO sensitivity issue). Black-Litterman model accepts (1) absolute return forecasts and (2) return differentials between asset classes. These are blended with RO returns to create a new set of (Black-Litterman) returns. So

if one (1) or (2) is changed, all returns change at least a little bit.

RO and Black-Litterman address MVO sensitivity to small differences in expected returns by anchoring expected returns to those implied by asset class weights of a proxy for the global market portfolio.

Additional constraints. To incorporate more realistic constraints, investor constraints (e.g. must have 30% for RE, short positions representing liabilities), or to get around MVO problems (input sensitivity, overconcentrated allocations).

Resampled MVO ('resampling'). MVO + Monte Carlo. Use Monte Carlo to find many possible frontiers ('simulated frontiers'), and then plug into MVO to get optimal weight sets - average the sets to get optimal weights. The asset allocation area graphs (chart below) are 'smoother' and more 'balanced' (larger allocations shrunk and smaller allocations grew). However, it also leads to (1) frontiers having 'bumps' where expected return decreases even as risk increases, (2) 'riskier' asset allocations are over-diversified, (3) optimal weights have estimation errors, (4) no theoretical foundation.



Non-normal returns and asymmetric risk preferences: mean-semivariance optimization, mean-conditional VaR optimization, mean-variance-skewness optimization, mean-variance-skewness-kurtosis optimization. MVO uses utility function which is expected return minus risk penalty based on variance (mean = first moment, variance = second moment). This works if returns are normally distributed (skewness = kurtosis = 0). But if returns are non-normal,

then need skewness and kurtosis. Moreover, investor preferences are skewed (negative worse than positive).

Unconditional v conditional inputs. Unconditional inputs = don't depend on current market or short-term market or business cycle. Conditional inputs depend on them. Unconditional inputs usually used for SAA, conditional inputs for TAA.

Frontier area graphs. Cash is usually safest, lowest volatility and lowest returns. Highest volatility and highest returns is the final corner portfolio. If portfolios are growing together, there may be a diversification benefit (particularly if peak of one corresponds to peak of the other).

Illiquid assets. PE, private RE, etc. Hard to diversify to eliminate idiosyncratic risk of illiquid asset classes. However, (1) there are no or few accurate indices, and (2) even if there were good indices, no low-cost passive investment vehicles for diversification (doesn't capture returns because of self-selection bias, and doesn't capture risk because of illiquidity) (e.g. while public equity indices pretty much completely diversify away company-specific idiosyncratic risk, private equity funds are dominated by company-specific idiosyncratic risk and much more costly to buy small chunks of private equity funds into an investable fund (though we can make 'diversified' theoretical PE return indices that would be diversified)). Options:

- Exclude less liquid asset classes from asset allocation decision + then consider RE funds, etc. to implement target SAA.
- Include less liquid asset classes + attempt to model inputs to represent specific characteristics (higher returns, desmoothed returns) associated with the likely implementation vehicles (e.g. RE funds). So, modelling how the PM experiences the assets - based on the behavior of the PE fund, RE fund, etc. but less representative of true underlying asset class.
- Include less liquid asset classes + attempt to model inputs to represent diversified true asset classes. Not just RE funds, PE funds but some kind of substitute for the underlying asset class. Often use listed real estate indices, listed infrastructure, or listed public equity having characteristics of private equity or real estate (e.g. public REITs) or other illiquid classes. This often violates mutual exclusivity of asset classes as they often or will include public equity securities already in other asset classes (thereby raising correlation between asset classes).

Risk budgeting. Allocates risk efficiently in pursuit of returns, by specifying total risk and divide to each allocation. **Maximize return per unit of risk.** Marginal contribution to risk, which is the partial derivative of the risk (e.g. total risk, active risk, residual risk) wrt portfolio holding (asset allocation

holdings, active holdings, residual holdings), MCTR:

$$MCTR_i = \beta_{i,p} \times \sigma_p$$

Beta considers i with respect to portfolio, i.e. with respect to all other assets. Absolute contribution to total risk ACTR:

$$ACTR_i = w_i \times MCTR_i$$

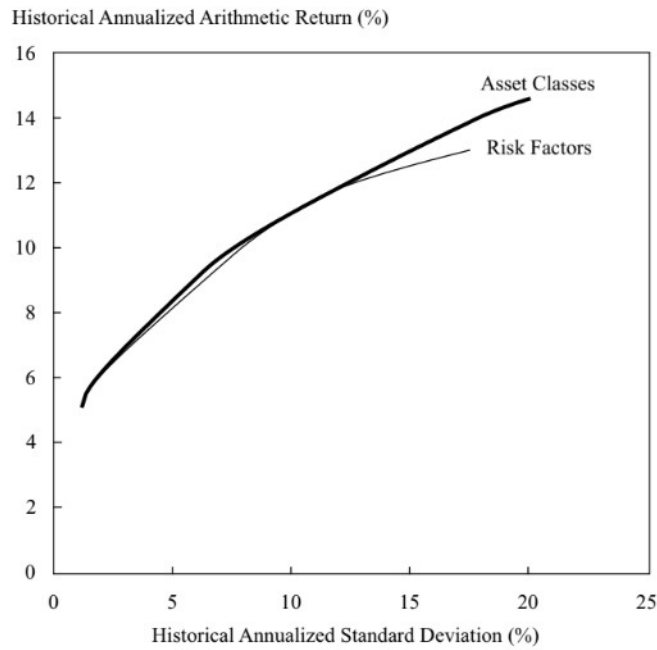
$$\sum_i ACTR_i = \sigma_p$$

In risk budgeting, optimal portfolio if

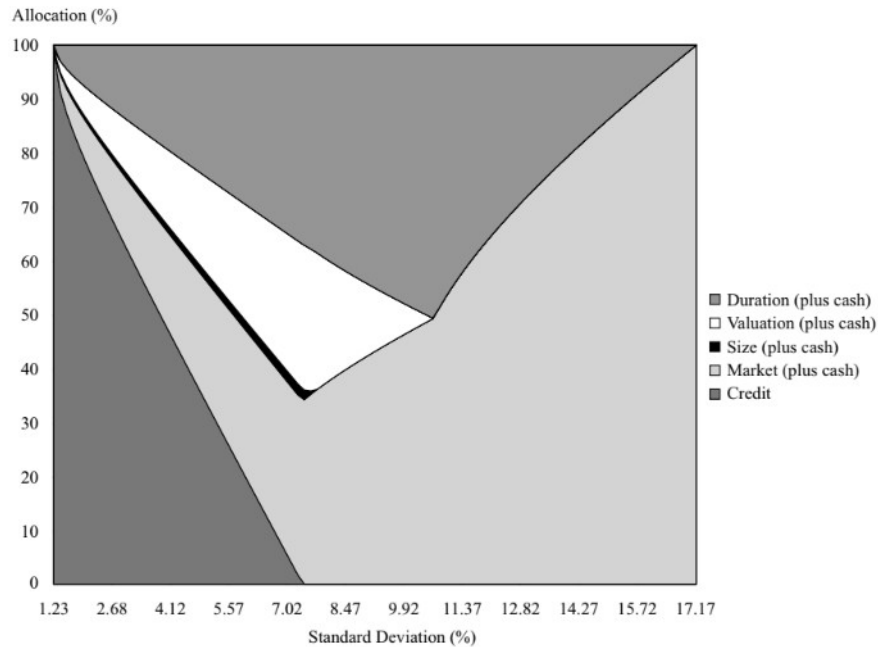
$$SharpeRatio = \frac{E(r_i) - r_f}{MCTR_i}$$

Note: in any optimization, e.g. MVO or anywhere, can use 'total return space' which includes r_f or 'excess return space' which subtracts r_f possible. Need to be consistent in calculations.

Factor-based asset allocation. Generally get lower average pairwise correlations when using factor-based models over asset classes, but Idzorek and Kowara (2013) show one is not superior over other. Can divide opportunity set (investable universe) by asset classes or factors. Efficient frontiers will be similar but not same (see below).



Asset allocation area graphs are similar (though because 'spaces' are different, they have to be conceptually remapped).



Liability-relative asset allocations. Liabilities: fixed or contingent. Quasi-liabilities: cash payments expected to be made in future for operational purposes but not legal liabilities. If liabilities (e.g. pension plan) can be hedged perfectly by assets, the discount rate for discounting the liabilities (i.e. $PV(\text{liabilities})$) can adopt the discount rate of the hedging assets.

Notes:

- $\text{Surplus} = MV(\text{assets}) - PV(\text{liabilities})$.
- $\text{Funding ratio} = MV(\text{assets}) \text{ divided by } PV(\text{liabilities})$.
- Basis risk in context of liability-relative asset allocation = degree of mismatch between liabilities and hedging portfolio.

Liability-relative asset allocations: surplus optimization. Single-period. Good for those with overfunded status and strong risk tolerance. Good for overfunded LT pension funds. Apply MVO to an efficient frontier based on surplus return volatility (or surplus risk). A correlation coefficient links assets and liabilities in the surplus volatility equation (where there are covariances).

Liability-relative asset allocations continued. Substitute asset return with surplus return. $\text{Surplus return} = (\text{change in asset value} - \text{change in liability value})$

divided by (initial asset value). Makes use of natural hedges between assets and liabilities. Then use

$$U_m = E(R_{s,m}) - 0.005\lambda\sigma^2(R_{s,m})$$

Liability return = PV(liabilities), accounting for expected changes in discount rate. Often model liabilities as (negative or inverse) bonds, but can also use factor models based on i (or r), employment, risk premiums (factor-based models can also be used to model assets). Graph surplus in USD v surplus risk (standard deviation) in USD.

Liability-relative asset allocations continued. Often, the asset-only mix is below new efficient frontier. Usually, find bonds are a big part of the asset mix as they hedge the bond-like liabilities. However, the bond proportion shrinks as surplus risk or returns rise, because they are best for conservative mixes (low returns) and because as surplus returns rise the liabilities are relatively small so hedging bond proportions shrink. The liability-relative allocation approaches the asset-allocation as risk or return rises because we want the highest returns only.

Liability-relative asset allocations: two-portfolio approach. Single-period. Most appropriate for conservative investors. Hedging and return-seeking portfolios. Two types: (1) when there is a surplus, and (2) a 'variants' approach when there is no surplus (or find an approach to maximize paying liabilities like liability glide-path rules + more sponsor contributions). No correlation coefficient links assets with liabilities; they are independent decisions. Note surplus optimization doesn't require a surplus.

Liability-relative asset allocations: integrated asset-liability approach. Single-period. Join asset and liability decisions and jointly optimize. Banks, long-short hedge funds, and insurance companies. Use when composition of liabilities might change, e.g. insurance company takes on new types of risk. The need to evaluate assets differently, may need different asset allocation. Asset allocation might constrain business decisions now. But can also be used to help optimize the business. Most comprehensive approach of three.

Surplus Optimization	Hedging/Return-Seeking Portfolios	Integrated Asset-Liability Portfolios
Simplicity	Simplicity	Increased complexity
Linear correlation	Linear or non-linear correlation	Linear or non-linear correlation
All levels of risk	Conservative level of risk	All levels of risk
Any funded ratio	Positive funded ratio for basic approach	Any funded ratio
Single period	Single period	Multiple periods

Liability-relative asset allocation: sometimes use factor approaches. Example: if inflation is uncertain, then include more inflation-linked bonds.

Use scenarios to test asset allocations. Simulations, historical, even studies, multi-stage simulations, reduce or raise expected returns, etc. To counteract specific issues, e.g. inflation, use something correlated like inflation-linked bonds or more equity.

Goals-based asset allocation. Important factors: (1) need to divide PF into sub-PFs for goals, (2) both taxable and tax-exempt investments are important, and (3) minimum expectations rather than mathematical expectations of liabilities of institutional investors.

Constructing sub-portfolios and overall portfolio. Choose PFs for each goal and add them up. If there's a surplus, pick a moderate goal and assign a PF. The calculate total returns and variances. Notes:

- With inflation = inflation goes into the cashflows, not the denominator in the NPV calculations.
- Liquidity can be defined as $1 - \frac{AvNoDaysToLiquidate}{TradingDaysInYear}$

To create modules (sub-PFs), start with CMEs for different asset classes. Should consider tax implications (for each of PF and even each asset and each asset operation like sale). Then specify times, probabilities, and then add constraints, like (1) liquidity suitable for each time period, (2) include strategies that address non-normal returns, (3) allowing drawdown control (because utility preferences are asymmetric). Sub-PFs should be reviewed regularly, e.g. if market is at top or bottom (but no need for systematic approaches like Black-Litterman because they explicitly incorporate quantitative views, e.g. equities should be up 2%).

Review goal-based PFs. Notes: (1) 'next 5 years' e.g. doesn't mean exactly within 5 years, so don't start subtracting 1 year each year, they are more 'general placeholders', (2) as PFs will generally have safety built into them, so have to rebalance - but note that rebalancing is complex and taxable.

Heuristics for asset allocation.

- 120 minus age = percentage of PF in stocks. Equity glide path. Or use other rules, e.g. financial to human capital ratios. But many funds appear to use a rule that is similar to or nearly equivalent to 120 minus age.
- 60-40 stock-bond heuristic. Skip optimization and use simple rules like 60% equities and 40% fixed income. This ratio is close to the global market portfolio. Passive investment style (Norway model), believing in market efficiency.
- Endowment model. Large allocations to alternative investments, including use of investment manager skill in equity selection. Also called Yale model. Use of active management to earn illiquidity premiums (e.g. private equity) due to long time horizons.
- Risk parity. Each asset (asset class or risk factor) should contribute equally to total PF **variance**. Controversial approach. Many variants. Most common is to use $w_i \times Cov(r_i, r_P) = \frac{1}{n} \sigma_P^2$ (no closed-form solution) to get weights by constraining asset classes so that each one (n) contributes an equal amount of risk (e.g. if 7 equity classes and 3 FI, then 70% of risk to equities and 30% to FI). So, if an asset class has low variance and low return covariances to other asset classes, then it must have a higher weight. Then take expected returns and divide by portfolio variance; if greater risk appetite then leverage up (which often works better than unleveraged investment in equity, because of 'leverage aversion'), if more risk-averse than the ratio then hold some cash. Big criticism: it ignores returns.
- 1/N rule. Theoretically not optimal. Does slightly better than expected however, possibly because avoids problems from estimation errors in inputs (which create problems for optimization methods).

Portfolio rebalancing in practice. To match SAA; some include TAA under rebalancing scope (but not in CFA L3). Note: in liability-relative allocation, need to adjust net durations as time progresses. If don't rebalance, there is a cost: expected utility loss from tracking error.

Rebalancing continued. Generally, target portfolio compound growth higher than simple average growths of portfolio asset classes, so better to rebalance. Moreover, portfolios tend to have higher returns (particularly over the long-term) because of 'safe' probabilities built into the portfolio. If don't rebalance, then risk levels will tend to drift to lower levels than client probably wants.

1.5 Asset allocation with real-world constraints

Constraints.

- Asset size. If too big, can't get enough of some assets. If too small, transaction costs can be prohibitive and some (e.g. HFs) have minimum investment sizes and they are illiquid. Asset managers disadvantaged from increasing scale because of organizational hierarchies, have to go outside core investment ideas, larger trade sizes and transaction costs, etc. However, asset owners advantaged by increasing scale: can negotiate lower fees and concentrate in assets that are good for large scales.
- If fund has too large a concentration in an investment fund elsewhere, then this exposes the investor fund and the investment fund to operational risks.
- Enough investments in investable universe. E.g. don't want to be too big a proportion of an AM or HF's AUM. There are about 10,000 HFs globally, with 60% below 100 USD million, and 40% around 1 billion USD. Is this enough? For PE and HFs, often have to invest tens of millions at least - so need at least this much.
- Expertise. If small fund, then might not have enough expertise for more complex investments and strategies (HFs, PEs, high-yield bonds). Even at large funds, have to develop this.
- Speed of investment. If too large too fast, this may be a problem.
- Liquidity. Banks need liquidity. Endowment funds can invest for LT and get illiquidity premiums.
- Time horizons. Closer to retirement, human capital becomes more bond-like. During working-age, retirement liabilities are LT bond-like, and approaching retirement they are cash-like. **Time diversification:** LT goals with higher-risk growth assets, ST goals with cash or lower-risk assets.
- Time horizons II. MVO assumes random walk for returns. So the dispersion widens with time, making LT investments risky. Actual asset returns are not as extreme because of some mean-reversion tendencies, but still have large spread over large time horizons.
- Laws and regulations. Must be certified investor to invest in some kinds of funds, e.g. HFs. Insurance companies: can be regulated capital or solvency ratios, there may be limits to investing equities. Pension funds: limits to investing in different assets (or minimum limits too), reporting, tax. Endowments and foundations: tax incentives to spend, minimum spending, often have minimum capitalization requirements for borrowing. Sovereign wealth funds: often governing legislation have restrictions, other restrictions (e.g. Sharia law, ESG - usually as constraints by having ESG be its own asset class or implementation after asset allocation).
- Taxes - see below.

Further points.

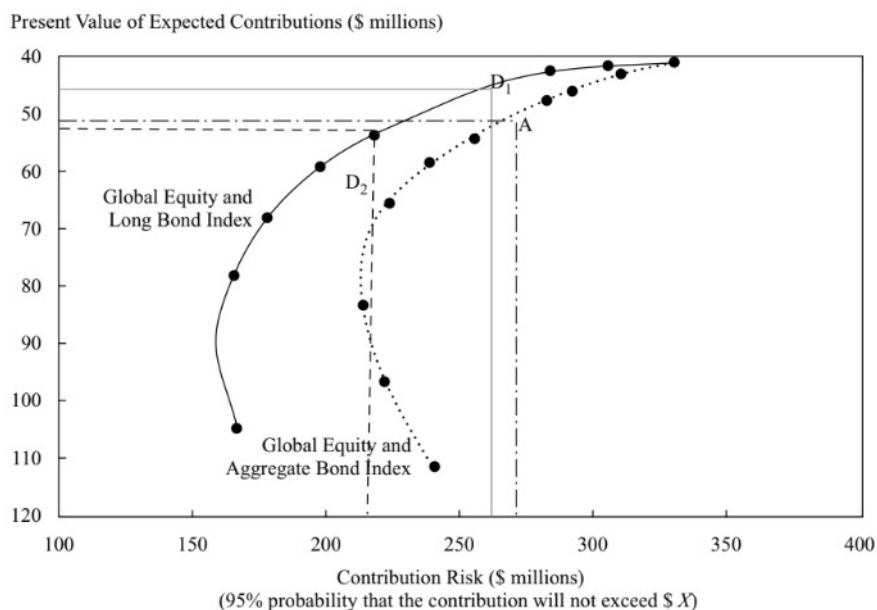


Chart above. For pension funds, efficient frontiers of PV expected contributions v contribution risk (95% probability that contribution will not exceed it). As PV expected contributions rises, contribution risk falls. But if PV expected contribution rises too much, then harder to invest and contribution risk might slightly rise.

Asset and liability matching. If liabilities are pension payouts, then treat these as LT bonds. Then want assets which match, such as bonds with long durations. Long durations mean that if interest rates change, if liabilities increase, LT bond assets will increase too. Additionally, if bond portfolio has a steep upward yield curve, this will lead to some extra profit.

Asset allocation for the taxable investor and after-tax portfolio optimization.

Taxes

- Taxes I. Different for each asset class. Interest income usually taxed the most (same as salary income), or at least more than dividends, though government bond interest often tax-free. Preferred share dividends usually taxed less than normal dividends. Capital gains usually taxed the least.
- Taxes II. Bonds: $r_a = r_p(1 - t)$ where subscripts a and p are after- and pre-tax; often, the capital appreciation portion is ignored. Equities: $r_a = p_d r_d(1 - t_d) + p_c r_c(1 - t_c)$ where subscripts d and c are dividends and capital appreciation, and p is proportion of returns. So, taxes reduce positive returns. They also increase negative returns (e.g. capital losses).

Some taxable assets have unrealized capital gains or losses; many ways to address this.

- Taxes III. Standard deviation for asset allocation: $\sigma_a = \sigma_p(1-t)$. So, after-tax volatility is less than pre-tax volatility, so larger asset class movements are needed to change after-tax portfolios, and wider rebalancing ranges. So, taxes reduce returns and standard deviations. The after-tax returns and standard deviations are fed into the MVO.
- Taxes IV. High-yield bonds are taxed like income (as are IG bonds), but they are more correlated with equities.
- Taxes V. Rebalancing ranges are wider for after-tax portfolios because of lower standard deviations (so, if two identical portfolios but one in taxable and one in tax-exempt account, then the one in the taxable account can be balanced less frequently due to a wider rebalancing range). Use $R_a = R_p/(1-t)$.
- Strategies. **Tax-loss harvesting** - trigger capital gains loss in one place to offset capital gains elsewhere in portfolio. **Strategic asset location** - locate or place tax-inefficient assets in tax-advantaged accounts like retirement accounts.
- Types of tax accounts. Tax-exempt. Tax-deferred: taxed upon distribution, so use $V_a = V_p(1-t)$, so higher investments are justifiable. Put everything but particularly taxable coupon bonds and dividend-paying equities in tax-exempt accounts (because they regularly produce coupons which is taxable at income rates), and put dividend-less stock in taxable accounts if can't put into tax-exempt accounts. Note, however, that tax-advantaged accounts reduce ST liquidity. Optimization is done by considering whether taxable, tax-exempt or tax-deferred.

Rebalancing: taxable v tax-advantaged accounts. In taxable accounts, have wider rebalancing targets because (1) after-tax volatility is lower, and (2) frequent rebalancing will trigger more taxes. Also, calculate the pre-tax rebalancing target by $AfterTaxRange * (1-t) = PreTaxRange$.

De-risking. Increase fixed-income proportions to better hedge liabilities. But there may be disincentives, e.g. US rules favor equities over bonds. Further, more FI means less risk and returns. As a matter of corporate governance, shareholders may want management to take a conservative view and focus efforts on core business.

More de-risking. Between IG bonds and HY bonds and equities, if there's limited space in the tax-exempt account, put HY bonds (which either have higher coupons and/or trigger larger capital gains) and equities in tax-exempt account. IG bonds have lower returns and have lower correlation with equities.

More de-risking. A shareholder who is also a retirement plan participant (i.e. works and has a stake in the company) would want their retirement plan (e.g. tax-exempt) to have 100% fixed-income because the person already holds equity in the company which is essentially a 'taxable account' - and this argument can be extended to pension funds operated by publicly-traded companies (workers there can buy shares of the publicly-traded company and have their retirement funds in the pension fund).

Revisiting SAA. Often revisit every 5 years, or if triggered by change in (or **anticipate** changes in):

- Goals (business conditions, business model, profitability, change in business risk, change in personal circumstances). For personal investors, it is a change in **risk appetite or risk capacity**.
- Constraints (time horizon, liquidity, asset size, regulatory constraints, change in inflow or outflows to fund, changes in donations, lump sum or regular payments). If less contributions and underfunded, need to rely more on returns (e.g. equities) rather than bonds.
- Beliefs (market expectations).

If some points at which SAAs should change are known, then can put in a policy to target or anticipate that in a 'glide path'.

Short-term shifts (TAA). TAA is predicated on investment returns being ST predictable, whereas SAA is random walk. TAA is not about security selection; alpha is generated by market timing and is limited to assets (not about liabilities). Often there are constraints around SAA for TAA, or tracking error or risk budgets. Evaluate Sharpe ratio, IR, t-statistic of average excess return over SAA portfolio, plotting efficient frontiers (because can get higher Sharpe ratios with lower frontiers). Higher transaction costs and probably taxes. Two types of TAA:

- Discretionary TAA. Depends on manager skill. Look at many different data points (GDP, VIX, etc.).
- Systematic TAA. Use signals with predictability, e.g. value, momentum. Or use quantitative rules. Valuation ratios (e.g. P/B , cash flow yield, roll yields, YTM, etc.).

GFC 2008. Prior to GFC, many pension funds moved from intermediate duration bonds to long duration bonds. During GFC, liabilities rise as corporate bond yields fall (because discount rates fall) while longer-duration bonds (which are generally higher-quality) increase a price a lot more because they are high-quality (in addition, the longer duration means a fall in rates means prices increase more). After, they switched from government bonds (since yields fell, their prices rose a lot) to corporate bonds (whose spread to gov bonds widened

and so were cheap) to lock in capital gains.

Behavioral biases.

- Loss aversion. Losses hurt more than good from gains. May have trouble maintaining ideal asset allocations during downturns. Address by (1) framing risk in shortfall probabilities (e.g. instead of PF will drop by 15% with 30% probability (which is about losing money), frame as there's a 5% chance of not meeting the goal (which is about the safety or goal), and (2) funding high-priority goals with low-risk assets (so that the investor feels more at ease, and might be willing to risk more for lower-priority goals).
- Illusion of control and hindsight bias. Overconfidence, hindsight bias exacerbate illusion of control. Examples: belief in superior skill or information, excessive trading, or doing something beyond normal which increases diversifiable risk. Address by starting with global market portfolio, and deviations must be fully justified.
- Mental accounting. Sub-optimal asset allocation. Goals-based investing does this. But 'provided each of the sub-portfolios lies along the same efficient frontier, the sum of the sub-portfolios will also be efficient'. An entrepreneur holding onto large shares of their company is also mental accounting (an overconcentrated position) - they should diversify beyond ownership control, tax, etc. Mental accounting exacerbated by endowment effect - that asset already owned is better than a replacement. Address by taking concentrated stock position to an aspirational goal, and more high-priority goals with diversified assets.
- Note: mental accounting can be used as an advantageous to get investor to think about goals-based allocation to help fully understand some risks, e.g. overconcentrated stock positions in their own company, compared to their steady consumption needs.
- Representativeness (recency bias). Return-chasing (overweighting assets with good recent returns), or overincorporating news. Address by objective allocation process and good governance.
- Framing bias. VaR and standard deviation might convey same information, but framed differently. Address by providing multiple measures like VaR and SD to see what investor is really thinking.
- Availability and familiarity bias. Biased towards whatever can be quickly recalled at that time. In asset allocation, 'home bias' is an example. Comparison with other people's portfolios is another example. Can be mitigated by beginning with the global market portfolio, allocation processes, good governance, diverse views.
- Prudence bias. Tempering extreme forecasts or being cautious.

- Anchoring bias. Being fixed to first impressions.

Good governance: (1) clear LT and ST investment objectives, (2) good division and delegation within investment process, (3) established processes for operations, (4) processes for approving SAA, (5) reporting framework, (6) periodic governance audits.

2 Portfolio construction

2.1 Overview of equity portfolio management

Roles of equities in a portfolio. Capital appreciation, dividend income, diversification, inflation hedge.

Dividend income. Companies with regular dividend - these are for conservative investors.

Diversification benefits. When mixed with non-equities. Note however that securities tend to be more positively correlated during economic downturns.

Inflation hedges. Companies which produce commodities or can pass on more of their costs are better inflation hedges. Equities therefore differ on inflation-hedging properties. Sometimes, equities are negatively correlated with inflation, though generally as a group they are positively correlated.

Considerations for adding equities. Risk, return, liquidity, time horizon, taxes, legal and regulatory, unique circumstances. Latter can include ESG; ESG equities can be chosen through negative screening, positive screening, best-in-class, thematic investing and impact investing, as well as 'ESG integration', i.e. ESG as a factor in equity selection.

Equity investment universe. Segment by:

- Size (small, medium, large cap) and style (value and growth through a scoring system looking at multiple factors; a single company can be both value and growth for different business lines). These divisions provide diversification across companies. Categories are flexible; can be defined differently by different investors.
- Geography. Developed, emerging, frontier markets. Note: foreign indices might provide less exposure than expected, e.g. Swiss index provides little exposure to Switzerland because the index companies are international. Note there is currency risk.
- Economic activity, i.e. industry. Main standards: GICS, ICB, TRBC, RGS. Some large companies might be in one more than one industry.
- Themes, e.g. ESG.

Income associated with owning and managing an equity portfolio.

- Dividend. Main income for equity PFs. Taxation is consideration. Special dividends (generally taxable). Optional stock dividends: investor can choose cash or shares when the dividend is issued.
- Securities lending (stock lending). Securities are known as 'stock loans'. Securities lender can generate additional cash by reinvesting cash collateral. Borrower compensates lender for any dividends. Note: borrower might short sell and reduce price of security - so this is a concern for the investor's long position.
- Ancillary investment strategies. **Dividend capture**: buy stock just before ex-dividend date, get dividend and then get a capital loss. **Write and sell options**: covered calls and cash-covered put (cash-secured put) have similar payoff profiles.

Costs of owning and managing an equity PF.

- Management fees. Often percentage AUM. For research + portfolio management (e.g. trade execution). Also include marketing costs.
- Performance fees. Generally 'upwards only' (PM gets if rises but doesn't return if falls), though sometimes can be reduced if poor performance. High-water mark: highest value, net of fees, that fund has reached; protects clients from paying twice for same performance.
- Administration fees. Processing corporate actions, measuring performance and risk, voting proxies - these can be admin or management fees. Other admin fees for third parties: custody fees (to keep money safe), depository fees (custodian complies with laws, leverage, etc.), and registration fees (register ownership of units in MF).
- Marketing and distribution costs. Part of management fees.
- Trading costs. Explicit costs such as trading commissions but also implicit ones like bid-offer spread, delays due to liquidity problems - these are part of the investment returns.
- Investment approaches. Index strategies incur fewer costs. However, even index strategies have implicit costs, e.g. predatory trading and front-running by buying stocks likely to be added to an index.

Shareholder engagement. Shareholders vote and interact with company management. Can get information (e.g. strategy, allocation of capital, corporate governance, remuneration of directors, ESG), improve corporate governance, profitability, etc. But costs time, pressures company management to focus on ST only, might lead to only selective transfer of information (and break insider trading rules), conflicts of interest (company might be invested in PM's portfolio).

Shareholder engagement might be through: activist investing, voting.

Voting is a right for the holder (not necessarily owner) of the shares. Empty voting. Borrow share from PM and then vote. But PMs often recall shares ahead of voting.

Considerations on whether to use active management in equity investments: (1) confidence and resources to outperform benchmark, (2) client preferences and beliefs - whether they think market is efficient, (3) whether there is any alpha left, e.g. if there's lots of stock coverage of that equity already, (4) suitable benchmark: benchmark stocks have to be liquid to allow PM to trade them, (5) suitable benchmark II: number of benchmark stocks has to be broad to generate alpha; many country or sector-specific investments are index-based, (6) client-specific mandates, e.g. ESG (more active), specific sectors (less active), (7) key person risk, (8) costs of active management - whether client is cost-conscious.

Brinson, Hood and Beebower (1986) find asset allocation dominates security selection in explaining return variability.

Benchmarks. Index for passive strategies has to be: rules-based, transparent (most important), and investable. Investable. Some indices like those for HFs are hard to replicate for normal investor. Some, like geometric average indices, are mathematically replicable but prices will drift and no longer be geometric averages if replicated. Indices are often 'free-float adjust', so only tradable shares are used. To keep costs lower, some index providers **buffer** their stocks (so if moving from small-cap to mid-cap as stock price rises, there's an extra buffer zone to keep it in small-cap index; but buffer zones can be for large-cap too: if company is smaller than x% of smallest company in large-cap index, then it gets migrated to mid-cap) and **packet** their stocks so that if enter buffer zone, part of the stock is transferred to next index (e.g. mid-cap) and then it stays in the buffer zone, then the rest is migrated.

Benchmarks: considerations when choosing. Make sure benchmark index has desired factor exposures (e.g. size, domestic, style, momentum, volatility). For style, note: (1) Growth stocks (aggressive, fast-growing) have high price momentum, high P-E, high EPS growth, (2) Value stocks (mature, stable): high dy, low P-E, low-P-B.

Benchmarks: construction methodologies. Exhaustive to selective (S&P 500, based on committee selection) investment universes.

Benchmark construction. Weighting.

- Market-cap weighting. Based on efficient markets. Note: CAPM global market portfolio is market-cap weighted. This GM portfolio is mean-

variance efficient. Market-cap weighting is like liquidity-weighting, because large-cap stocks have highest liquidity. Most common market-cap weighting is free-float weighting (remove closely-held shares): requires analytical judgment. Free-float adjusting gets rid of treasury and closely-held shares (which exert a disproportionate effect when outstanding shares are traded) and also gets rid of traded shares which have low liquidity and have high trading costs.

- Price-weighting (DJIA, Nikkei). Like a PF of one share of each company. Stock splits are complicated (affect both numerator and denominator). PMs rarely use: almost nobody trades with a PF with same number of shares held in each stock of the benchmark.
- Equally-weighted. The dollar value of each stock is the same (though different number of shares), and is rebalanced to maintain this - it is equivalent to equal DOLLAR VALUE and equal RETURN PERCENTAGE weighting. Each stock given same weight $1/n$. Least-concentrated PFs and stable, but it's naive. Imparts a small-cap bias, so it is more volatile than cap weighting. Indifferent to factors and randomizes factor mispricing. Requires constant rebalancing (theoretically, every second). Small-cap bias could lead to liquidity problems - buying lots of small-cap, but also not being able to buy enough of small-caps. Advantage over market-cap weighting: overpriced stocks will be sold to bring their weight back to $1/n$ and underpriced stocks will be bought.
- Fundamental characteristics weighting. Based on inefficient markets. Sales, income, dividends, etc. Delinks weight from market value, theory is that market will eventually converge to fundamentals.

How concentrated an index is. Concentrated indices are usually not diversified. HHI measures stock-concentration in PF:

$$HHI = \sum_i^n w_i^2$$

Ranges from $1/n$ to 1.0. HHI estimates the 'effective (equivalent) number of stocks' in equal weights:

$$\text{Effective number of stocks} = \frac{1}{\sum_i^n w_i^2} = 1/HHI$$

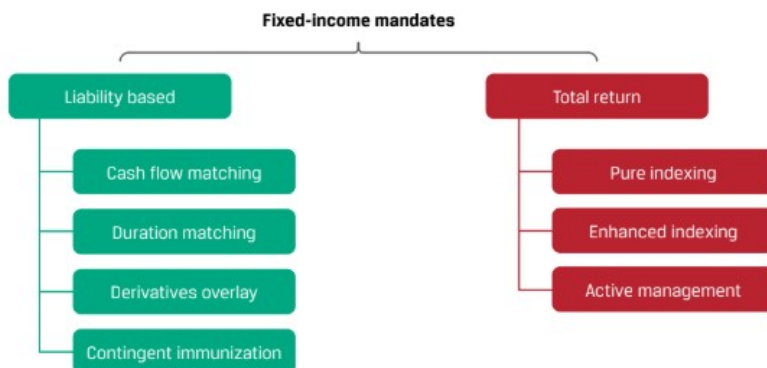
Rebalancing and reconstitution. Reconstitution = addition or deletion. Rebalancing = reweighting. Turnover = reconstitution + rebalancing. Reconstitution leads to stocks added increasing in price as index-followers want to buy it, and stocks deleted falling in price - speculators will front-run.

Note: adding IG bonds to S&P 500 won't necessarily reduce PF ST risk, because the correlations are uncertain.

2.2 Overview of fixed-income PM

Roles of fixed-income in PFs.

- Diversification. Due to correlation primarily. However, note: interest rate changes are responsible for most price changes. So, calculate excess return (subtract r_f). Credit spread gives diversification - depending on sector, geography, etc. Correlations change with time, and depending on market conditions (e.g. during stress, government bonds and HY bonds not correlated anymore and HY bonds and equity correlations will increase). Volatilities can also change throughout time and volatilities affect correlations (and part of the correlation formula).
- Regular cashflows. Fixed-income investments. Can be used for dedicated goals or liabilities. Longer maturity bonds are better for regularity of cash flows, for funding future liabilities.
- Inflation hedging potential. Bonds with floating-rate coupons, inflation-linked bonds. Note: inflation-linked bonds are less volatile because volatility linked to real rather than nominal rates, good for LT investors, and give superior risk-adjusted returns.



Fixed-income mandates:

- Liability-based mandates. For asset-liability management (ALM) = liability-driven investments (LDI). **Cashflow matching** - match future liabilities precisely with fixed-income cash inflows. For predictable and/or near-term liabilities, the possibility of defeasance - should eliminate reinvestment and interest rate risk; complex and expensive. **Duration matching** - matches duration of assets and liabilities (i.e. assets and liabilities affected similarly by interest rate changes). Might use futures contracts for mandates. For complex liabilities, cheaper than cashflow matching. **Contingent immunization** - combine matching (= immunization) techniques with active management when assets exceed liabilities.

- Total return mandates. Track and outperform fixed-income benchmark. **Pure indexing** which attempts to replicate bond index ('full replication') which targets active return and active risk are both 0, but trading costs and fees mean usually below benchmark. **Enhanced indexing approach** tries to outperform benchmark but still remains close to benchmark (but allows sector, quality, etc. bets). **Active management** tries to outperform benchmark and can vary a lot from benchmark.
- FI mandates with ESG considerations.

FI PF measures. EXHIBIT 8.

Macaulay durations and convexities. Convexity becomes more valuable when interest rate volatility increases. Since LT bonds have higher convexities, markets will then pile into LT bonds, leading to yield declines and possible inversion. (1) increase with maturity, (2) convexity = duration squared (approx), and (3) coupon-paying bonds have higher convexities than non- or low-coupon bonds (of the same duration).

Modified duration of bond PF:

$$AvModDur = \sum_j^J ModDur_j \left(\frac{MV_j}{MV} \right)$$

Convexity of bond PF. Can ignore for small changes. If derivatives are present, big changes possible. Negative convexity (e.g. callable bonds, MBS) can lead to big changes. PFs with high convexities have low YTM - investors pay premiums.

$$AvConvex = \sum_j^J Convex_j \left(\frac{MV_j}{MV} \right)$$

Effective duration and convexity of a PF. Modified duration measures a bond's price sensitivity to a change in YTM, assuming that the bond's cash flows **do not change** as interest rates move. Effective duration measures the same price sensitivity but **accounts for the possibility that a bond's cash flows may change** when interest rates shift. Use effective duration and convexity for FI with embedded options - but don't need to use for FI without options.

Spread duration. PF's sensitivity to changes in credit spreads. If expecting credit spreads to narrow, then increase spread duration of bond PF (and/or reduce average credit rating of bond PF).

Duration times spread (DTS). Modifies spread duration so that it is based on a proportional percentage basis.

PF dispersion. Variance of times to receipt of cash flows. Macaulay duration is weighted average of times to receipt, while PF dispersion is variance.

This affects convexity. Higher PF dispersion leads to higher convexity.

Note: interest rates and credit spreads are negatively correlated, during good and bad economic times.

Relative value. PMs will rank securities along dimensions such as valuation, fundamentals, supply and demand, key rate durations, DTS, etc. for PM inclusion.

Bond market liquidity. Much of it is OTC, so there are search costs. Illiquid bonds = premium for holder but harder to sell. If want bonds but want lots of liquidity and don't want to buy single bonds, buy fund (e.g. mutual fund, ETF) shares or derivatives - but these have their own costs.

Total return swap (TRS): based on equity, commodity or bond index. TRS receiver: cash flows from underlying and appreciation, and pays MRR + predetermined spread. TRS payer: will be compensated for depreciation in index or default losses. A cheap way of accessing these underlying securities. However, TRS receiver doesn't own anything, and there are transaction costs. A model of FI returns. Assumes cashflows reinvested at YTM.

Expected FI returns = coupons + rolldown return + change in price due to investor's expected view of benchmark yield + change in price due to investor's view of yield spreads + change in price due to investor's view of FX value changes.

Coupons. Annual coupon payment divided by current bond price (NOT the par price).

Rolldown return = (bond price at end of horizon period - bond price at beginning of horizon period) divided by bond price at beginning of horizon period. Note: bonds priced at premium to par will fall in capital value, while bonds priced at discount will rise.

Note: coupons + rolldown return = 'rolling yield'.

Views of benchmark yields = $(-\text{ModDur} \times \Delta \text{Yield}) + (0.5 \times \text{Convexity} \times (\Delta \text{Yield})^2)$. This is about both YTM and yield volatilities. Use effective duration and effective convexities.

Views of yield spreads = $(-\text{ModSpreadDur} \times \Delta \text{Spread}) + (0.5 \times \text{Convexity} \times (\Delta \text{Yield})^2)$. Reflects general market conditions and idiosyncratic risk, credit migration. If no other information, use ModDur for ModSpreadDur. This is usually less relevant for government securities.

Views of FX changes $R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1 = \sum_i^n \omega_i(1 + R_{FC,i})(1 +$

$R_{FX,i}) - 1$ where $\sum \omega = 1$ and is the weights of the foreign assets. Combines both views of benchmark yields and FX values.

Leverage. Derivatives and borrowing, structured financial products.

$$r_p = \frac{PFReturn}{PFEquity} = \frac{r_l(V_E + V_B) - (V_E + V_B)}{V_E}$$

where r_l is the return on leveraged funds and r_b is the borrowing rate. Rearranging,

$$r_p = \frac{(r_l \times V_E) + (V_B \times (r_l - r_B))}{V_E} = r_l + \frac{V_B}{V_E}(r_l - r_B)$$

Leverage: futures. Sell treasury futures if, e.g. if bullish on corporate bonds (i.e. spread will fall) but bearish on treasuries (i.e. rates will rise).

$$Leverage = \frac{NotionalValue - Margin}{Margin}$$

where margin = margin deposit.

Leverage: swaps. Buy receive-fixed, pay-floating e.g. if fully invested, and believe rates will fall. Because interest rate swaps are equivalent to long-short (i.e. floating and fixed) bonds, they are leveraged products.

Leverage: repos. Note: repo rates, e.g. '5%' are annualized. Dollar interest = principal x repo rate x ($days/360$). Lender is protected by collateral + haircut on collateral (collateral value exceeds principal by 1-3%). Different kinds of repos (for general bond collateral, specific collateral, whether to use a management service).

Leverage: security lending. Similar to repos but open-ended time period. This allows parties to short sell specific securities. Security lender gets collateral (generic bonds or cash) and can reinvest to get even more returns, but some of this **rebate rate** is given to security borrower: rebate rate = collateral earnings rate - security lending rate. If the security being lent is in high demand however the rebate rate can be negative.

Risks of leverage. If PF value falls, leverage automatically increases. If everyone's PF falls at same time, fire sales. There can be downward spirals, which can be triggered by market crashes when credit lines, repos, may be less available.

FI PF taxation.

- Tax-loss harvesting. Delay realizing capital gains, bring forward capital losses as early as possible. Allows accumulation of pre-tax gains.

- Collective investment schemes, e.g. MFs: interest income taxed at final investor level when MF receives - even if MF reinvests or pays it out or doesn't pay out the interest income immediately.
- US: pass-through treatment of capital gains on MFs. Net capital gains taxed each year, even if no sale. UK: no pass-through on MFs; net capital gains only when asset is sold.

Liability-driven investing. LDI: liabilities are a given and assets are managed. Asset-driven liabilities (ADL): assets are given, liabilities are structured to manage interest rate risk - e.g. leasing company with short-term contracts (leases to customers) which finances itself by issuing short-term debt (i.e., match asset and liability maturities). ADL: company has assets and liabilities, it's about how to manage the liabilities.

Liability Type	Amount of Cash Outlay	Timing of Cash Outlay	Examples	
Type I	Known	Known	Traditional fixed-income bond with no embedded options	MacDur, ModDur, money duration, and the PVBP can be used to measure the interest rate sensitivity
Type II	Known	Uncertain	Callable and putable bonds Term life insurance policy (timing of death unknown)	
Type III	Uncertain	Known	Floating-rate note-interest payments depend on future interest rates Inflation-protected securities-amounts of interest and principal payments tied to inflation	Effective duration needed to estimate interest rate sensitivity. Calculated using a model for: • Uncertain amount and/or timing of the cash flows • Initial assumption about the yield curve
Type IV	Uncertain	Uncertain	Property and casualty insurance (weather events difficult to predict)	

Liability classification. Residential mortgage loans, despite looking like Type I, are usually Type IV because of prepayment options and because default possibilities. CoCos are Type IV.

Managing interest rate risk of multiple liabilities. Even if have cash to buy back and retire liabilities, this could incur lots of transaction costs (many different parties own them), will drive up the price of liabilities (by creating demand for these bonds). So, often just manage the liabilities.

- Cash flow matching. Accounting defeasance or in-substance defeasance, where a company sets aside enough high-quality assets to repay the liability. Raises credit rating. However, this can lead to holding lots of securities and therefore reinvestment risk for low-yielding investments (usually high-quality bonds). Do Example 2.
- Duration matching.

Types of FI portfolios: laddered (even cashflows), bullet (concentrated cashflows), and barbell (concentrated in 2 cashflows). Barbell has high convexity as cashflows very spread apart, bullet has low convexity, and laddered has relatively high - given same cashflow yield.

$$\text{Immunized PF convexity} = \frac{MacDur^2 + MacDur + Dispersion}{(1 + CFYield)^2}$$

Laddered PFs: balances reinvestment and price volatility risks. Reinvesting ST bonds as they mature into LT bonds maintains PF duration. Because it has lower convexity compared to barbell, ladder has less reinvestment risk (because cash flow is spread out). Barbell = have to constantly reinvest near term interest income, so it has higher reinvestment risk.

Note: bonds nearing maturity are high-quality collateral because won't move much even if interest rates are volatile. Use for collateral or repo contract to get extra cash. So, this is an advantage of laddered PF: there are such bonds frequently.

Other options for laddered FI funds: (1) Fixed-maturity corporate bond ETFs to build laddered PFs. These are ETFs of e.g. 50 IG bonds, having same duration, convexities, properties. But more liquid. (2) MFs. More diversified.

Good laddered FI PFs: (1) high-quality bonds, (2) liquid bonds, (3) yield curve diversification (i.e. maturities of different years).

2.3 Asset allocation to alternative investments

Allocations to alternatives. Higher returns, greater diversification. Overall, often an increase in risk-adjusted returns. Real assets generally reduce risk, VC and PE increase returns. Some HFs are risk-reducing (arbitrage strategies), some HFs are return-enhancing (activist).

- Capital growth - public equities, private equity, private real assets. PE has more capital growth than HF. Note: timber is capital growth = trees + land; public RE and long-short HFs are not good capital growth.
- Income generation - FI or real estate
- Risk diversification - for equity-centered PFs, real assets and 'absolute return HFs'. For FI PFs, private credit.
- Safety. Government bonds, gold.

Asset Class		Role			
		Capital Growth	Income	Diversifying Public Equities	Safety
Fixed Income and Credit	Governments		M	H	H
	Inflation-Linked		M	H	H/M
	Inv.-Grade Credit		M	H	M
	High-Yield Credit		H	M	
	Private Credit		H	M	
Equities	Public Equity	H	M		
	Private Equity	H	M	M	
Real Estate	Public Real Estate	M	H	M	
	Private Real Estate	M	H	M	
Real Assets	Public Real Assets (Energy, Metal, etc.)			H	
	Private Real Assets (Timber, etc.)	H	H	H	
Hedge Funds	Absolute Return		M	H	
	Equity Long/Short			M	

Notes: H = high/strong potential to fulfill the indicated role; M = moderate potential to fulfill the indicated role.

Roles of different assets in a multi-asset PF:

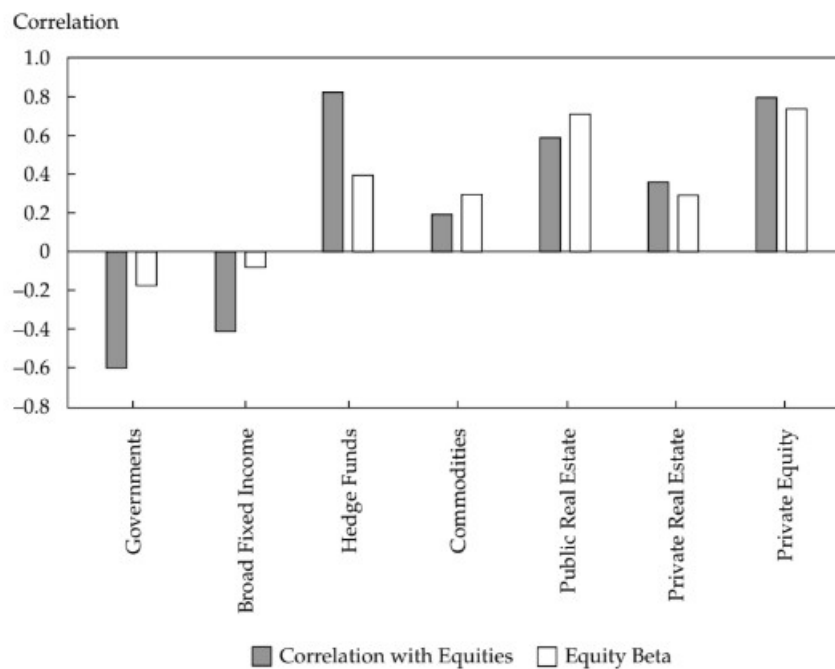
- Private equity. Return-enhancer due to illiquidity risk. But limited diversification to public equities - they're both equities.
- Hedge funds. Long-short are equity-like returns and moderately diversify equity. Arbitrage and event-driven strategies have little correlation to traditional asset classes. Most HF arbitrage strategies have 'short volatility' risk. Opportunistic strategies also have little correlation to traditional assets.
- Absolute return hedge funds hedge out systematic risk and rely on alpha generation. So, greater potential to diversify public equity risk than PE or private real estate. Equity beta that is less than PE or private real estate. High potential to diversify public equities.
- Real assets. Physical asset has high correlation with inflation. Timber - growth (tree + land) and inflation-hedging (if timber price weak, just leave for a few more years). Commodities: inflation-hedging. Farmland: own farm directly and hire farmers (higher risk-return), or lease to farmer (more like real estate). Infrastructure: high correlation with inflation.

- Commercial RE. Strategies: core (ownership and manage property), opportunistic (develop, construct, sell). Inflation protection. Diversified exposure.
- Private credit. Distressed investment and direct lending (capital to people that cannot access traditional lending; some unsecured others secured). Direct lending: income-producing assets, assume default or recovery risks. Distressed investment are more like equity. Risky. Illiquid. No secondary market.
- 'Public' credit, i.e. bonds. Private credit diversifies bonds; private credit generally on floating rates while bonds are usually fixed rates (so, private credit prices are LESS sensitive to rates than bonds); the lender profile differs. Private credit is appraisal-based, while bonds are valued more regularly.

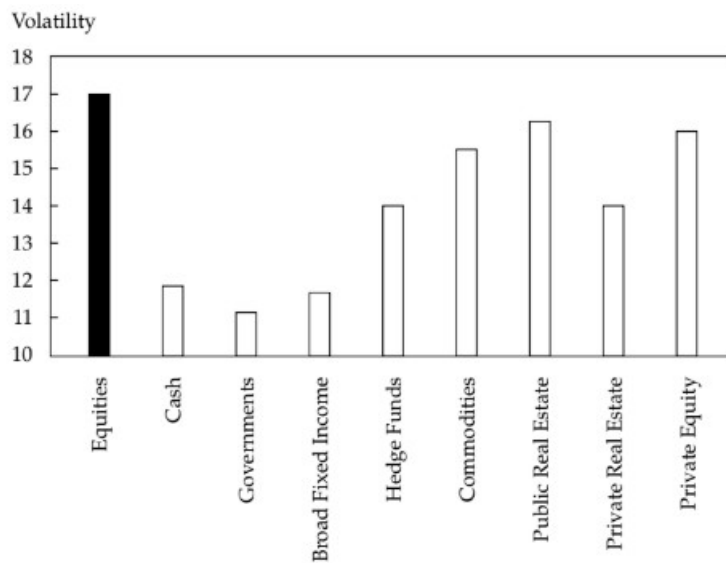
Diversifying equity risk with alternative assets: alts versus bonds.

- Alt returns have to be (1) unsmoothed, (2) databases have to be removed of backfill and survivorship bias, (3) alt funds have very low correlations to each other (because of idiosyncratic strategies) and so an HF or PE index might have VERY low volatilities and can't be used for comparison to a single particular PE or HF fund.
- So, PE and HF indices have very low volatilities because of (1) smoothed returns, and (2) low inter-fund correlations.
- LT. While government bonds have lower correlation and lower volatility than alternative assets, and can therefore reduce returns volatility when added to equity PFs, over LT horizons, the returns are low. Low returns compounded over LT means comparatively VERY low returns.

Note: bonds and equities are usually negatively correlated but correlation can become positive during high inflation periods. Below are betas (directional relationship) and correlations to equity (strength of that relationship due to equity), with alts being UNsmoothed:



Below is 70% equities + 30% other given class portfolio return volatility:



Traditional (functional) approaches to asset classification: Based on roles. Three types:

- Growth, income, risk diversification, safety.

- Liquidity. Some alternative assets are liquid, e.g. public REITs. But most are illiquid. Can do asset allocation with equity, FI, and RE. Then decompose equity into public and private equity, FI into public and private credit, RE into public and private RE.

	Equity & Equity-Like	Fixed Income & Fixed Income-Like	Real Estate
Marketable/Liquid	Public Equity Long/Short Equity Hedge Funds	Fixed Income Cash	Public Real Estate Commodities
Private/Illiquid	Private Equity	Private Credit	Private Real Estate Private Real Assets

- Macroeconomic regimes. Inflation environment, economic growth environment, etc. Growth assets: public and private equities (but: high correlation between these assets!). Inflation-hedging assets: real assets, inflation-linked bonds. Deflation-hedging assets: nominal bonds (when inflation is low or negative).

		Inflation Environment		
		Deflation	Moderate Inflation	High Inflation
Economic Environment	High Growth		Public Equity Private Equity High-Yield Bonds Private Credit	Real Estate Commodities
	Low Growth/ Recession	Government Bonds		Inflation-Linked Bonds Gold

Benefit of traditional approach compared to risk-based approach: good for liquidity and manager selection.

Risk-based approaches to asset classification. Risk factors (e.g. size, value, liquidity, etc.) instead of traditional asset allocation. Example: conditional factor model. Risk factors are assets with a common sensitivity:

- Equity market return: representative of the general direction of global equity markets, and investors may also refer to this as the best market proxy for “growth”
- Size: excess return of small-cap equities over large-cap equities
- Value: excess return of value versus growth stocks (negative factor sensitivity = growth bias)

- Liquidity: the Pastor–Stambaugh liquidity factor, which is a market-wide liquidity measure based on the excess returns of stocks with large sensitivity to changes in aggregate liquidity (less-liquid stocks) versus stocks with less sensitivity to changing liquidity (more-liquid stocks)
- Duration: sensitivity to 10-year government yield changes
- Inflation: sensitivity to 10-year breakeven inflation changes obtained from the inflation-linked bond markets
- Credit spread: sensitivity to changes in high-yield spread
- Currency: sensitivity to changes in the domestic currency versus a basket of foreign currencies
- This framework can easily be extended further to other risk factors, like momentum or volatility.
- Traditional approach. Advantages: easy to communicate (describing asset classes by roles is more intuitive, easier to therefore figure out the mandate for each alternative asset), based on liquidity profiles and operational ease (easier and therefore cheaper) (e.g. third party PMs are based on traditional asset class classifications). Disadvantages: over-estimation of portfolio diversification (e.g. PE and public equities in 'growth assets'), obscures primary risk drivers (e.g. FI includes HY bonds and gov bonds but each has different risk characteristics).
- Risk-based approach. Advantages: leads to better risk factor identification and integrated risk framework (better risk quantification). Disadvantages: sensitive to historical data set (risk factors change through time), implementation becomes complicated (liquidity planning, manager selection, rebalancing), selecting risk factors and how to define them.

Alternative asset investment considerations.

- Risk. MVO uses SD for risk, but SD doesn't often adequately capture risk of alternative assets which are illiquid and with non-normal returns; some use options whose risks are hard to model in MVO. Might get around this by artificially increasing SD of alternative assets. Note: short funds have unlimited downside, unlike equity funds which are bounded on the downside by 0.
- Return expectations. Very idiosyncratic. Can use a 'building blocks' approach: begin with r_f + estimate return for factor exposures + assumed manager alpha - fees - taxes.
- Investment vehicle. Most alternative investments through private limited partnership LP where they are limited in downsides whereas GP takes unlimited downside risk. Otherways: FoF, private partnerships, direct

investments, SMAs (high minimum requirements, operational challenges) or 'funds of one', mutual funds and UCITS (undertakings for collective investment in transferable securities). Note MFs and UCITs have regulatory burdens.

Liquidity.

- Usually there is a private placement memorandum (PPM), e.g. for private limited partnership LPs, HFs, PE funds, etc. which governs investment and redemption conditions. The PPM may contain 'liquidity terms', i.e. when capital may be withdrawn. Even if can't redeem shares, e.g. of private equity fund, can sell those shares subject to conditions, though generally sold at a discount because secondary market is small and security is unique.
- Private equity, private credit, private RE and real asset funds usually 'call' capital after start up. So, they aren't fully funded until a couple of years in. Several capital calls per year is usual. So, investors have to be ready for capital calls - and these aren't predictable. Neither are drawdowns and distributions.
- Liquidity risks. Equity HFs may have '**side pockets**' of illiquid assets which are not part of general 'liquidity terms' so can't be liquidated for some time. So, the investor has to wait for these to be liquidated. Event-driven HFs, e.g. merger-arbitrage, have longer liquidity terms to better enable the strategy. Relative value HFs: restrictions in liquidity terms. Managed futures HFs: often more relaxed liquidity terms. Levered funds: lenders have first rights, which complicates liquidity.

Note: Consider '**vintage year diversification**' for PE funds, so that capital is committed to PE funds over a number of years, instead of being concentrated. PE fund returns often depend on vintage year, so want diversification.

Fees. Admin fees. Usually 0.5%. Management fees, incentive fees. Private equity management fees. Based on committed, not invested, capital. This results in a J-curve of IRRs if not all committed capital is invested early.

Taxes. Most HF investors have tax-exempt status. Endowments, SWF, etc.

Other considerations. If small investor, can't access large funds and may have to settle for FoFs. If large investor, still problems: (1) as top-tier funds may be capacity-constrained, and large investors may have to make public disclosures which are not favored by HFs, (2) need to have access to the PMs to properly evaluate investment strategies and evaluate quickly when to leave, (3) whether there is sufficient in-house expertise and resources to oversee third-party funds and/or manage own, (4) ESG and other considerations.

Suitability.

- Expertise. Information asymmetry between LP and GP.
- Investment philosophy. If investor thinks markets are efficient, then hard to use alternative assets which are active management.
- Transparency. Alternative asset funds like HF funds are often very opaque. No standardization of reporting of alternative asset funds. Increases fraud probability.
- Fund value. In many cases, use independent administrators to calculate NAV and process cash flows and fees - common in HF funds but not so much in private equity and credit, real asset funds. Some funds will have third party auditor.

Note: SMAs are more transparent and can control capital flows better because manager doesn't have discretion, but not suited for alternative strategies (lack of resources and expertise) and manager can't invest alongside SMA (reducing interest alignment).

Asset allocation continued. Above, noted that one approach to asset allocation is to divide into broad asset classes and then do detailed tuning. This is about detailed tuning, the second step.

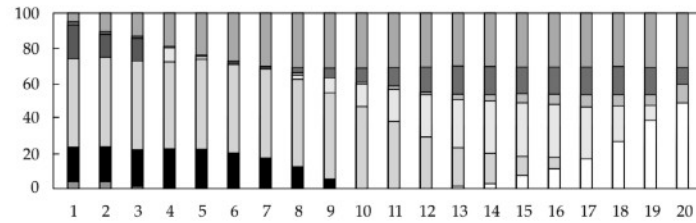
Statistical challenges: (1) appraisal-based valuations underestimate issue of smoothed returns - look for autocorrelated returns and if yes then unsmooth, (2) non-normal returns (skew, excess kurtosis) is even bigger problem for alternative assets (particularly private RE and equity market-neutral HFs, and distressed HFs, which have negative skew and fat tails). To address (2), use historical returns (but often not enough time points), time-varying volatility models (stochastic volatility), regime-switching models (e.g. combine normal distributions with different average returns, volatilities, correlations to capture skew and fat tails), extreme value theory.

Use one or a mix of techniques:

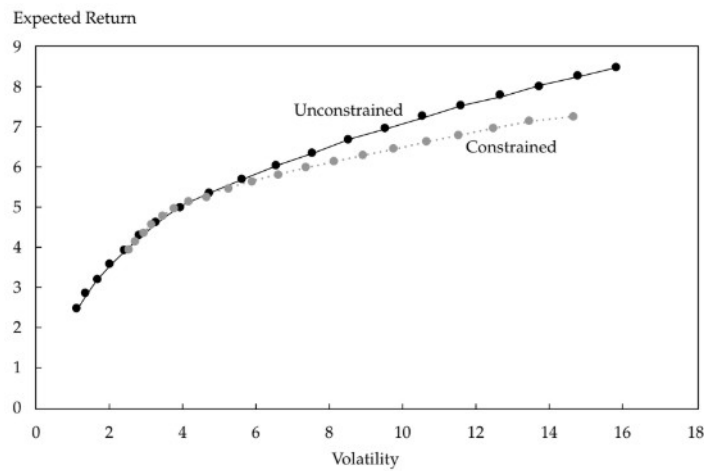
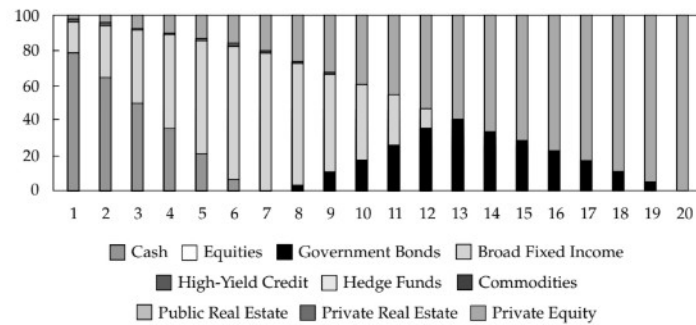
- Monte Carlo simulation. Relax assumption of normal returns. Can set variables, which time series model (e.g. random walk, mean-reversion), which distributions, which volatilities and correlations and whether they change through time, whether there's active management or just passive management. Incorporate skew and fat tails by combining two or more normal distributions, regime 1 over X years of history when equity did well, regime 2 over Y years of history when bonds did well (high volatility regime).
- Optimization techniques. MVO and Mean-CVaR.
- Optimization: MVO. MVO over-allocates alternative assets because SDs are underestimated, so use minimum or maximum constraints (e.g. x% of

alternative assets). Incorporate CVaR to enhance asset allocation. Can get many different possible portfolios depending on the return target.

A. Constrained Portfolios



B. Unconstrained Portfolios



- Optimization: Another type of MVO is mean-CVaR optimization to improve means versus absolute losses (CVaR) - this could significantly change optimal PFs especially if negative skew and fat tails. Mean-CVaR requires longer historical series to establish the higher moments. The CVaR risk = 'tail risk' with -7% being 'better' for CVaR optimization than -10%.

- Risk-factor approaches, or risk factor-based optimization. Note: 'global equity' is often a proxy for the risk factor of 'growth' returns. Start with expected risk factor movements: e.g. assume growth rises, then duration and value expected factor returns will be negative (as rates rise nominal duration factor returns fall, and returns rising leads to value factor returns rising); if concerned about inflation, then assume positive to inflation risk factor (i.e. expected inflation is positive). When optimized, equity diversifies well with duration and value, so stocks with high duration and value factors will actually rise in proportion in the optimized PF!

Note that correlations, whether in Monte Carlo, MVOs or risk-factor approaches can change through time.

For beginner investors, use MVO (simple) with Monte Carlo simulations to show different possibilities. For more sophisticated investors, use mean-CVaR optimization (which has more sophisticated view of risk - left-tail risk).

If investor doesn't have the resources for in-depth monitoring and analysis, then risk-factor based approach not appropriate.

Liquidity planning (e.g. payments to beneficiaries, PF rebalancing, capital calls to PE funds). For funds, use a liquid forecasting model or pacing model, e.g. Capital contribution = rate of contribution x (Capital commitment - already-paid-in-capital). Not all committed capital may be called. Distributions are a proportion of NAV; NAV increases as capital is called and falls as D is made. Make assumptions about pattern of capital calls and distributions, and growth (IRR) of fund.

$$D_t = RD_t(NAV_{t-1} \times (1 + G))$$

$$RD_t = (t/L)^B$$

$$NAV_t = (NAV_{t-1} \times (1 + G)) + C_t - D_t$$

where RD is rate of distributions at time t, L is the expected life of the PE fund, G is growth rate, C is capital contribution and D is distribution, B is a 'bow' parameter like 2.5 (gradual to 100%) or 5.0 (starts slower but accelerates to 100% much faster). Lower D's early in fund life allows NAV to grow faster and more.

Liquidity planning: preparing for unexpected. Need to prepare for capital calls, often within 30 days of notification. Sometimes not all capital called. Want to keep funds ready in liquid but yielding accounts. During bear market, capital calls faster and distributions slower, and GP might exercise option to extend fund life - leaving distributions very late. Use stress test to model.

Liquidity planning: bad CME with rising inflation. Little cash inflow. Look at liabilities over next year (capital calls, spending needs). Look at immediate liquidity (cash, government bonds within SAA range), as well as other liquidity

(equities, credit, HFs, etc.). Note that selling assets to get to low end of SAA range needs to consider the downside stress test results. Move funds into cash and government bonds to the extent possible to increase liquidity, shorten the duration of bond portfolio as interest rates will rise with rising inflation. Real estate is defensive (steady rents, low volatility of capital), so increase allocation (or just leave, as PF total value will decline anyways).

Liquidity planning: investing in private RE with multiple capital calls over next years. Invest 100% in REITs, not cash nor cash + REITs. REITs have most similar return and risk characteristics and maintain SAA of plan. Cash won't meet LT objectives of plan (opportunity cost).

Monitoring. Must monitor both PMs individually and the fund as a whole. Have to monitor all the time, not just at investment maturity - e.g. to be ready for situational changes.

Monitor these: key person risk (lower for quantitative processes), alignment of interest (fees, compensation, AUM, key professional committed capital, etc.), style drift (make sure PM is using strategies they are good at), risk management (whether leverage is being used), client and asset turnover, client profile, service providers (third party PMs, auditors). If investing in fund which will have a big influx of funds soon, be careful: big influx could dilute near-term performance if funds remain underdeployed and there is insufficient capital to invest them; further, might lose control and might have larger capital calls.

Performance evaluation.

1. Fund quarterly reporting = lag + volatility is too smooth. Stale pricing from quarterly reporting. Betas, correlations, Sharpe ratios will not be what they should be.
2. Investment vintage, and vintage years.
3. Performance evaluation can be done wrt IRRs, time-weighted returns (TWR), multiple on investment capital (MOIC; current value + distributions divided by total invested capital). Example: use MOIC if return of investment is too fast (artificially high IRR) or use IRR if investment time is long (artificially high MOIC).
4. Benchmarks: (1) public index + premium, custom benchmark (e.g. 5% + equity index) - but subjective criteria, can't exactly match situation (won't match PE-type risk, return, liquidity), pricing issues, and (2) rank or compare to peer group representative - but each PE and HF have idiosyncratic strategies and therefore properties, different methodologies used by PE and HFs for definitions, weighting and valuation; and ranking and comparison mean depends on what other managers do. No benchmark is perfect, but these are ones used in industry.

5. Peer evaluations are not a good substitute for market price indexes because it is difficult to identify a peer group with the same investment strategy as the manager being evaluated. Furthermore, the lack of standards for constructing such benchmarks results in the same manager's performance being assessed in very different ways by different evaluators.
6. Qualitative analysis based on PM's initial thesis: what were PM's goals? How PM planned to add value? What was exit strategy? Holding period?
7. To make a judgment, (1) use more than one criteria, and (2) wait until most or all investments have been returned to investor.
8. Although it is difficult to assess the performance of an investment program that is not fully invested (such as a private equity investment or a direct real estate investment), investors must still monitor market conditions to ensure the investment program remains appropriate.

Notes:

- PE has higher returns than HFs.
- REITs: real estate asset depreciation can be used to offset some of the income from it, reducing taxes to the investor.
- SMAs: high minimum investment because tailored service, own service with high overhead. Need operational expertise.
- UCITS: high liquidity, but low risk-return. Regulatory burdens.

2.4 Private wealth management

Definitions.

- Net worth. Assets minus liabilities which are easy to measure, and excludes nontangibles and claims which are hard to measure.
- Aggregate wealth. Total value of all assets.
- Personal property (exclude here primary residential property, which is a real asset), financial wealth, real wealth, productive wealth, liquid wealth, investable wealth (assets for investment).
- Investable net worth. Sum of liquid assets, such as savings and investment accounts (but NOT checking accounts for day-to-day needs, which are transaction accounts), minus ST liabilities such as credit card debt.
- Economic net worth. Includes PV human capital and vested pensions.

Global wealth.

- Countries with economic growth opportunities. Technological advancement, free market, business privatization, business formation (entrepreneurship), deregulation.
- Sources. Entrepreneurship and asset price appreciation = main source of wealth creation. Privatization of government assets, IPOs, financial innovation and financial product development, sudden fame, are others.
- Mix. China has rising debt due to increasing access to credit and to fuel growth. Poorer economies tend to have more nonfinancial assets versus financial assets, and low debt.
- Wealth growth. North America: business formation and entrepreneurship lead to economic wealth, leading to wealth; increasing productivity growth, financial market returns (from equity-based compensation), professional athletes and artists; but wealth increase has been uneven. Europe: inheritances dominate wealth (generally RE). Asia-Pacific: high economic growth (and economic transformation from agrarian to industrial and service), integration into global commerce, financial development. Latin America: growth, but unequal.
- Inequality. Both income and wealth inequality. Use the Gini coefficient: 0 for perfect equality, 1 for perfect inequality. Technology, demography affect LT Gini.

Institutional versus individual investors. Note institutional investors are usually highly sophisticated: MORE sophisticated than individual investors. Institutional investors face objective and quantifiable risk (e.g. volatility, beta) and measure performance by comparison to benchmarks, whereas it is probability of 'failure to meet a goal' for individual investors.

Economic balance sheet. Human capital:

$$HC_0 = \sum_1^N \frac{p(s_t)w_{t-1}(1+g)}{(1+r_f+y)^t}$$

where $p(s_t)$ is probability of survival to year t , w_{t-1} is income, r_f is the **nominal** riskfree rate, y is the premium associated with occupational income volatility (dying or getting fired, or having uneven income).

Lifetime wealth accumulation. If have stable salary when young, balance with equities. If entrepreneur, fixed-income assets and investments in industries less correlated to their fields.

Individual investors. Considerations.

- Inflation. Equities beat inflation over the LT but not a good hedge in ST. Commodities are an inflation hedge in the ST but don't beat inflation

over LT. Real estate is hedge against inflation as the rise is positively correlated.

- Risks. Return volatility, drawdown risk, risk of not achieving goals.
- Risk. Risk tolerance (a bit subjective; the return per risk) and risk capacity (objective; amount of capital that can be put at risk without affecting current and future consumption), and risk perception (very subjective, depends on investment and situation).
- Goals. Planned (e.g. retirement, philanthropy) and unplanned (e.g. medical issues).

Taxation and inflation. **Tax drag** = (value with no taxes at time T minus value with annual taxes at T) divided by (value at time T no taxes minus value at time 0). It is the taxes paid, and lost compounding from paying taxes. Tax drag is larger than accrual tax rate; this means FI, which is usually accrual-taxed, is tax-dragged a lot. For simplicity, assume entire investment income return R is taxed each year, then the future value interest factor is:

$$FVIF_T = (1 + R(1 - t_x))^T$$

with the real version being

$$FVIF_T = \frac{(1 + R(1 - t_x))^T}{1 + r_{inflation}}$$

and for deferred taxed, e.g capital gains,

$$\begin{aligned} FVIF_T &= (1 + R)^T - ((1 + R)^T - 1) \times t_{CG} \\ &= (1 + R)^T (1 - t_{CG}) + t_{CG} \end{aligned}$$

where the second is rearrangement of the first formula, and because the second formula's first term is an overtaxation of the 1 from $1 + R$, the principal is added back with t_{CG} .

Cost basis on capital gains. The cost on which capital gains are assessed; subtract off cost to buy it. Selling price minus cost basis = capital gains. B = cost basis (how much the asset has appreciated until now) divided by CURRENT MV. If $B = 1$, there is no cost basis (it was probably purchased recently and has no 'embedded' tax). If $B = 0.8$, then it has appreciated 20% until the current time, and so has a lot of embedded tax. Low cost-basis = more tax.

$$FVIF_{CG,B} = (1 + R)^T - ((1 + R)^T - 1) \times t_{CG} - (1 - B) \times t_{CG}$$

where the last term is like a 're-basing' before the returns compound. Rearranging,

$$= (1 + R)^T (1 - t_{CG}) + t_{CG} - (1 - B) \times t_{CG}$$

$$\begin{aligned}
&= (1 + R)^T(1 - t_{CG}) + B \times t_{CG} \\
&= Value((1 + R)^T(1 - t_{CG}) + B \times t_{CG})
\end{aligned}$$

where B is cost basis proportion of current (i.e. at time $t = 0$ before time T) market value. If something was just purchased, e.g. \$2 million house, then $B = 2/2 = 1$.

Note! Taxes are levied on **nominal** price appreciations! This is already reflected in the formulas. Dividing income and capital appreciation:

$$\begin{aligned}
FVIF_{INC,CAPITAL} &= (1 + (R_{INC}(1 - t_x)) + R_{CAP})^T \\
&- \frac{R_{CAP}}{R_{INC}(1 - t_x) + R_{CAP}} \times ((1 + (R_{INC}(1 - t_x) + R_{CAP})^T - 1) \times t_{CG}
\end{aligned}$$

Income = dividends.

Investment policy statements.

- Use this to figure out **capital sufficiency analysis** whether there's enough capital to fulfill goals and objectives. Capital sufficiency analysis triggered if large (e.g. more than 5%) change.
- Specifies SAA, TAA or both.

Parts of an IPS

- Background. Assets, values, accounts, people, wealth sources, core values. Make sure to get: taxes, existence of pensions.
- Investment objectives. ST and LT goals (if they conflict, find a primary goal). Overall PF returns and risk objectives. Specific goals. Cash flows and liquidity.
- Investment parameters. Risk tolerance, time horizon, asset class preference, liquidity preferences, other investment preferences (ESG, passive or active, special requests), constraints (tax mitigation strategy, ESG constraints), PF asset allocation (including asset target ranges).
- Portfolio management. Discretionary authority. Full discretion: rebalance and replace fund managers without prior approval. Nondiscretionary: recommendations but need prior approval. Otherwise, authority to make certain changes. Frequency and method of rebalancing: time-based (easy to understand, simple rules but unresponsive to sudden changes, making small adjustments costly, have to transact at exact times and incurs losses), threshold-based (rebalance when deviate from specific percentage, but complex, might over-react). Also: timeline of PF, derivatives use, third-party funds (and how to benchmark), PM investment preferences.
- Duties and responsibilities.

- IPS appendix. Details that change more frequently than in main IPS. Ranges of PF behavior and returns and quantify risk, CMEs.

Note: if young and working in technology start-up, a PF with technology equity is risky. So, diversify, even if it's FI.

2.5 PM for institutional investors

Institutional investors. Distinguished from retail and high-net worth investors by: (1) scale, (2) LT horizon, defined by liabilities, (3) regulatory framework, more complex, (4) governance framework, well-defined, (5) principal-agent issues, must be managed.

Overview of investment policy.

- IPS helps minimize principal-agent problems by providing clear guidance on day-day management of assets.
- Risk tolerance is expressed differently for different IIs. DB pensions: surplus volatility (SD of asset returns in excess of liability returns). SWFs: probability of investment loss or purchasing power loss over a time period. Endowments: return volatility. Banks and insurance companies: VaR, CVaR, scenario stress tests.

IPS objectives section: whether in perpetuity, what level of risk (e.g. 'reasonable') and real return rate (often = annual spending policy).

Major approaches.

- Norway model. Nearly all equities and FI. Passive. Almost no alternative assets. Low-cost, simple.
- Endowment model, popularized by Yale. Lots of alternative assets (PE and HF). Very actively management, lots of external PMs. Good for LT horizon, high risk tolerance, small liquidity needs, skill in outsourcing. Expensive.
- Canada model. Lots of alternative assets. Relies more on INTERNAL PMs. Use a passive 'reference PF' which is public equities and public FI consistent with institution's objectives, but SAA has targets for PE and HFs and is actively managed to get a 'total PF' where risks are kept constant even as securities enter and leave the PF.
- Liability-driven investment (LDI). Generate returns sufficient to cover liabilities. Lots of long-duration FI.

Investment Approach	Description
Norway Model	Traditional style characterized by 60%/40% equity/fixed-income allocation, few alternatives, largely passive investments, tight tracking error limits, and benchmark as a starting position. <i>Pros:</i> Low cost, transparent, suitable for large scale, easy for board to understand. <i>Cons:</i> Limited value-added potential.
Endowment Model	Characterized by high alternatives exposure, active management and outsourcing. <i>Pros:</i> High value-added potential. <i>Cons:</i> Expensive and difficult to implement for most sovereign wealth funds because of their large asset sizes.
Canada Model	Characterized by high alternatives exposure, active management, and insourcing. <i>Pros:</i> High value-added potential and development of internal capabilities. <i>Cons:</i> Potentially expensive and difficult to manage.
LDI Model	Characterized by focus on hedging liabilities and interest rate risk including via duration-matched, fixed-income exposure. A growth component in the return-generating portfolio is also typical (exceptions being bank and insurance company portfolios). <i>Pros:</i> Explicit recognition of liabilities as part of the investment process. <i>Cons:</i> Certain risks (e.g., longevity risk, inflation risk) may not be hedged.

Pension funds

DB plans. Consider service, salary, mortality, discount rates, cyclicalities of business, DB plan size versus business size, how old people in plan are, how quickly can switch out of plan, longevity risk, tax statuses, etc. Funded ratio (= vested benefit index) = fair value of plan assets divided by PV DB obligations. Going below 100% is **shortfall risk**.

DB plans that are overfunded. Potentially supports company's share price, company has to pay fewer contributions in the future. If high employee turnover, then funded ratios generally increase. Note: if employee part of DB pension quits before retirement, they might only get what is vested at that point.

DB plans that are underfunded. Options: (1) LDI suits plans that are slightly underfunded, (2) grow assets aggressively - also suits plans want to reduce contributions over time and can tolerate higher uncertainty and volatility, (3) invest defensively (less return volatility) and increase contributions.

DB plan objectives: exceed discount rate, which is too low. Use detailed Asset Liability Management studies every 3-5 years, which are Monte Carlo simulations of returns and liabilities under different scenarios, including stress tests, and calculating different measures.

DB plan risk tolerance. If underfunded, aggressive (i.e. high) discount rate, old members close to retirement, long duration, high correlation of returns with company earnings, company is not doing well financially, then low tolerance. HOWEVER, underfunding can also lead to higher risk tolerance.

DC plans. DC plan sponsor takes liquidity risk of plan if participants exit

quickly. Usually have life-cycle options (i.e. automatic options): (1) participant-switching life-cycle options which switch asset mixes to more conservative mixes as they get closer to target dates and (2) participant-cohort option which pools assets for cohorts with similar target dates.

Pension fund external constraints. Laws and regulations for transparency, fees, auditing, reporting, fiduciary duties (to reduce principal-agent problems), funding status, how much equity of the company's equity a pension fund can own if the pension is for that company, minimum diversification, minimum contributions for people who are not sophisticated - because the government may have to pay public pensions. Highly regulated as government charges plans and pays retirees of terminated plans and tax benefits are costly to governments. Regulates discount rates. If plans don't follow, then lose tax benefits as an example.

Pension fund asset mixes. Equities - hedge inflation. FI - hedge interest rate risk (but don't do well in inflation). Real assets - hedge inflation. Over the last decade or so, more funds have moved equities to FI because of funding ratio volatility concerns. Many funds have invested more in alternative assets. Some countries, like Australia and Netherlands, have strong home biases.

Note: If moving to higher-return assets in portfolio, then **discount rate applied to liabilities** can be increased, reducing liabilities. At the same time, asset returns will rise, leading to increased funding status.

Note: If move assets to alternative assets from traditional equities and FI, then there's manager selection risk.

Note: LOW correlation between DB pension fund results and company's operating results suggest low risk tolerance.

Note: ERISA of US requires DB pension plan appear on balance sheets.

Sovereign wealth funds.

- Usually established by legislation and need good governance. Santiago Principles - a set of good practice guidelines.
- Often invest abroad to diversify against domestic assets. This raises international tax issues.
- Liabilities. Unlike pension funds, banks, endowments - SWF liabilities are unclear.

Type	Objective	Examples
Budget stabilization funds	Set up to insulate the budget and economy from commodity price volatility and external shocks.	Economic and Social Stabilization Fund of Chile; Timor-Leste Petroleum Fund; Russia's Oil Stabilization Fund
Development funds	Established to allocate resources to priority socio-economic projects, usually infrastructure.	Mubadala (UAE); Iran's National Development Fund; Ireland Strategic Investment Fund; Temasek (Singapore)
Savings funds	Intended to share wealth across generations by transforming non-renewable assets into diversified financial assets.	Abu Dhabi Investment Authority; Kuwait Investment Authority; Qatar Investment Authority; Russia's National Wealth Fund
Reserve funds	Intended to reduce the negative carry costs of holding reserves or to earn higher return on ample reserves.	China Investment Corporation; Korea Investment Corporation; GIC Private Ltd. (Singapore)
Pension reserve funds	Set up to meet identified future outflows with respect to pension-related contingent-type liabilities on governments' balance sheets.	National Social Security Fund (China); New Zealand Superannuation Fund; Future Fund of Australia

SWF 5 types. But some funds can be mixes.

- Budget stabilization funds. Objective: capital preservation. Usually ST and support government budget (make up for shortfalls), diversify away from government's main revenue sources. Usually invest in government bonds and debt (stable, diversify against cyclical government budget revenues). Example: Russia's Oil Stabilization Fund.
- Development funds. Objective: support national economic development. Doesn't invest in 'assets' in PF sense.
- Savings funds. Objective: maintain purchasing power. LT horizons. Take non-renewable assets and turn them into real and financial assets for future generations, e.g. Norway Government Pension Fund Global. Also includes government investment holding companies, e.g. Temasek. Invest in equities and alternatives. Usually follow endowment model.
- Reserve funds. Objective: max rate of return. Excess central bank FX reserves. Get a return higher than FX reserves. Usually invest in high-grade debt, equities. ST horizons.
- Pension reserve funds. Fund using fiscal surpluses. LT horizons. Fund unfunded or contingent pensions, health care, social security. Invest in equities, alternative assets. These funds have an accumulation phase and a decumulation phase. Usually follow endowment model.

University endowments and private foundations

Foundations: community foundations (public donations), operating foundations (individuals; e.g. for museums), corporate (businesses), and private grant-making foundations (individuals; specific purpose).

Usually subject to (1) following modern PF theory and therefore diversify, and (2) fiduciary duty.

Taxation of gifts and donations. Often gifts to endowments and foundations, income and capital gains for the endowment or foundation, and payouts if receiving institution is tax-exempt.

University endowment spending policy: tension between investing and spending. Based on capacity for fund-raising, university operating budget, and ability to borrow. Rules for spending:

- Constant growth rule. Regardless of how well the fund does. Fixed amount annually to university, adjusted for inflation by $1 + r$. Can be floors and caps.
- Market value rule. Percentage of fund given to university. But then it fluctuates each year.
- Hybrid (Yale spending) rule. Weighted average of constant growth and market value rules. $S_{t+1} = w \times S_t(1 + \pi) + (1 - w) \times SR \times AUM_{average}$, where S is spending amount.

Private foundations. Generate real return (above CPI of 5%) with reasonable volatility (10-15% annual SD) over 3-5 years typically, with secondary objective of outperforming policy benchmark. Similar to university endowments. But different: (1) board members in endowments are investment professionals, but grant experts in foundations; (2) allows impact investing; (3) foundations have greater liquidity needs; (4) in the US must spend all donations in that year (flow-through rule) and spend at least 5% of assets per year to remain tax-exempt; (5) hybrid rule only; (6) as they don't have as many income sources (e.g. tuition) as universities, PF is more conservative; (7) don't necessarily need principal-protection, and instead can have limited lifetime foundations; (8) foundations have to support the entire organization and are different from universities; (9) foundations usually don't get further donations which is different from endowments.

Larger foundations (and endowments) allocate more to alternatives.

If an endowment pays a small proportion of a university's budget, then there's less volatility for the fund.

Banks and insurers

After GFC, systematically important financial institutions (SIFIs) began to be directly or indirectly regulated: (1) higher capital requirements, (2) limit dividends and repurchases of common equity, (3) make subordinated debt and preferred shareholders less able to assert claims in bankruptcy and restructuring. Derivatives, prop trading, off-BS items became more regulated - reporting, reserve requirements, stress testing, etc.

Commercial banks have higher cost of funds and lower liquidity because of wholesale funding (large, off-balance sheet promise to lend) and contingent commitments - so they need large capital reserves. Retail banks have lower cost of funds and better liquidity because retail deposits are low cost and stable.

Insurers. Publicly-listed, or mutual companies. Often have **general account** for general policies and **separate accounts** for specific or variable policies. General accounts have **reserve PFs** (regulated, and conservatively managed) and **surplus PFs** (more aggressively managed). Life insurers have longer investment horizons than P&C insurers.

Both types of insurers: two-part liquidity plan of internal and external components. Internal liquidity: cash in BS for operations, investment portfolio income and principal repayments. External liquidity: issue bonds in capital markets and access bank credit, buy and sell repos for ST liquidity.

Insurers and interest rates. High or rising rates mean life insurers face cash outflows as policies are surrendered and people look for higher yields elsewhere.

P&C insurers face uncertainty re timing and size of payments. Lots of cash and ST FI.

IPS should specify market, credit, interest rate risk appetites. Specify liquidity, FX risk, regulatory requirements, tax considerations as well. The IPS can just say will monitor these risks, follow regulations, and consider tax consequences.

Balance sheet. Can measure liabilities, e.g. deposits and insurance policies. Use probabilistic methods to model outcomes. Use **LDI** unlike pensions, SWFs, endowments and foundations.

Calculating modified durations. Assets = liabilities + residual equity:

$$A = L + E$$

$$\Delta A = \Delta L + \Delta E$$

Divide by E,

$$\frac{\Delta A}{E} = \frac{\Delta L}{E} + \frac{\Delta E}{E}$$

$$\frac{\Delta A}{A} \left(\frac{A}{E} \right) = \frac{\Delta L}{L} \left(\frac{L}{E} \right) + \frac{\Delta E}{E}$$

Rearranging,

$$\begin{aligned} \frac{\Delta E}{E} &= \frac{\Delta A}{A} \left(\frac{A}{E} \right) - \frac{\Delta L}{L} \left(\frac{L}{E} \right) \\ \frac{\Delta E}{E} &= \frac{\Delta A}{A} \left(\frac{A}{E} \right) - \frac{\Delta L}{L} \left(\frac{A-E}{E} \right) \\ \frac{\Delta E}{E} &= \frac{\Delta A}{A} \left(\frac{A}{E} \right) - \frac{\Delta L}{L} \left(\frac{A}{E} - 1 \right) \end{aligned}$$

So, high leverage (A/E) means a small change in asset or liability leads a big change in equity. Let y be 'reference yield', i.e. yield on assets. Let i be yield on liabilities (i.e. how much liabilities 'cost').

$$\frac{\Delta E}{E \Delta y} = \frac{\Delta A}{A \Delta y} \left(\frac{A}{E} \right) - \frac{\Delta L}{L \Delta y} \left(\frac{A}{E} - 1 \right)$$

and rearrange

$$\frac{\Delta E}{E \Delta y} = \frac{\Delta A}{A \Delta y} \left(\frac{A}{E} \right) - \frac{\Delta L}{L \Delta i} \left(\frac{\Delta i}{\Delta y} \right) \left(\frac{A}{E} - 1 \right)$$

Modified duration of asset W is

$$D_W^* = -\frac{\Delta W}{W \Delta r}$$

So,

$$D_E^* = D_A^* \left(\frac{A}{E} \right) - D_L^* \left(\frac{\Delta i}{\Delta y} \right) \left(\frac{A}{E} - 1 \right)$$

which is actually

$$D_E^* \left(\frac{E}{E} \right) = D_A^* \left(\frac{A}{E} \right) - D_L^* \left(\frac{\Delta i}{\Delta y} \right) \left(\frac{L}{E} \right)$$

where duration of equity and assets are defined with respect to y (the reference yield) and liability duration is defined wrt to i (liability cost). Generally, sensitivity of yield changes:

$$\frac{\Delta i}{\Delta y} = 0.9$$

The equity-to-assets ratio = capital ratio = equity capital ratio = equity ratio. Generally around 10-20%. To specify duration surfaces as in EXHIBITS 24 and 25, specify E-A ratio and sensitivity of yield changes.

Exhibit 24: Duration of Shareholders' Equity as a Function of Asset and Liability Durations (Given Equity/Assets = 20% and Sensitivity of Yield Changes = 0.90)

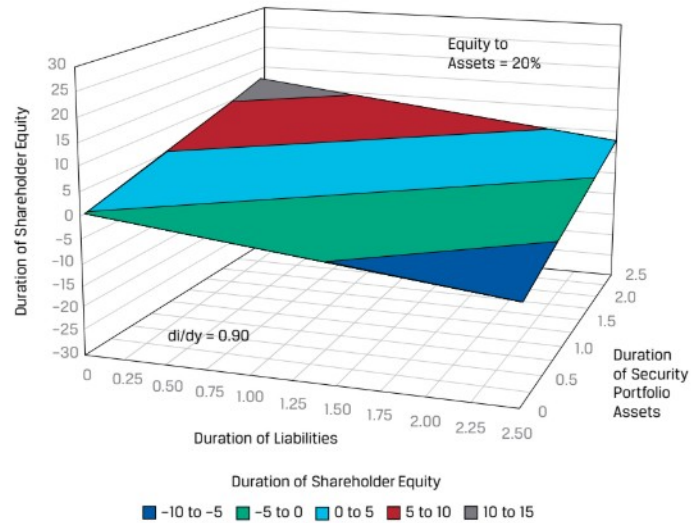
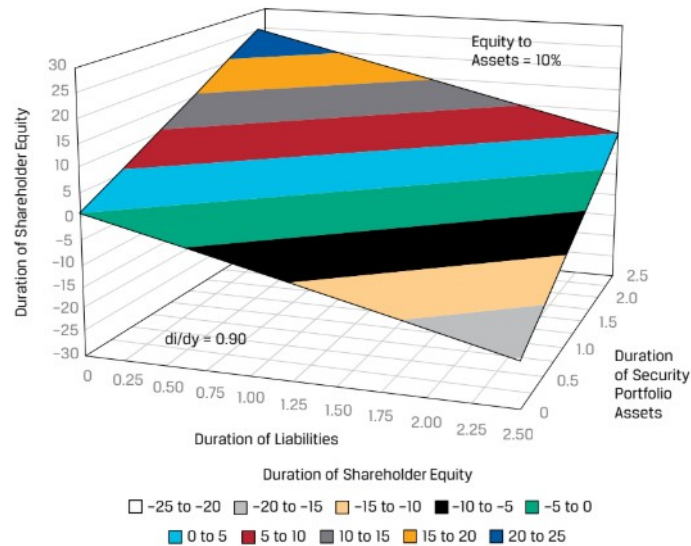


Exhibit 25: Duration of Shareholders' Equity as a Function of Asset and Liability Durations (Given Equity/Assets = 10% and Sensitivity of Yield Changes = 0.90)



Notes: (1) equities (and preferred stock, which is like equity) have very high durations, (2) fixed-rate mortgages have very high durations (because mortgages last a long time), (3) most deeply-subordinated debt is perpetual so has high duration.

Low capital ratios mean high leverage, which means high equity durations, which means high sensitivity to interest rate changes.

Banks often 'climb yield curve' by borrowing ST and lending LT, but not necessarily true: if D_L^* is very low, and D_A^* is high, then D_E^* is extremely high and extremely volatile. To reduce asset duration:

- Cash, increase deposits at central banks, FX reserves
- Increasing floating rate loans, including credit card floating rate loans
- Securitize fixed-rate mortgages and sell them off

To increase liability duration:

- Issue LT debt
- Issue deeply-subordinated debt (typically LT debt, increases duration if callable bonds)
- Issue preferred stock

ALSO, use futures and interest rate swaps to reduce asset-liability duration mismatches.

Volatility. Square the above, the LHS is overall measure of bank's balance sheet risk.

$$\sigma_E^2 = \left(\frac{A}{E}\right)^2 \sigma_A^2 + \left(\frac{A}{E} - 1\right)^2 \sigma_L^2 - 2 \left(\frac{A}{E}\right) \left(\frac{A}{E} - 1\right) \rho \sigma_A \sigma_E$$

where σ_E is actually $\sigma\left(\frac{\Delta E}{E}\right)$ and so forth. ρ is correlation between percentage value changes of assets and liabilities. When correlation is -1, σ_E is large (e.g. when assets fall, liabilities rise). When correlation is +1, σ_E is 0 (e.g. when assets fall, liabilities also fall). Notes from EXHIBIT 27.

- Balancing asset and liability durations often leads to increased ρ .
- Increasing common equity increases σ_E and reduces ρ . The latter because equity returns don't correlate well with liabilities.
- Increasing derivatives use reduces σ_E, σ_L and increases ρ .
- More surrender penalties. Reduces σ_L .
- More prepayment penalties. Increases ρ . When interest rates are rising, offset rising liabilities with rising equity values.
- Variable interest rates = shorter duration. Fixed rates = longer duration.

Further notes.

- Another good way of buffering financial uncertainty: ability to raise capital.
- When reducing liability SD, note this usually reduces earnings and EPS. Also, increasing shares will reduce leverage but reduce EPS denominator. However, these kinds of changes will reduce volatility and therefore discount rate.
- Variable rates on corporate loans and variable rates (by using derivatives) on government securities are different. The latter has lower volatility and greater liquidity. It is also better for regulatory capital. Also, during a recession, the corporate spreads will be high, so can switch from government securities to corporate loans to get higher profit in future.
- If changing assets (e.g. from regulatory capital to securitized car loans), then might want to see its interaction with liabilities and change liabilities too.
- Pay-fixed, receive-floating interest rate swap is a **synthetic liability** because duration increases. It is liability, where the liability becomes smaller as rates rise and liability increases as rates fall. So, trading FRNs for fixed-rate loans + pay-fixed, receive-floating swaps = asset duration increases + liability duration increases.
- 'Plain vanilla' derivatives are easy to value and easy to unwind. While regulatory capital (for banks, insurers) estimates penalize derivatives use (stress tests assume worst for derivatives), if use is limited then OK.
- If interest rates rise, annuities might be surrendered more. Reduces liability volatility.
- Asset-backed securities have short durations because (1) early payment of principal, (2) often floating rates, and (3) prepayment option.
- Corporate debt often has longer duration than government securities because of higher coupon rates, but this is mitigated by callability.
- Corporate debt has shorter durations than government securities of similar maturity because former usually has higher coupons. Asset-backed securities have even shorter durations. Shorter durations = better if rates are about to rise.
- Most corporate debt is callable by the issuer, reducing duration.
- Asset-backed securities often have prepayment risk. Borrower can refinance if rates fall, leading to sudden payment of principal. Asset-backed securities often on floating rates too.
- If think interest rates will rise, then buy puts on government securities (if have any), buy swaptions with right to be a fixed-payer and floating-receiver.

2.6 Trading costs and electronic markets

Explicit costs: broker commissions, transaction taxes, exchange fees. Implicit: bid-ask spread, market impact, delays, opportunity costs.

Best bid = inside bid. Best ask = best offer = inside ask. Bid-ask spread = inside spread.

Effective spread

Midquote price as benchmark. So, for buy orders, (trade price - midquote). For sell orders, (midquote - trade price). Effective spread = buy effective spread + ask effective spread = $2 \times$ buy or sell effective spread. Multiply effective spread $\times$ trade size = effective spread transaction cost. If sold price above bid price, or bought price below ask price, then there was **price improvement**; there's no price improvement if sold price = bid price.

An order that fills at a price outside the quoted spread has an effective spread that is larger than the quoted spread. Such results occur when trade execution prices are worse than quoted prices.

Effective spread is not accurate if large trades are split into many small ones - there is **market impact**, **delay costs** (slippage; e.g. if trade size is large so can't be 'put onto the market' quickly because of liquidity), and **opportunity costs** (e.g. if trader is too aggressive - should have paid slightly higher earlier instead of much higher later; hard to calculate but depends on time period chosen). **Implementation shortfall** compares the actual trading costs to hypothetical trading costs as if there were no market impact, delay costs and opportunity costs by using a **prevailing price** = decision price = arrival price = strike price, which is the midquote at the time of trade **decision**.

Volume-weighted average price VWAP. For buy orders, Trade VWAP - VWAP benchmark. For sell orders, VWAP benchmark - trade VWAP. VWAP benchmark is volume-weighted average of all trades occurred from start of order until order completion. VWAP transaction cost = multiply buy or sell VWAP spread $\times$ transaction size. Popular because easy to interpret and biased towards 0 because VWAP benchmark so trade VWAP = VWAP benchmark when (1) trade size is large, it dominates market, and (2) trades took place at same rate as other trades in market.

Trade VWAP = your own trades' VWAP.

Notes.

- To calculate quoted spreads, take lowest ask minus highest bid. Ask and bid order sizes associated with the prices don't matter.

- Average quoted spreads - just take simple averages.

Electronic trading

Reduced costs and bid-ask spreads. However, bond markets are still largely OTC - they are larger than equity markets and traded less. However, this has allowed more venues, so now there is **market fragmentation** because trade same instrument in multiple venues, leading to different price and liquidities. Use 'super books' and smart-routing and other analysis to figure out best deals.

High-frequency traders buy and sell quickly, often within ms. Low-latency traders hold onto the positions but buy fast - includes parasitic traders.

News traders try to trade ahead of everyone else; they only need to be right in a proportion of times since trading costs are lower than gains.

Electronic dealers carry inventory and provide liquidity. They cancel orders if their stocks are in the news to avoid losing to news traders.

Electronic front runners. Use AI to try to predict market movements. Dealers can't do for client orders submitted and not yet marketed (illegal - like insider trading); but can study client patterns and trade before client submits. Some buy-side traders randomize time of trades to try and foil.

Electronic **quote matchers**. They 'get ahead' of the order book by 'leaning on the trade' by e.g. buying slightly higher than bid price, and if price falls they can reliably sell to the highest bidder (at a slight loss) by looking at the order book before buying, and if the price rises then make a profit.

Electronic trading systems. Speed (low latency) is important because it allows traders to (1) take a trading opportunity before others, (2) make a trade opportunity available to the market before others, and (3) cancel orders quickly. Want simple (e.g. MS-DOS), not complex, operating systems.

Electronic trading important (don't have to monitor all the time, can react quickly) for (1) advanced orders, e.g. pegged limit (pegged at a fixed distance from the benchmark) orders = floating limit orders - often hope to be last to supply liquidity to large trader at a good price; (2) trading tactics, which generally involve multiple orders or sequential orders that 'sweep' for hidden liquidity (e.g. dark pools) with a single instruction, e.g. **IOC = immediate or cancel** (send IOCs sequentially to exchanges to see if dark pools or hidden orders will 'bite'); (3) algorithms are automated to fill small orders efficiently, e.g. spread large orders to minimize market impact (e.g. VWAP algorithms which trade more when forecast market activity is high).

Note: midsread order = limit order pegged to midquote.

Strategies.

- Hidden orders. Use hidden orders to thwart quote matchers. But can be discovered by small-order pinging to test the market, and if the order is accepted, other traders can see it but can't see the actual order size.
- Leapfrogging orders to maintain trading position - best bid or ask. Usually do this when bid-ask spreads are large.
- Flickering quotes. Traders briefly flash order publicly and then cancel, sometimes repeatedly, prompting other traders who don't want to reveal their positions to submit hidden orders so that the original traders will then hit that hidden order.
- Arbitrage. Must race for arbitrage opportunity. Can (1) take liquidity on both sides (i.e. marketable orders on both legs of arbitrage trade) which is costly but safe; (2) offer liquidity on one side and then take liquidity on other side - cheaper than (1) but riskier; (3) offer liquidity on both sides, one at a time but risky and exposed with sequential trading (if well-informed traders are present, then first offer will fill quickly but not second). Offer liquidity = make an offer; take liquidity = buy or sell against an existing order. Taking liquidity is costly because it's a marketable order - has to be filled immediately.
- ML and extraordinary regimes. ML good for repeated patterns but traders close out if regimes change or high volatility or new market environment and ML no longer useful.

Electronic trading has reduced trading transaction costs, but now technologies for them are expensive and often investment is just to beat competition, which leads to higher barriers to entry and erodes reduced transaction costs.

Electronic trading also increases systematic risks, e.g. programming errors. Runaway orders, fat finger errors (accidentally larger order than intended), overlarge orders (which can cause feedback loops if the algorithm trades a large volume which changes prices and market conditions; so, add trading limits), malevolent order streams (deliberately disrupt markets, DoS attacks, etc.).

Potentially abusive trading.

- Front-running based on tips - these are illegal. Increases transaction costs for others.
- Trading for market impact. Incurs transaction costs, but moves the market.
- Rumor-mongering. Usually illegal, but not usually illegal to report on one side of issue.

- Wash trading. Using common accounts to artificially boost liquidity, to attract investors and given them confidence about liquidity.
- Spoofing or layering. Hidden low-ball (real) order, and then trick market by placing standing limit orders just above the low-ball order that suggest price will fall, making everyone else believe the price will fall.
- Bluffing - often target momentum traders. Use pump-and-dump, short-and-distort.
- Gunning the market. Trigger stop-losses by selling a lot (which pressures the price even more), then buy them up at low prices.
- Squeezing and cornering. Buy resources used for settling contracts, then withhold from market, triggering defaults, then sell resources at very high prices. Examples: short-squeezes (holding large proportion of lendable stocks and bonds for repos and jacking up the price), buy up commodity and commodity futures and then demand commodity from short-sellers who have to buy the commodity at high prices, and from futures short positions who have to buy the commodity or offsetting futures at high prices. Usually illegal - futures long positions cannot demand physical delivery unless valid business reason.

Measuring market impact. If the trade price (i.e., the settled price) of a trade is 10, and then the second trade price because the order size is large, is 12, then the market impact of the second trade is $12 - 10 = 2$.

2.7 Case study in PM: Institutional investors (SWF)

Here, focus on LT institutional investors such as pension funds, SWFs, endowments. Not banks or insurance companies which are more liability focused and have regulatory constraints. An important distinguishing factor between these funds and banks and insurers is use of illiquid alternatives.

Financial risk management. Note: nonfinancial risks can be e.g. reputational risks. When making investments, hiring external PMs, etc.:

- Both top-down (by CIO and board) and bottom-up risk analysis (by individual PFs by analysts).
- PF-level risk and asset-class level risk. Can't have granular risk analysis of some assets like alternatives. So need macro-risk, at PF-level.
- Return-based risk-estimation approaches: based on history of whole PF (but biased because relies on historical regimes). Holdings-based approaches are more costly and time-consuming and sometimes not possible (e.g. HFs are opaque).
- Both absolute and relative risk. Absolute risk is about SDs, VaRs, draw-downs. Relative is about versus other benchmarks and tracking error.

- Both short-term risk metrics (e.g. volatility, VaR, CVaR) and long-term risk metrics (use Monte Carlo simulations with CMEs and calculating benefits, contributions).
- Both quantitative and qualitative risk assessments. But beware of biases in qualitative assessments.
- Pre-investment risk assessment (due diligence) and post-investment risk monitoring.

Monte Carlo can help determine LT tradeoffs of maintaining purchasing power and loss or drawdown.

IIs need liquidity for: (1) payouts, (2) capital calls on illiquid investments, (3) rebalance PFs. however, liquidity increases during bad times: (1) need higher payouts, (2) capital calls increase, (3) more rebalancing needed.

IIs get liquidity from: (1) inflows (e.g. donations, government savings, pension contributions), (2) distributions from illiquid investments, (3) investment income and selling assets. IIs lose liquidity during bad times: (1) less inflows, (2) distributions reduced, (3) assets fall in value. However, there is some cash and investment income.

Note: due to the **denominator effect**, AUM of II can fall a lot in a crisis, and the illiquid investment percentage (which is subject to a lag in valuation) will soar. This will create a liquidity crunch.

Illiquid assets have low apparent volatilities. To address, (1) use public market proxies, or (2) unsmooth (many techniques, such as GLM, to increase volatility and therefore beta). Using (1) or (2), use MVO; but this doesn't consider the illiquid nature (e.g. hard to rebalance) and MVO is single-period so it assumes investors keep PF weights constant (rebalanced perfectly) and ignore drawdowns and uncertain cash flows.

Direct investment or not. Instead of being an LP in a fund, go direct. But disadvantages: no limited liability protection, have to pay for in-house expertise, usually direct investments are bigger (to reduce admin costs) which makes them riskier and less liquid, smaller in-house teams have more trouble with deal-sourcing, and hiring deal sourcers can also lead to problems.

Managing liquidity risk.

- Establish liquidity risk parameters and guidelines. In IPS or internally. Should include uncalled capital commitments.
- Assess liquidity in current PF and how it could evolve. Liquidity reports showing different assets, percentages, how much with external PM, what are legal terms (e.g. HF lockups).

- Model cashflows. Inflows, outflows, balance. Forecast.
- Stress test cashflows and projections. This is an art - no accepted standard way of doing this.
- Have an emergency plan. What to liquidate in a crisis, and in what order.

Enterprise risk management. Board decides risk tolerance and formulate into SAA as asset-only (SWF, endowments, foundations) and asset-liabilities (pension funds, banks, insurers). Board also approves risk parameters and guidelines into IPS. Management implements SAA and monitors specific risks, including by investment type (public equity, private equity, etc.) as well as FX risk, rebalancing.

Note: CVaR = expected tail loss ETL.

Note: Risk factor analysis = divide risk by asset-class, e.g. equity, FI, etc.

ESG. Affects SWFs the most, whose asset PFs are effectively 'mini-economies'.

UN-backed Principles for Responsible Investing (PRI), environment costs will be reflected in investments. Remediation costs will be lower than impact costs. PRI's Inevitable Policy Response (IPR) project aims to prepare financial markets for climate policy risks likely to emerge in short and medium term. Forecasts by 2025 a forceful, abrupt and disorderly response because of delayed action. PRI says finmarkets inefficiently value climate transition risks. Task-force on Climate-Related Financial Disclosures (TCFD) recommends financial organizations estimate implications into business using scenario analysis.

New climate technology opportunities. This includes climate adaptation (e.g. new crop types, sustainable water), climate mitigation (e.g. clean energy, materials for clean energy). Climate adaptation for those organizations that are too small to make a difference.

Other aspects of ESG, e.g. labor, communities, are hard to estimate.

Social license to operate. Consider before investing. Consider talking to all stakeholders. Example: invest in building a dam because it's renewable power but dam-building displaces locals who might protest and sue. Another example (just transition): increasing renewable power displaces coal miners who face unemployment.

Notes from case 1:

- Direct investment, instead of fund investing, means there's more liquidity because can pull out independently.
- Look at other investments, make sure not too much correlation in PF.

3 Performance measurement

3.1 PF performance evaluation

Performance evaluation: (1) measurement, (2) attribution, (3) appraisal (e.g. was performance achieved by skill or luck?).

Performance measurement. Typically versus a benchmark. Have to calculate performance of both PF and benchmark, and the difference is the **excess return**. Also measure risk: ex-post (returns, SD) and ex-ante (VaR).

Performance attribution. Explain how performance achieved: what portion of returns was due to active management and rest. Can explain absolute returns and relative returns. Can look at excess returns. Can also attribute risk.

- Requires accounting for all of the PF's return and risk (completeness), account for investment process (e.g. can't evaluate sector effects of an individual stock-picker, whose sector effects are just a by-product).
- **Macro attribution:** 'fund sponsor decisions' (examples of fund sponsors: endowment or DB pension plan sponsor), including TAA AND manager selection decisions (who might use TAA) and impact on fund returns. **Micro attribution:** individual PM's allocation and selection decisions impact on fund returns.
- Can be returns-based, holdings-based, or transactions-based.
- Returns-based. Least accurate but easiest, and vulnerable to manipulation. Look at entire PF returns. Use for HFs, where underlying holdings of HFs unclear.
- Holdings-based. Best for PFs with little turnover. Best for passive strategies. If transaction price is different from closing price, create problems for attribution (timing or trading effect).
- Holdings-based: timing or trade effect - the residual caused by ignoring transactions. For example, in a daily holdings-based attribution, securities are included at the end of the day they are purchased and excluded at the end of the day they are sold. If the transaction price is significantly different from the closing price, the attribution analysis can differ significantly from the actual performance.
- Transactions-based. Most accurate but most time-consuming. Uses holdings + transaction prices + all transaction costs.

Return attribution. Ex-post. Use a benchmark (but can just compare two portfolios to see how they differ). Quantifies active decisions of PM. Decompose into asset class allocation + security selection. See EXHIBIT 2, p12 for calculation of portfolio return R and benchmark return B . Excess return = $R - B$.

- Arithmetic attribution. Use if addressing non-sophisticated audience. Explains excess return, the difference between PF and benchmark. For one period, excess return = sum of attribution effects. However, for multiple periods, if the sub-periods are linked geometrically, excess return $\neq$ sum of attribution effects, so will need smoothing.
- Geometric attribution. Requires no smoothing across sub-periods. Excess return = sum of attribution effects in final and sub-periods.

Equity return attribution (this is a bottom-up approach): **Brinson-Hood-Beebower (BHB)** model. Allocation + selection + interaction effect = $A + S + I$. Allocation effect:

$$Allocation = A = \sum A_i = \sum (w_i - W_i)B_i$$

where w is asset weight of PF, W is asset weight of benchmark. Selection effect:

$$Selection = S = \sum S_i = \sum W_i(R_i - B_i)$$

Interaction effect:

$$I = \sum I_i = \sum (w_i - W_i)(R_i - B_i)$$

Equity return attribution (also a bottom-up approach): **Brinson-Fachler** model. BHB has flaw: overweight in negative market i that outperformed the overall benchmark (i.e. $B < 0$), then effect should be positive but it's not. BF solves substituting allocation A with:

$$A = B_{sector} - B = \sum (w_i - W_i)B_i = \sum (w_i - W_i)(B_i - B) = \sum A_i$$

$$\sum (w_i - W_i)B = \sum w_i B - \sum W_i B = B - B = 0$$

since

$$\sum w_i = \sum W_i = 1$$

Essentially, B_{sector} is now 'compared' or 'normalized' to B . A_i is defined differently in BF than BHB at the segment level, but at the fund level, allocations are the same! Selection + interaction is same as BHB.

Note: underweighting an asset class or segment which has lower benchmark returns gives a boost to the portfolio returns! See q11, p75.

Equity return attribution: factors. Using factors instead to decompose attribution of excess returns. Most common: Carhart 4-factor model.

Factor	Factor Sensitivity			Contribution to Active Return		
	Portfolio	Benchmark	Difference	Factor Return	Absolute	Proportion of Total
	(1)	(2)	(3)	(4)	(3) × (4)	Active
RMRF	0.95	1.00	−0.05	5.52%	−0.28%	−13.30%
SMB	−1.05	−1.00	−0.05	−3.35%	0.17%	8.10%
HML	0.40	0.00	0.40	5.10%	2.04%	98.40%
WML	0.05	0.03	0.02	9.63%	0.19%	9.30%
A. Factor tilts return =					2.12%	102.40%
B. Security selection =					−0.05%	−2.40%
C. Active return (A + B) =					2.07%	100.00%

FI return attribution. 'Decisions made wrt to credit risk and positioning along the yield curve'. Three major approaches.

- Exposure decomposition (duration-based). Top-down. Easy to do, easy to understand. The key exposure to decompose is duration. Group FI by different ranges of duration. Then do Brinson-type attribution.
- Yield curve decomposition (duration-based). Top-down or bottom-up. Very complex. Decompose into % total return = % income + % price return, where % price return $\approx -Duration \times \Delta YTM$. Divide % total return into: yield, roll, shift, slope, curvature, spread, specific, residual.
- Yield curve decomposition (full repricing based). Bottom-up, by security then combine into PF. Very precise but very costly and time-consuming, needs lots of details. Look at expected interest rate changes for each security and each year.

Example of FI return attribution: exposure decomposition.

- Compare government versus corporate and short-mid-long durations. See EXHIBIT 6, p23.
- If more emphasis on long-duration bonds, then PM probably was bullish on interest rates FALLING.
- If more emphasis on corporate bonds versus government bonds, then PM probably expected corporate spreads to narrow.
- FI total interest allocation = duration effect (aggregate movement of curve) + curve effect (change in shape of curve). Note: if weight heavily to long-duration, then must weight less relative to benchmark on other durations in sum.

- If negative benchmark returns across all durations, the interest rates went up across the term structure.
- If PM overweights long-duration bonds, and the interest curve flattens (bearish flattening, not a bullish flattening), the long-duration bonds will experience a positive curve effect because their interest rates rise less and therefore their prices fall less compared to the benchmark. Shorter-duration bonds will also experience a positive curve effect because while interest rates rise more, there's less of the shorter-duration bonds in the PF (basically, a short position on shorter bond maturities).
- Return attribution = total interest allocation + sector allocation (corporate-gov bond mix relative to benchmark; e.g. how much greater % of corporate bonds) + individual bond selection.

Example of FI return attribution: yield curve decomposition (duration-based).

- Duration effect is completely different from curve effect. Don't 'combine' the two as a yield effect! The curve effect is about relative weighting of durations; the duration effect is the overall total duration.
- If overweight corporate bonds relative to government bonds, and longer maturity bonds versus shorter ones than the benchmark PF, then generally have higher yields. Positive yield % for return attribution.
- If yield curve is flatter at long end (generally true), then overweighting longer maturities will give lower roll yields. Negative roll % for return attribution.
- Shift. Higher PF duration than benchmark, and if interest rates rise, then shift % for return attribution is negative.
- Slope. Flattening term structure with overweighting of long durations means positive slope % for return attribution.
- Spread. Overweighting the corporate sector versus benchmark and if interest rates rise, corporate spreads rise, so negative spread % for return attribution.
- Specific spreads. For specific individual security selection.
- Residual. Unaccounted for because duration and convexity are only estimates.

Return attribution: **macro attribution** (i.e. at fund sponsor level). Set a specific benchmark (fund sponsor's benchmark, e.g. 75% to large cap value equities, 25% to small cap growth equities). Then calculate allocation, selection + interaction wrt to each segment of the **benchmark** (aggregated for each third-party PM). Use Brinson-Fachler allocation + selection + interaction. Allocation = fund sponsor's allocation return. Selection + interaction = fund sponsor's

manager selection return. This is about how the fund sponsor managed allocation + manager selection.

Return attribution: **micro attribution**. Same as macro, except decompose each **PM's segments** individually and do return attribution (allocation, and selection + interaction). Then ADD (i.e. arithmetic Brinson approach) the segment allocations and segment selection + interactions across the total fund. Note: this will give different allocations and selection + interactions than if the individual PM allocations and selection + interactions were added in macro attribution (if the PM has more than one segment, but they will be the same as macro attribution if the PM only has segment), though will be broadly similar; the macro and micro attribution fund total attribution will be the same but allocation, selection + interaction will be different. Allocation = PM's allocation return. Selection + interaction = PM's security selection return. This is about how the PM contributed to the total fund evaluation.

Return attribution: similar to macro and micro attribution, can apply to countries, industries, etc. as segments - even down to individual securities (if the individual security returns are different, probably transaction costs and trading timing will explain differences). For individual securities, the 'allocation' is the choice of different securities and 'selection + interaction' is the transaction and timing costs. The segments used should reflect the investment approach actually used.

Return attribution: use to determine whether it's the PM or the strategist at fault.

Exhibits 11 and 12. Differ. For value portfolio manager, fund sponsor allocation decision is:

$$(78\% - 75\%)(0.32 - (-0.03)) = 0.01$$

This decision is mathematically independent of the value PM's; the fund sponsor's selection + interaction is the fund sponsor's manager selection decision - and the PM's allocation + selection + interaction decisions affect the fund sponsor's selection + interaction. But for the value portfolio manager, the PM allocation decision is:

$$(20\% - 25\%)(1.52 - (-0.03)) + (58\% - 50\%)(-0.28 - (-0.03)) = -0.1$$

because the PM underweighted (20%) the positive return and PM overweighted (58%) the negative return. The PM however makes good individual security bets, raising the PM's selection + interaction, which outweighs the PM's allocation decision, and this is reflected in the fund sponsor's selection + interaction return.

Risk attribution. Can be absolute (use volatility) or relative (use tracking

risk).

Investment decision-making process (bottom-up, top-down, factor-based) versus type of risk attribution analysis, i.e. return target (relative, absolute). An absolute return target is T-bills + 5%, a relative target is outperforming an index.

Investment Decision-Making Process	Type of Attribution Analysis	
	Relative (vs. Benchmark)	Absolute
Bottom up	Position's marginal contribution to tracking risk	Position's marginal contribution to total risk
Top down	Attribute tracking risk to relative allocation and selection decisions	Factor's marginal contribution to total risk and specific risk
Factor based	Factor's marginal contribution to tracking risk and active specific risk	

- Bottom-up: take each position's marginal contribution to tracking risk (TR) and multiply by active weight = position's contribution to TR.
- Top-down: (1) if relative, then look at relative allocation decisions, then add individual security selection, (2) if absolute, then look at factor's marginal contribution to TOTAL risk and SPECIFIC risk.
- Factor-based: (1) if relative, look at factor's marginal contribution to TRACKING risk and ACTIVE specific risk, (2) if absolute, then look at factor's marginal contribution to TOTAL risk and SPECIFIC risk.

Liability-based benchmarks. Focuses on the cash flows that assets must generate. Often includes bonds and shares. Compare the fund's funded status with the funded status had the benchmark been used, or compare volatility of funded status had the benchmark been used.

Asset-based benchmarks. Types of benchmarks:

- Absolute return benchmark. Minimum target, e.g. 5% or 5% above market index or T-bill. Market-neutral long-short equity funds often have absolute return benchmarks.
- Broad market index.
- Style index.
- Factor-based model indices, e.g. Carhart. Uses historical returns of the portfolio and may not use styles (might use e.g. momentum and other factors).

$$r_p = a + b_1 F_1 + b_2 F_2 + \dots e_p$$

where the betas are sensitivities to factors.

- Returns-based benchmarks. Sharpe-style analysis and weights. Uses factor models, but force the factor sensitivities to be non-negative and sum to 1, using historical results of the portfolio, to give a style-based result e.g. 60% small-cap value and 40% large-cap growth. There are no restrictions on sensitivities in factor-based models, which mean negative betas (sensitivities) which are short positions and sum of betas greater than 1 which is leverage - this is not a PASSIVE benchmark portfolio. We want passive benchmarks. This benchmark views 'style on a continuum' because it's a blend of styles (unlike style benchmarks).

$$r_p = a + w_1 R_1 + w_2 R_2 + \dots e_p$$

where w are weights of a hypothetical portfolio with R being returns (e.g. of a style index). So there are no sensitivities unlike factor models here.

- Manager peer group or universe. E.g. take a group of manager returns and take the median.
- Custom or strategy benchmarks. Built around professed fund strategy. Custom-built so expensive and have to rebalance to maintain.

Benchmark selection. Benchmark and returns should correlate, i.e. their factor sensitivities should be similar = benchmark quality test. Other factors important to benchmark selection:

- Unambiguous
- Investable. Must be able to buy it physically, e.g. shares are traded.
- Measurable. Must be published regularly and updated. Must be replicable.
- Appropriate - consistent with investment process. Consistent with PM's investment style or expertise. If there are constraints (e.g. liquidity) imposed are not consistent with the PM's usual way of investing, might not be appropriate.
- Reflects current investment opinions. PM should understand the securities in the benchmark and have opinions about them. Benchmark should have a clear thesis.
- Specified in advance.
- Accountable - PM and fund sponsor should be willing to be held accountable to the benchmark.

Decomposing the benchmark. Let P = PF return, B = benchmark return, M = (broad) market index return.

$$A = P - B$$

where A is active return.

$$S = B - M$$

where S is the PF's style return. Then,

$$P = M + S + A$$

If the style index is skipped and we just use a market index, then the PM's return performance true added value will be obscured. This is an example of a **'misfit active return'**.

Using a peer group benchmark. This often results in PMs clustering performance around the median. So, the benchmark can influence the PF returns.

Alternative investment benchmarks. Illiquid, HFs and PEs often use leverage, use IRR instead of returns for comparison, opaque, lack of investable indices make this difficult.

- HFs. Often use T-bill + spread, or broad market index + spread, because many HFs are market-neutral. However, even if market neutral, often leveraged, which magnifies gains - so the spread should be revised up. Moreover, T-bill and broad market don't correlate well with HF returns. Further problems: HF strategies are very diverse so finding indices is hard, HF indices that are available often have survivorship and backfill bias (backfill bias = when new HF is added to index, import entire history to revise index), and HF return data is self-reported, self-selected, and smoothed. Smoothing leads to high HF allocations in portfolios!
- RE, particularly private RE. Many offered, many levels of detail and segments. But often heavily weighted towards large cities. They are also only a subset (a sample) of that RE segment, don't include RE transaction costs (which are high), based on appraisals and so smoothed, contain varying amounts of leverage, some use IRRs while others use returns.
- PE. Usually use IRR. Note: valuation methodologies of different indices can differ, IRRs can be easily influenced by stage of fund. There is a semi-standard 'PME' public market equivalent methodology for converting PE IRR to public equity index comparisons (includes capital calls and distributions). PME has many variations.
- Commodities. Based on futures, not physical delivery. Replicates physical delivery through cost-of-carry + fully (100%) margined positions. Note: because every long futures position has a corresponding short, market cap of a futures contract is ALWAYS 0.
- Managed derivatives. No market indices, benchmarks specific to strategies.
- Distressed securities. Use indices of distressed security funds. But very heterogenous, stale pricing.

Performance appraisal. Was attributed performance due to skill or luck (markets). Performance attribution identifies and quantifies sources of added value, while appraisal determines whether added value was skill. Skill = add value on risk-adjusted basis.

Appraisal measures: ratios.

- Sharpe: excess return divided by its SD. Total risk. Weakness: indifferent between upside and downside volatility, unlike Sortino.
- Treynor: excess return divided by its beta. Systemic risk in a well-diversified PF.
- Information: excess return over benchmark return, divided by $\sigma(r_P - r_B)$.
- Appraisal (Treynor-Black):

$$AR = \frac{\alpha}{\sigma_e}$$

where the RHS variables can be from any factor regression, but usually use Jensen's alpha (excess returns above CAPM prediction), and σ_e from PF's residual or non-systemic risk (= standard error of regression). Note: $\sigma_P^2 = \beta^2(\sigma_m^2) + \sigma_e^2$.

- Sortino ratio. Useful for appraising capital preservation. Ex-ante version is

$$SR_D = \frac{r_P - r_T}{\sigma_D}$$

where T is the target (MAR = minimum acceptable return), and σ_D is the semi-standard deviation. Numerator: no screening; denominator: screening. Penalize only for harmful, downside deviations (upside deviations are positive surprises). Sortino ratio useful if there's skew (e.g. positive skew for commodity funds which follow trends, and negative skew for HFs as they write out-of-the-money call options).

- Capture ratios. Useful for evaluating consistency and manager's type of participation in the markets. Divide return history into when the BENCHMARK was up (time period over time period), and when it was down. Then upside capture ratio = (PF returns) divided by (BM returns), and downside capture = (PF returns) divided by (BM returns) too. The returns are geometric averages (using $1 + x\%$ and then reconvert to time period changes) of those time periods (e.g. if $n = 5$ time periods of up, then returns for up use $n = 5$). Note: ratios will be negative whenever PF and BM returns are of opposite signs - that's OK.

$$CaptureRatio = UC/DC$$

- Drawdowns. Useful for stress tests. A PM who experiences large drawdowns unsuitable for short-horizon investors. Based on a chosen time

period, the drawdown begins with the first of a continuous set of negative returns and ends when cumulative returns are 0%. This is the duration of the drawdown, and the drawdown is the cumulative returns at the trough. Note: asymmetry in time (need more time for recovery) because proportionally larger gains needed to get out of a trough than to fall into one.

For a final appraisal, need to look at many factors, including consistency, potential risks, compliance, investment process (e.g. whether country allocations are deliberate decisions or just by-products).

3.2 Investment manager selection

Before selecting manager: (1) decide whether to hire PM, (2) draft IPS, (3) know roughly SAA.

Evaluate in three steps: universe (manager universe, suitability, style), quantitative analysis of record (attribution and appraisal, capture ratio, and drawdown), qualitative analysis of investment process. Qualitative analysis: (1) investment DD - philosophy, process, people, portfolio; (2) operational DD - accuracy of track record, whether record reflects all risks, back office procedures, terms and conditions, firm profitability, culture, monitoring, investment vehicle suitability. See EXHIBIT 1, p83.

Manager universe. Don't want too narrow or too broad a set to evaluate. Don't use historical performance to exclude potential managers yet. Identify and use a benchmark. Search is determined by IPS needs. Use a combination of: (1) third-party categorizations, (2) returns-based style analysis - but time-consuming, (3) holdings-based style analysis - even more time-consuming, (4) manager experience.

Errors in manager selection: should do cost-benefit to see which way to bias. Type I (false positive; explicit costs): hiring or retaining manager who underperforms. Type II (false negative; opportunity costs): not hiring or firing a manager who outperforms. Decision-makers prefer to avoid Type I to Type II, as it's clearer to see Type I. Avoid short-termism (trend following) and identify behavioral biases to improve manager selection.

In an efficient market, the bad managers and good managers will not be very different. In an inefficient market, good managers can exploit the market. In an efficient market, there is therefore a larger opportunity cost to not hiring a good manager.

Quantitative aspects of manager selection. Quantify manager's self-reported risk exposures as a starting point. Then augment analysis using returns-based style analysis (RBSA) which is top-down, doesn't require much additional data, can compare across managers, valid even for complicated strategies, and not

subject to window-dressing, but is imprecise and illiquid securities can underestimate risk. Also augment with holdings-based style analysis (HBSA) which is bottom-up, precise, and comparable across managers and time, but is time-consuming, subject to window-dressing, getting unstale data from HFs and PEs can be challenging, high turnover can compromise the HBSA results. Then take these results and look for their risk and style exposures, but these should be: meaningful (should contain important sources of risk), accurate, consistent, timely. Risk and style analysis most useful for public equities, and only a starting point for private markets.

HBSA is subject to window-dressing because it is a snapshot in time, which can be manipulated just before reporting time. RBSA is harder to window-dress because it uses data over a long period of time.

Other tools for manager selection.

- Capture ratios. How has the manager performed in “up” versus “down” markets?
- Drawdowns and drawdown ratios. Does the return distribution exhibit large drawdowns?
- Active share versus tracking risk. Active share is

$$AS = \frac{1}{2} \sum |StrategyW_i - BenchmarkW_i|$$

and if AS is 0, same as benchmark; if AS is 1, then no holdings in common with benchmark. (1) AS low, tracking risk low = closet indexer. (2) AS low, tracking risk high = sector rotation. (3) AS high, low tracking risk = diversified stock picker. (4) AS high, high tracking risk = concentrated stock picker.

		Active Share	
		Low	High
Tracking risk	High	Sector rotation	Concentrated stock pickers
	Low	Closet indexer	Diversified stock pickers

Manager selection: investment philosophy. Make assumptions about market, e.g. indexers and passive strategies assume market is efficient enough and so only capture returns through systematic risk premiums. Active strategies assume either (1) behavioral inefficiencies - caused by other traders e.g. trend following or loss aversion - they are temporary, and (2) structural inefficiencies from regulations and rules.

Investment philosophy should be articulated, credible, consistent, not change over time except for good reasons, and have capacity. **Capacity**: how much

assets can be absorbed by the manager, firm, market, before returns get diluted - including number of opportunities, whether trades can be done timely, etc. Overall, liquidity is REPEATABILITY or sustainability of the strategy. Capacity is part of the investment process.

Manager selection: investment personnel. Expertise, depth, experience, key person risk, incentives, turnover.

Manager selection: teams or individual, and how teams make decisions and run meetings and address different perspectives. Team decisions can be better or worse than individuals. Behavioral biases with teams: groupthink, authority bias, aversion to complexity (Parkinson's law of triviality).

Investment decision-making process has 4 elements:

- Signal creation - idea generation. Observe signal early enough for investment. Has to be unique strategy or information, timely, and interpreted differently from other managers.
- Signal capture - idea implementation. Turn the investment idea into a position.
- Portfolio construction. How investment positions are implemented as part of PF and how is the risk addressed, e.g. derivatives risk, liquidity risk, use of stop-losses, etc.
- Monitoring the portfolio.

Operational due diligence. Analyzes integrity of manager's business and evaluate risks by looking at its policies and procedures, infrastructure, how fees are calculated and collected, whether it uses soft dollar commissions, how many other accounts it manages, what technology it has, which third-party service providers it uses, risk policies, ownership structure, AUM, future plans, asset base, legal and compliance, compensation, terms and conditions, etc.

Separately managed accounts (SMAs). Customizable. Investor directly owns securities in PF: this is a safety feature, as other investors can't pull out of fund and create a run. Moreover, asset ownership is clear legally. Tax efficient, transparent. However, high-cost, bigger tracking risk (because SMA is unique), investor might micro-manage.

Management fees. Most of it is variable cost: human capital, and bonuses are large.

AUM fees, or ad valorem fees.

Performance fees. Can be symmetrical (share in performance but lose when fund

loses) or semi-symmetrical (symmetrical within limits) or upside only. Symmetrical can be dangerous to the management firm, as it increases bankruptcy risk. Upside or bonus-type performance fees are like manager's call option, which incentivizes **higher risk-taking!** (increased volatility increases value of long call option).

Performance fees may incentivize manager to hold onto assets until sellable at profit even if the client benefits from selling assets at loss and investing elsewhere.

Management (AUM) fees reduce returns but don't change SDs. Symmetrical-type performance fees reduce returns AND reduce SDs - returns become biased to zero.

Notes

- Without incentive or performance fee, fund managers might not work once they get the assets because the assets tend to be 'sticky'. Once investors allocate their assets to the manager, the manager often does not need to generate the same level of returns to retain the assets as he or she did to attract them.
- If long and short positions are paired, this is generally good even though it's a leveraged position: prices generally converge.
- Closed-end funds and ETFs are most liquid vehicles, as they are listed. Open-end funds slightly less liquid, though provide daily liquidity and priced at end-of-day NAV.
- Hard locks, stop-losses are absolute. Soft means evaluate at that point.
- Do EXHIBIT 8, p113 and EXHIBIT 1, p129. Calculate billed fees and **net active returns**. Notes: (1) 'beyond base fee' means it's not symmetrical incentive fees, (2) on p129, at the breakeven active return, suddenly the standard fee without sharing is imposed, but it's not consistent with the sharing calculation + base fee - however, note that it says 'including less than or equal to', so if the funds use these set buckets then perhaps it's consistent.

Panel A. Sample Fee Structure					
Standard fee					0.50%
Base fee					0.25%
Sharing*					20%
Breakeven active return					1.50%
Maximum annual fee					0.75%

Panel B. Numerical Examples for Annual Periods					
	Active Return				
	$\leq 0.25\%$	1.00%	1.50%	2.00%	$\geq 2.75\%$
Billed fee	0.25%	0.40%	0.50%	0.60%	0.75%
Net active return	$\leq 0.00\%$	0.60%	1.00%	1.40%	$\geq 2.00\%$

*On active return, beyond base fee.

3.3 Overview of GIPS

GIPS based on fairness and full disclosure. 5 objectives: (1) promote investor interests and confidence, (2) accurate and consistent data, (3) worldwide acceptance of single standard, (4) promote fair, global competition, (5) promote industry self-regulation globally.

GIPS has 3 chapters: (1) GIPS standards for firms (compete for business), (2) GIPS standards for asset owners (don't compete for business), (3) GIPS standards for verifiers. Read sections of (1) referenced in CFAI text.

GIPS standards for prospective: clients (segregated accounts) and investors (pooled accounts). Prospective = expressed interest in.

GIPS report. To claim compliance with GIPS, must use this standardized report.

GIPS compliance. All fee paying discretionary accounts must be included for measurement. Only for whole firm, not some PFs. Discretionary SMAs assigned to **at least one** composite. Consultants, software companies, measurement providers can't be GIPS compliant (they have to follow GIPS standards for verifiers if the client wants GIPS compliance, and the verification can be GIPS compliant, but the consultant, software company, etc. cannot claim GIPS compliance itself).

Terms. Composite = PFs managed according to similar mandate, objective, or strategy; must be documented in firm's policies and procedures. Segregated account = SMA. Pooled funds = broad distribution funds (open to public, e.g. mutual funds, UCITS) + limited distribution funds (PE, HF).

Conflicts with laws. If conflict, firms must comply with laws and regulations and disclose the conflict. Encouraged to comply with both.

Fundamentals of compliance. GIPS 1.A.

Definition of firm: held out to the public as a firm, subsidiary, division as a distinct business entity (has discretion over assets and autonomy in decisions) if represented to clients that way. However, GIPS encourages firms to use the broadest, most meaningful definition.

Total firm assets = fair value of assets (non-discretionary + discretionary; fee-paying + non-fee-paying) but excludes advisory-only assets and uncalled committed capital. Changes in firm, e.g. restructuring **cannot affect historical performance**.

Definition of discretion. All discretionary, fee-paying SMAs must be included in at least one composite. Discretionary, fee-paying pooled funds too. Discretion: generally, if manager can implement an investment strategy.

Non-discretionary PFs to be excluded from composites. Some accounts have some many restrictions placed by client or investor that it's a judgment to determine whether it's discretionary or non-discretionary.

Valuations. Valuations are market-based. Use quoted active market prices, but if not then quoted inactive market prices but if not, then model using market-based inputs, but if not then subjective unobservable inputs. GIPS requires documenting and using composite and pooled-specific valuations.

Returns. Includes cash and cash equivalents (even if the firm doesn't control the cash investments - e.g. in a custodial bank directed by the client), accrued income on bonds, and anything else that has interest income. Subtract transaction costs (including legal, financial, advisory, banking fees for private market investments). Transaction costs include trader commissions and bid-ask spreads. However, transaction costs exclude custody fees (for safekeeping investment assets). If transaction costs cannot be reasonably extracted from bundled fees (all-in fees), then must count the entire all-in fee = transaction cost (so, reduce gross-of-fees by all-in fee).

Time-weighted returns (TWR). Removes effect of external cashflows. GIPS 2.A., 2.A.1-39.

GIPS requires use of TWR. GIPS TWR is monthly in most cases, but quarterly for private market investments. True TWR requires portfolio be valued at

certain times.

$$r_t = \frac{V_1 - V_0}{V_0}$$

The TWR methodology removes the effect of contributions and withdrawals from the return calculation, allowing performance evaluation to focus on value added by investment decisions controlled by PM.

If there are small external cash flows (includes returns from cash and cash equivalents) and the PF is valued daily, then divide into sub-periods where the external cashflow occurred, and link them geometrically. Don't need to value portfolio at those subperiods, can estimate. If there's a cashflow mid-month, then if the returns are calculated daily, then calculate with the cashflow already included at the beginning of the day for that day, and then geometrically link the daily returns - see p152 example. This eliminates the external cashflow effect.

If cash flow during the month and PF not valued daily, messes up calculation - so adjust by (1) dividing into sub-period returns. If returns are not calculated daily, then adjust using (2) Modified Dietz or (3) Modified IRR. Modified Dietz:

$$r_{MD} = \frac{V_1 - V_0 - \sum CF_i}{V_0 + \sum (CF_i \times w_i)}$$

$$w_i = \frac{CD - D_i}{CD}$$

where D is number of calendar days (CD) to the time cashflow CF_i occurs. Note D_i assumes cashflows occur at END OF THE DAY. Doesn't require valuation of PF on day of cash flow. Modified IRR: calculate using iteration, the r from:

$$V_1 = \sum (CF_i \times (1 + r)^{w_i}) + V_0(1 + r)$$

(1)-(3) will give slightly different answers.

TWR. GIPS requires firms to formulate and document composite-specific and pooled funds-specific policies for external cashflows, and to use them. GIPS recommends (1). If the cash flow is 'big', defined by the firm for each composite in its policies, then it must use (1) or break into sub-periods and then can use (2) or (3). Answers differences modest if flat or steady slopes, but will be different from true TWR if volatile (e.g. V-shaped returns).

Money-weighted returns (MWR). Can use MWR instead of TWR if (1) firm has control over external cash flows, AND (2) PFs are closed-end, fixed-life, or fixed-commitment, ie the fund has control over cashflows OR (3) illiquid investments are a significant part of the investment strategy, ie it is hard to value daily. **Must be calculated at least annually** but can't take periods less than one year and annualize (this is essentially a forecast).

Composite TWR. GIPS 3.A, 3.A.1-14.

Non-discretionary segregated and pooled funds must not be included in any composite. Non-fee-paying discretionary SMAs and pooled funds MAY be included in a composite (including the investment firm's own assets) but with explanatory notes about what it is.

Composite TWR. **Must be calculated at least monthly.** To calculate TWR for a composite: (1) individual PF using beginning-of-period values and then take a weighted average, (2) individual PF using beginning-of-period values and external cashflows (e.g Modified Dietz) and take a weighted average, or (3) aggregate return method.

Beginning-of-period values, i.e. (1) and (2). Do this for individual PFs (can use Modified Dietz for external cashflows). Weighted average: take the returns of individual PFs and then, whether Modified Dietz or not, weight by one of: (a) beginning assets,

$$r_C = \sum \left(r_{pi} \times \frac{V_{0,pi}}{\sum V_{0,pi}} \right)$$

or (b) beginning assets + weighted external cashflows

$$r_C = \sum \left(r_{pi} \times \frac{V_{pi}}{\sum V_{pi}} \right)$$

$$V_p = V_0 + \sum (CF_i \times w_i)$$

(3) Aggregate return method. Don't decompose by individual PFs, just lump all PFs together into the composite, and then use (1) or (2) - but generally, Modified Dietz.

PFs with restrictions and constraints. If not material to strategy, then include in a composite. If material, include with other constrained PFs in a separate composite, remove some assets and include the rest in a composite, or exclude as non-discretionary. PM defines what is 'material', and this can include prohibitions on use of derivatives, must hold low-basis holdings because of tax consequences, client makes huge withdrawals frequently (i.e. external cashflows change a lot).

Models, histories, scenarios. These are NOT part of any composites. But can be 'supplemental information'.

Composites. Can be asset classes like 'equities' and 'fixed-income'. Can also be styles. Can be broad or detailed, can be based on benchmarks or not, can be based on different risk levels. Usually start by looking at SMAs and then create composites, and then see if pooled funds fit the composites; if not, iterate.

Adding and removing PFs from composites. Firms must document in policies and procedures, and should be composite-specific. Generally add PF to composite and start measuring the next month after receiving funds, unless delay in deploying funds or other reasons. Generally remove PF from composite to the full period before the termination notice (not actual date of termination).

Changing composites. Can move PF from one composite to another, but must be in accordance with firm policies and procedures, HOWEVER the historical performance remains with the original composite! Two examples that are allowed under GIPS: (1) client makes changes, e.g. change assets or remove constraint, (2) strategy changes over time.

Temporary changes. If large cashflow change, then temporarily remove from composite. The large cashflow has to be directed by the client and impairs the ability to adjust the PF. Can also use **temporary new accounts** to add whole or parts of investments until they are normalized, i.e. cash inflows are invested, transfer equivalent cash + securities to the temporary new account and liquidate and treat as external outflow in the composite. If a portfolio falls below a stated minimum value for a composite with a defined minimum value, then it temporarily becomes non-discretionary. Firm defines what the minimum value is.

Misclassifications. If find out retrospectively that there was a problem (e.g. large external injection of cash and no time to equitize during a downturn and this shows up as 'good' performance) with an account or fund being part of a composite, then can remove the fund from that composite for that time as being non-discretionary.

GIPS reports. Must be provided to prospective clients and investors. Must be updated every year. GIPS 4.A.1-12, 4.A.18. Two types: (1) GIPS composite report, (2) GIPS pooled fund report. At least 5 years INITIALLY (unless fund has not existed for that long) and to 10 years. GIPS recommends the whole record.

GIPS composite report. Composite returns, benchmark returns, number of PFs (if 6 or more), amount of assets in the composite, total firm assets, internal dispersion if six or more PFs, 3-year annualized SDs in some cases.

Dispersion of returns. How much PFs differ within a composite - if large dispersion, suggests composite defined too broadly. GIPS explains multiple methods of measuring dispersion, including ranges and simple weighted SDs. Weighted SDs:

$$SD_w = \sqrt{\sum ((r_i - \bar{r})^2 \times w_i)}$$

$$w_i = \frac{V_{0,i}}{V_{0,T}}$$

$$\bar{r} = \sum (w_i \times r_i)$$

The weighted SD is using weights of assets at the beginning of the period.

Portability. If a firm acquires the firm with the GIPS performance recorded fund. Requirements: (1) have documents of ALL PFs in PREVIOUS composite, (2) strategy must remain substantially intact and independent, (3) continuity (no break in track record, e.g. 3 months). Then the acquirer gets the record. There's a 1-year grace period to get GIPS compliant after acquisition.

GIPS. Third-party verification of GIPS compliance and implemented firm-wide. GIPS recommended. Only one report and result; no partial, no composite-specific results. However, can elect to have detailed 'performance examination' or scrutiny for specific composites. Verifier follows GIPS standard for verifiers, and takes samples (and might need larger sample if aberrations encountered) for detailed calculations. Verifiers have to: determine whether PFs are classified appropriately, whether interest income, dividends, and other flows are classified correctly, whether PF movements in and out of composites is acceptable.

Notes:

- 'Timely and consistent basis' as defined by firm. If a firm receives securities and they are to be invested but it takes time, then GIPS performance begins on a timely and consistent basis, e.g. when funds are fully invested.
- There can't be - even temporarily - a 'miscellaneous' composite. Composites must have some kind of similar mandate and-or strategy.
- Firm can set minimum asset level for a composite, where it is removed if below X and added if above Y and X and Y can be different.
- Under GIPS there's no min or max number of PFs in composite.

3.4 GIPS

1.A Fundamentals of compliance.

- Must not present performance-related information which is false or misleading, even if not referencing GIPS standards.
- Cannot say calculation methodologies are 'in accordance', 'in compliance', 'consistent' or similar with GIPS. However, can say performance 'calculated in accordance with' GIPS or similar if GIPS-compliant firm reports SMA or pooled funds.

- Make reasonable efforts to provide GIPS composite or pooled fund report to prospective client or investor once they become prospective clients or investors, must update every 12 months at least, if still prospective client or investor.
- Must correct material errors and update current and former verifiers, current clients and investors but not former clients and investors.
- Must notify the CFA of GIPS compliance claim, and updated annually.
- If a firm wants to be verified, must assess the verifier's policies of independence and assess the verifier's independence.

2.A.1-39 Input data and calculation methodology.

- For TWR calculations, must VALUE PFs in composites at time of large cashflows.
- For TWR calculations, VALUE pooled funds in a composite, whenever there are subscriptions to or redemptions from the pooled fund.
- Net-of-fee calculations must subtract investment management fees (actual or modeled). Must reflect performance fee clawbacks.

3.A.1-13 Composite and pooled fund maintenance.

- Need composites for segregated accounts, but only need to include pooled funds (which are already like composites) if they fit one of the definitions of composites. Don't have to make composites containing only one or more pooled funds - and these can be terminated.
- Changes to composite definitions can't be applied retroactively.

4.A.1-12 and 18. Composite TWR report.

- Presentation and reporting requirements.
- Committed capital, advisory funds.

4 Ethics

4.1 Code of ethics and SPCs

CFAI encourages firms to adopt Code AND SPC as part of their code of ethics. Code and SPC aimed at individual professionals; Asset Manager Code is for firms to help interpret the Code for firms.

4.2 Guidance for SPC

SPCs: read recommendations.

SPC 1(A): Knowledge of the law

- Use more strict version - SPC or law or regulation
- CFAI encourages reporting of violations but doesn't require it.
- If problem discovered, then legal counsel (+ independent legal counsel if company counsel provides unethical advice) and supervisors, and follow firm policies.
- Recommendations: stay informed, review procedures regularly, maintain copies of laws and regulations; familiarize with laws and regulations of distribution area, ask legal counsel if unsure. Encourage firm to adopt code of ethics, provide information on applicable laws, establish procedure for reporting violations.

SPC I(B): Independence and objectivity

- Small gifts are fine, but nothing bigger which would compromise independence and objectivity.
- Receiving extra consideration from clients is less of an issue but still needs to be reported to the employer, to make a determination. Employer might then monitor the employee to ensure that accounts are fairly treated.
- Some analysts have coped with interference in investment analysis recommendations from e.g. executive members of the firm, by using vague language, but clients expect clear recommendations.
- CFA members personally liable for maintaining independence and objectivity. Recommendations have to be in clear language.
- If CFAI member and seeking corporate financial support for CFAI activities, be careful that it doesn't threaten image of independence from clients.
- Applies to performance measurement analysts too.
- For issuer research, must still be diligent, unbiased. Best to negotiate flat fee rather than incentive fee structures. And disclosure nature of relationship and compensation.
- Modest hotel, remote flights - can let issuer pay, generally.

I(C): Misrepresentation. Can't knowingly misrepresent.

- Can't guarantee results.

- Must not misrepresent external PM's as their own work.
- Don't use performance benchmarks that are not relevant, where a benchmark is used. If no benchmark used, state reasons why to avoid misrepresentation.
- Don't change pricing and valuation providers solely on basis of higher prices and valuations.
- Social media communications rules = same as traditional communication means.
- Key omissions = misrepresentations. Models as results = misrepresentations.
- Plagiarism = misrepresentation. Even if taking idea (e.g. quant model), cite. This is ethics, not law.
- Citing quotes to 'leading analysts' or 'investment experts' without naming them is misrepresentation. Omitting statistical caveats = misrepresentation. Copying proprietary algorithms = misrepresentation. Multiple sources without acknowledgment = misrepresentation. Oral presentation without acknowledgment = misrepresentation.
- Prohibits guarantees or assurances of portfolio performance.

I(D): Misconduct. The 'professionalism' provision.

- Misconduct if it affects member's professional life.
- Includes actions that 'reflects adversely' on the profession.
- Omissions (e.g. bad due diligence) are also misconduct.
- CFAI aware this provision can be abused to ruin another member's reputation.
- Recommendations: get firm to adopt code of ethics, list of violations, check employee references before hiring.

I(E): Competence.

- Knowledge, skills, abilities.
- Depends on member's needed skills.
- Passing an industry or licensing exam alone isn't necessarily enough.

II(A): Material nonpublic information. Check to see if information is public already.

- Private company reports and materiality. From p220: To be considered material, needs to be clear indication that the general investing public relies on the information (e.g. from that particular bank's reports). It has to be distributed to the public or highly valued by the public (if it was so highly valued, then everyone would be clients of that bank's reports).
- Information is 'material' if disclosure probably has an impact on price of security. Quality of information, how different it is from public information, reliability (traders aren't necessarily reliable sources), timing, source of information, how clear the effect is on the price.
- Can ofc use inside information for confidential due diligence, etc.
- Mosaic theory. Gather lots of public information and come to a conclusion that would have been nonpublic information had it been disclosed by the company. Allowed. Should document work under V(C).
- Report from well-respected analyst is material and nonpublic, using e.g. mosaic theory. But doesn't have to be disseminated - public can do the same research themselves or buy the report.
- In normal prop trading, if taking an issuer public, and market making that issuer's securities, don't stop trading - instead just passively take contra side of trades, to prevent market signalling. For risk-arbitrage trading, stop all activity, as it's not passive.
- Failing to prevent use of such information is also SPC violation.
- Doesn't matter if the information was actually used for trading; all that matters is that the information was transmitted.

II(B): Market manipulation.

- Information and transaction based manipulation. It is about intention.
- Information manipulation often occurs together with V(A): diligence and reasonable basis.
- Exaggerating earnings to force a company to do something is SPC violation.

III(A): LPC. Reasonable care + duty to client.

- Legal (not SPC): duty of fiduciary if in direct or indirect control of funds, though government contracts are still primary. Enhanced position of trust. Not in SPC.
- SPC applies primarily to the governing agreed services.

- Client is beneficiary of assets. Sometimes there is no beneficiary, so it's just the plan mandate. Sometimes the beneficiary is society as a whole. Sometimes unclear who the beneficiary, so just go with independence and objectivity.
- If conflict and can't do LPC, disclose.
- Soft commission (use client brokerage to buy research services by paying more than usual and extra benefit not going to client) and directed brokerage (but still search for best price and best execution).
- Submit to each client **at least quarterly**, assets and transactions of that quarter. Must let know client where assets are stored, and keep it separate from other assets.

III(B): Fair dealing

- When disseminating investment recommendations, change in recommendation and communicating that to existing clients is more important than new recommendations.
- Addressing oversubscribed issues
- Don't have to communicate recommendations to all customers - but at least those who have expressed an interest. Selection should be based on suitability and known interest.
- If get odd-lot sizes, then distribute as best as can, with minimum lots by allocation interest.

III(C): Suitability

- More for decision-makers, advisory relationship on buy-side. Not really for sell-side.
- If managing fund to index, follow the index as there's no 'client' for SPC purposes.
- IPS should be updated at least annually and before big changes, including personal situation.
- If client wants to make a strange trade, discuss and update IPS but client doesn't want to, in some cases can use new unmanaged account.
- No need to explain issue-by-issue details, but rather explain to client context and impact on entire portfolio.

III(D): Performance presentation. Fair, accurate, complete.

- If presentation is brief, allow details on request.
- GIPS compliance is best way to comply with III(D).

- State if clear, gross of fees, net of fees, after tax, simulated results, etc. = part of complete disclosure.

III(E): Preservation of confidentiality.

- Maintain confidentiality even after client relationship ends.
- Members should know electronic and information policies of firm.
- Can submit confidential information to the PCP
- If suspect client of financial crime, then talk to compliance and legal department.

IV(A): Loyalty. Watch out for non-compete clauses.

- Follow firm policies and procedures.
- If member's independent business competes with employer, must notify employer about clients, duration of services, and compensation.
- Integrity of capital markets and clients are more important than employers. So, if there's conflict, may need to consider reporting and whistleblowing reporting to authorities.
- If member is independent contractor, abide by agreement. To determine whether IC or not, look at employer's degree of control.
- Employee buyouts of employer are consistent with IV(A) because it rests on the permission of the employer.
- Includes breaking informal promises to the employer.
- Can recruit from within company before finishing work - not a violation of duty of loyalty.

IV(B): Additional compensation arrangements. If client gift to analyst or PM then also triggers I(B).

- This is NOT about bonuses. This section about is employer VERSUS employee interests. Bonuses align these parties' interests.
- This is NOT about compensation when employer and employee incentives are aligned, e.g. if a employee promotes employer's services on a talk show and the talk show provides gifts to the employee for the guest appearance. The employee is being rewarded for what the employer wanted, and there's no potential conflict between employer's clients. So, neither employer nor clients are being disadvantaged.
- Even if non-monetary compensation, e.g. gym benefits.

IV(C): Responsibilities of supervisors.

- Promote actions of all employees (CFAI members or not) with code and SPC under supervision.
- After a problem is addressed, have to check the whole system to ensure no new problems have been created.
- Must develop or ensure that the compliance policies and procedures are adequate. Regular trainings. Periodic reviews of policies and procedures. Periodic reviews of actions.
- Depends on size of organization. Catch the most likely problems.
- Have to investigate strange or abnormal developments.
- Need to keep written records.

V(A): Diligence and reasonable basis.

- Develop procedures and policies for determining if source, model, external PM with diligence and selecting on reasonable basis.
- If using quant models, test a wide range of scenarios before using results. If making quant models, then must understand more deeply and be able to explain it and have good assumptions and data ranges. Quant models have to be evaluated piecewise and as a whole.
- Review external PMs with diligence too.
- Past experience with and friendships are not sufficient justifications for diligence and reasonable basis.

V(B): Communication with clients and prospective clients

- Applies only to client-facing members.
- Clear communication. Appropriate, timely, accurate, complete.
- Set out nature of work to be done and expectations and costs. Best practice is written form.
- Costs. Reasonable detail. Don't have to spell out numbers. Don't have to disclose firm's costs, but have to state how much the firm is paying for third-party services. Can't assume client is sophisticated enough to know - have to provide.
- Investment process. Disclose positive and negative influences on recommendations. Newly identified significant risks and limitations.
- Significance of particular investment depends on client's particular portfolio. Particular investment 'behaves' differently for different PFs.
- Members must inform clients of external PMs and their specialization and diversification.

- Investment communications. Not just written. Be fair.
- Significant risks and limitations. Note: use of leverage is almost always important here. Others: complex financial instruments, counterparty risk, country risk, sector or industry risk, security-specific risk, credit risk, liquidity and capacity. Should provide balanced scenario testing.
- Methodological changes should be highlighted.
- Must state assumptions.
- Should have review methodology for reviewing firm's research, analysis, and recommendations to be published. Maintain records of research.
- Distinguish fact from opinion.

V(C): Record retention.

- Generally firm's, not individual's, responsibility.
- **In the absence of regulation**, CFAI recommends keeping records for at least 7 years.

VI(A): Avoid or disclose conflicts.

- Should avoid even appearances of conflicts of interest.
- Need to also disclose change in compensation or incentive structure of employer to clients which creates a conflict of interest (e.g. change towards incentivizing ST performance over IPS).
- Includes disclosures of conflict to employers so that employers can decide.
- If conflict is about securities, then beneficial ownership: direct or indirect pecuniary interest in securities, vote power, power to dispose or direct position of security or interest.
- Conflict can be about not securities, e.g. fee structures.

VI(B): Priority of transactions

- Can't disadvantage client. However, if member selling stocks for good reason e.g. for children's tuition, then not a violation.
- If member has a beneficial ownership in a client account, might be subject to preclearance or reporting under employer procedures or under local law.
- If member has beneficial interest accounts in funds where the member is a client of those funds, then if there is a hot issue, the member must get the hot issue for its own fund first before directing beneficial interest accounts in the client funds to get the hot issue. Otherwise, there may be fewer hot issues for its own clients.

- Should have employer rules about limiting participation in equity IPOs, private placements. Should establish blackout or restricted periods, reporting requirements, disclosure of holdings to employers.

VI(C): Referral fees. For recommendation of services and products.

- Inform employer, clients and prospective clients.
- Inform of nature of benefit, and estimated monetary value.
- Provide updates to employers at least quarterly. What kind of compensation (e.g. one-time, ongoing).
- Applies even to different branches or subsidiaries of the same company.

VII(A): Participating in CFAI programs

- Can't compromise integrity or reputation of CFAI programs.
- CFAI encourages reporting violations.

VII(B): Reference to CFA

- Must not misrepresent or exaggerate. Directly or indirectly.
- Must not misrepresent holding the CFA designation.
- The CFA designation must always be used as an adjective, i.e., "the entire research team is made up of CFA charterholders."

4.3 Application of the code and SPC

Sovereign investment corporation

- Sovereign funds usually can't promise high returns - they have multiple mandates usually, e.g. local economic development and high returns.
- Can't personally buy into investment before employer to 'entice' employer to invest.
- Hiding corrupt 'deal fees' as operating expenses = misrepresentation.

Castle biotechnology

- An analyst publishing a 'buy' recommendation on a stock he is selling short has a conflict of interest issue.
- If an analyst's bonus is linked to gross IPO values, analyst not independent and objective under I(B) for issues which his company is overseeing the IPO.
- Can use PERSONAL social media to contact former clients after ending employer relationship, if no noncompete agreement and did not use previous employer's materials or information about clients.

- Signing something on behalf of the client is a violation of misconduct I(D)
- client has to sign themselves.
- Must deal with account-holder client, not someone (even if e.g. spouse) of client - otherwise, violation of LPC under III(A).

Tie-in agreement: underwriter of securities requires investors interested in the issue to buy more shares of a new share of a new issue in the secondary market as a condition of buying shares in the IPO. (Those investors who plan to buy 'additional shares' on the 'first day of trading').

Lionsgate Limited

- Breaking informal promise to employer is violation of IV(A)
- Recruiting within firm before finishing is not a violation of IV(A).
- If misrepresent roles of people in investment decision making process, this is misrepresentation I(C) violation.
- If supervisor finds something wrong, have to initiate an assessment.
- Supervisor 'investigation' doesn't have to be formal.

Gabby Sims

- See p246: fee arrangements are different from referral fees.
- See p247: if a director is using help from someone who has hired the director's son, then if the director is using that person to help the director's business, there's a CI that should be disclosed or avoided.
- See p254: if making a change to investments, and client says to do right away something that is contrary to their IPS and analyst or PM thinks the client doesn't understand, then have to provide more detailed information such as the cost of taking that action and discuss IPS.

Extra points

- If there are decent reasons (e.g. good returns and good diversification) for keeping an external PM until a review is complete - even if reviews flag items that should be addressed - can keep that external PM without reducing allocations.
- Suitability of trade allocations can't just be reviewed at the end of each month; they have to be reviewed BEFORE the trades.
- Interest should be paid to accounts which were wrongly or over-allocated shares and therefore shares removed from that account. Interest should NOT be taken, ever, out of accounts which were wrongly or under-allocated shares and therefore shares added.

- IV(C) says decline in writing to accept supervisory responsibilities until a firm adopts reasonable procedures to adequately exercise responsibility. (CFA members should be able to regulate themselves, but need help with regulating line reports?).
- If independent researcher, then have to disclose accepting payments in plain language prominently.
- Conflicts of interest have to be disclosed, whether or not the member thinks they're material. But only have to do those which are 'reasonably expected' to impair judgment.
- Blaming an incorrect cause (a coverup) is misconduct and misrepresentation.
- Benchmark performance under III(D) must be accompanied by gross-of-fees and taxes information.
- Reasonable and adequate basis doesn't apply for personal investments.
- Professionalism = refers to the misconduct I(D) section.
- Not a violation to distribute IPO subscriptions to discretionary accounts that normally participate in IPOs first - don't have to go to all other accounts first (assume that IPSs are such that it's rare that a discretionary account not normally participating in IPOs would be interested).
- If buying an exclusive subscription where there's material nonpublic information but the publisher says not to disseminate until public (and the publisher hasn't said to use or not to use material nonpublic information), then can keep but not use in analysis. Don't have to terminate relationship.

4.4 Asset manager code of professional conduct

Voluntary. Ethical and professional responsibilities of firms that manage assets on behalf of clients. Covers all employees of firm. Requires compliance procedures, which depend on size and nature of the firm. Global.

Compliance. Must comply with each principle and provision of the Code to say that a firm is in compliance with it. Statements of partial compliance are prohibited. Must notify CFAI of claim of compliance online. CFAI nor does anyone else verify compliance.

Loyalty to clients

- Place client interests above own. Includes compensation aligned to client interests
- Preserve confidentiality.

- Refuse business or accept gift which could affect independence, objectivity or loyalty to clients. Defined what 'minimum value' gifts are.

Investment process and actions

- Manage client assets with reasonable care and judgment. Balance risk and return.
- Don't distort prices or trading volumes with intent to mislead.
- Deal fairly and objectively with clients. Allows 'side letters', 'sidecar' or 'tag-along' investments with clients if similarly-situated clients get the same offer.
- Reasonable and adequate basis for investment decisions. Understand the securities and strategies thoroughly, and their risks.
- Managing to a specific strategy or style or mandate: (a) take only investment actions consistent with objectives and constraints, (b) disclose about proposed changes to strategies, styles, etc. If PM makes a material change, client should be able to redeem without undue penalties.
- Managing separate accounts: (a) take into consideration client's situation and risk tolerance, (b) determine whether an investment is actually suitable for the client.

Trading

- Don't act or cause others to act on material nonpublic information.
- Priority of investment to clients. Employee investment in IPOs and private placements should require approval.
- Use soft dollars only for investment decision-making, NOT for firm management. Must disclose soft dollar policy.
- Seek best execution for all client transactions, given the entire situation. If client directs brokerage, then PM should get written acknowledgement that this might lead to restraints and ultimately less than best execution.
- Policies to ensure fair and equitable trade allocations among clients.

Risk management, compliance and support

- Policies to implement AMC.
- Appoint compliance officer. According to the recommendations of Section D(2) of the Asset Manager Code, where possible, the CCO should be independent from the investment and operations personnel and should report directly to the CEO or the board of directors.

- Ensure portfolio information provided to clients is accurate and complete and arrange for independent third-party confirmation.
- Maintain records. Records that show compliance, that show violations. Records of investment decisions. Seven years at least or if law or regulation specifies a shorter time.
- Employ qualified staff and have adequate resources and infrastructure for investment processes.
- Business-continuity and disaster-recovery plans. Includes communication with clients during extended disruptions and alternative trading and offsite backup. Test periodically. According to the guidance provided in Section D(6) of the Asset Manager Code, the level and complexity of business-continuity planning depends on the size, nature, and complexity of the organization involved. Bornelli is a large firm with hedge fund investments and it should have alternative plans for monitoring, analyzing, and trading investments if primary systems become unavailable.
- Firmwide risk management procedures and policies. Risk management function should be separate from investment decision-making and might have to be outsourced, though PMs evaluate the risks in their portfolios in investment decision-making.

Performance and evaluation

- Performance information should be fair, accurate, relevant, timely, and complete. GIPS fulfills this.
- Fair-market prices to value client holdings and use good-faith methods for valuations for which no market quote is available. Particularly when calculating AUM. If independent directors, let them do it. Or use a third party.

Disclosures

- Communicate with clients regularly and timely.
- Disclosures are clear.
- Include material facts in communications about anything.
- Disclose: conflicts of interest, past disciplinary action, investment process, fees and how they're determined and calculated (minimum: net-of and gross-of-fees returns and unusual expenses) (be specific, itemize, and provide average or expected fees), amount of soft commissions and how used to benefit client, regular and ongoing performance reporting, how client assets are valued (methodology), shareholder voting policies and how they were voted, trade allocation policies, review or audit results, significant personnel or organization changes, risk management processes and details on risk metrics being used.

- According to the recommendations and guidance of Section F(4d) of the Asset Manager Code, managers must disclose to prospective clients the **average or expected expenses or fees** clients are likely to incur, and to existing clients the **actual fees and other costs charged to them**.
- The Asset Manager Code (AMC) requires disclosing the “performance of clients’ investments on a regular and timely basis.” The AMC recommends that “managers should report to clients at least **QUARTERLY**, and when possible, such reporting should be provided within 30 days after the end of the quarter.” The AMC also states that “at a minimum, Managers should provide clients with gross- and net-of-fees returns.” Because quarterly reporting is the recommended minimum, managers may choose to provide more timely performance to clients.
- For disclosure of client performance: even if lock-up periods, reporting should be provided within 30 days after each quarter.

Further points

- If PM is on board of a company which issues stocks which the PM can buy as part of his or her fund, then this is a business relationship conflict of interest.

5 Derivatives and risk management

5.1 Options strategies

Put-call parity.

$$p_0 + S_0 = c_0 + \frac{X}{(1+r)^T}$$

Put-call-forward parity, where $F_0(T) = S_0(1+r)^T$:

$$p_0 + \frac{F_0(T)}{(1+r)^T} = c_0 + \frac{X}{(1+r)^T}$$

Synthetic long forward = long call + short put at same X. Synthetic short forward = sell call and buy put at same X. Note American-style options and dividends will complicate calculation.

Synthetic long call (protective put) = long stock + long put. Synthetic long put = short stock + long call. Covered call = long stock + short call.

Greeks. Signs reversed for short positions.

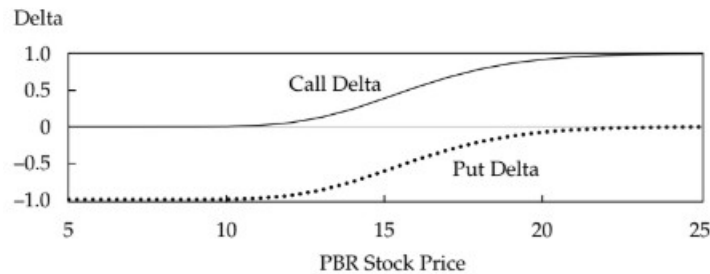
- Delta. Positive for long calls, negative for long puts. 1 for long stock, -1 for short stock.

- Gamma. Greatest near-the-money and gets smaller deeper in or out of the money. Positive for both long calls and long puts. So, as underlying price rises, long call price accelerates up to +1 delta (i.e. accelerates but accelerates at a progressively slower rate), and long put falls slower to 0 delta (i.e. insensitive to underlying price changes). **Highest when expiry is closer in time than farther in time.** Gamma is bell-shaped around the option strike price.
- Vega. Volatility. Positive for long calls and long puts. Per 1% increase in underlying price volatility. Long or shorting a stock: 0 vega. Vega only for options.
- Theta. Time decay, usually per day. Negative for long calls and long puts.
- Time value. Different from theta. Option value = intrinsic value + time value. Theta is how fast time value decays per day.

Covered calls. Can increase yields, reduce exposures, realization of stable target price.

Notes: (1) Writing a call option is same as betting that you think the volatility is lower than the implied volatility. (2) Decompose American (but not European!) options into time value + intrinsic value. (3) Note distinction between option and portfolio deltas, as well as option and portfolio gammas.

Option deltas. General rule: 0.5 for ATM long call and -0.5 for ATM long put. Generally, call delta minus put delta = 1 for same underlying:



ATM covered call, ATM protective put, long stock + short forward all have same position (portfolio) delta.

Cash-secured put. Put option writer deposits cash equal to X, providing assurance to put buyer. In a covered call, the owner sells a call option which is covered by the shares in the covered call, providing assurance to the call option buyer. The cash-secured put is a similar idea.

Greek calculations. Note deltas for calculating the value of the position only apply for small changes in the underlying price; for larger changes, deltas will

change with gamma. Thetas and vegas will also change the value of the position. Value of the position begins with the option price, then make adjustments based on greeks.

Covered calls are a hedging strategy. The premium from selling the call option reduces the breakeven point and maximum loss. This can be seen by calculating the position (portfolio) $\text{delta} = (1 \times \text{No.shares}) - (\Delta_{\text{call}} \times 100)$ if there are 100 shares on the option. The position delta will be less than the No.shares, which means it is less risky on the downside. Portfolio delta will also fall if a long put is purchased, which is less risky.

Protective puts. Similar to insurance.

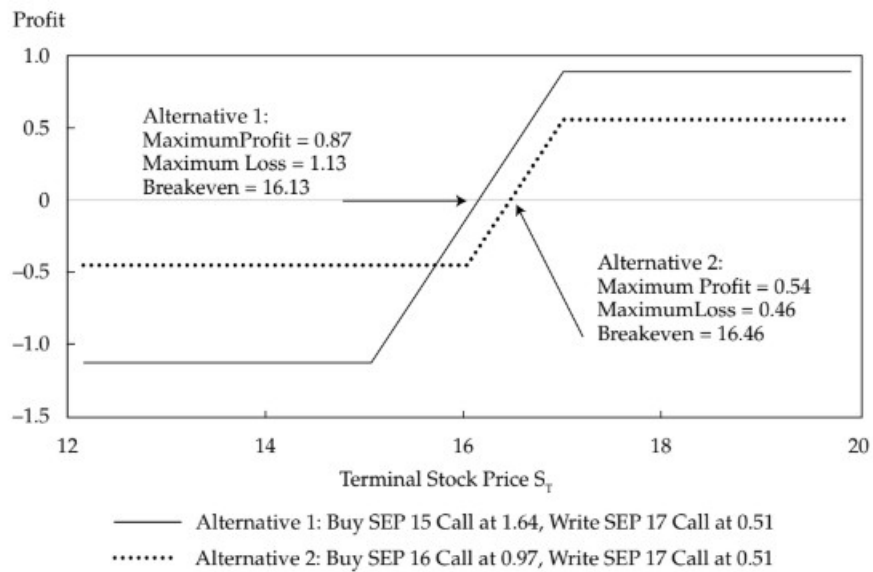
USD/EUR long call option at $X = USD1.25/EUR1$ = option to buy euros, is the same as EUR/USD put option at $X = 1/1.25$.

Symbolic convention: $USD/EUR : USD \rightarrow EUR$ on a long position. For options, futures, etc. A long futures position in USD/EUR means obligation to sell USD for EUR.

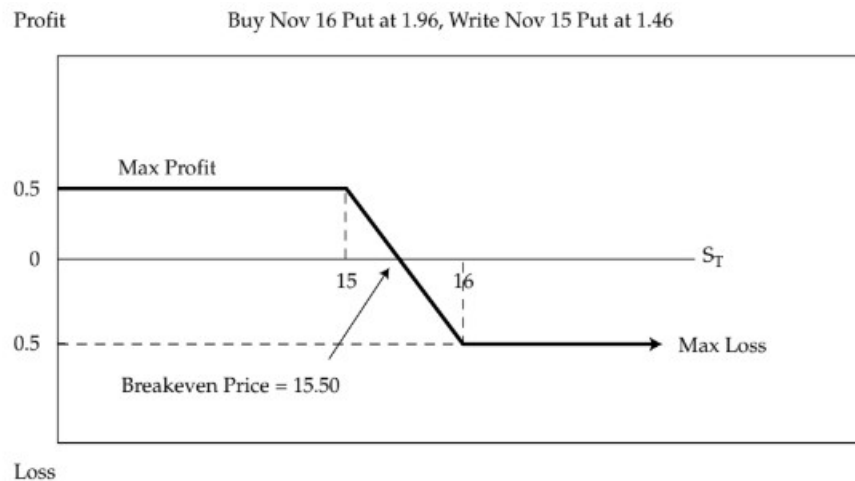
Option spreads and combinations.

Bull spread - more valuable as underlying price rises. Bear spread - more valuable as underlying price falls. Debit spread - initial net cash outflow to establish the position and usually net long (long position is more costly than short position, e.g. long call position at low X and short call position at high Y). Credit spread - opposite of debit spread.

Bull spreads. Buy the lower X option and sell the higher Y option. (1) Call bull spread - debit spread. (2) Put bull spread - credit spread. For (1) call bull spreads: $\text{max loss} = \text{net premium} = c_L - c_H$, and breakeven price $= X_L + c_L - c_H$. When comparing call bull spreads, like below, look at theta, vega, delta, gamma.



Bear spreads. Buy the higher Y option and sell the lower X option. (1) Put bear spread - debit spread. (2) Call bear spread - credit spread. Put bear spread breakeven = $X_H - p_H + p_L$.



Refining spreads. Example. Buy a call option, and then see price rise. Then if don't think price will rise much further, write a call option. Can make the most money by increasing position delta before price rises, and then reducing position delta before price falls.

Straddles.

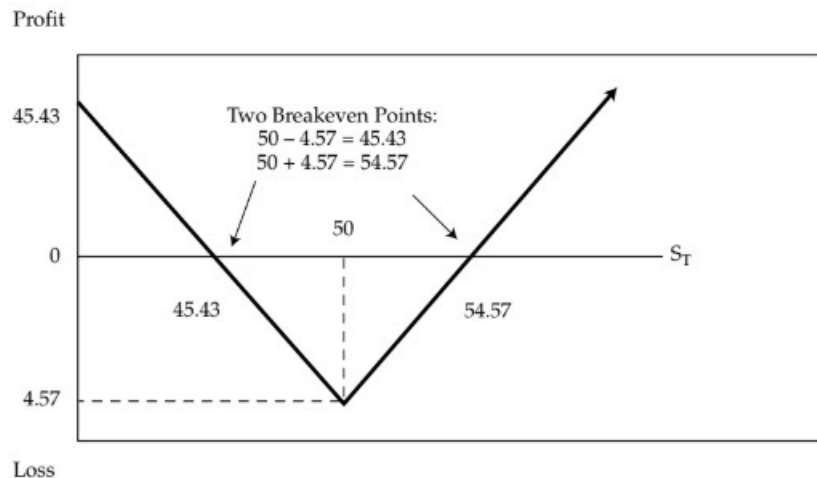
Long straddle = buy BOTH puts and calls at same X. Low delta and high gamma, but once underlying price swings, delta will RAPIDLY become more like +1 or -1 (from low delta, because deltas of call and put cancel each other out; but then when underlying price moves, put delta will rise to 0 and call delta will rise, so the straddle delta rises very quickly). Straddles express long volatility around X. Long straddles are expensive because buy BOTH calls and puts, so price has to move in one direction so much to make up for the position cost.

Short straddle = sell at same X. Opposite of long straddle: express short volatility around X, i.e. hope for limited movement around X, so that premiums from put and call is greater than the payout on the call or put.

Straddle breakeven points. There are two of them:

$$S_L = X - c_0 - p_0$$

$$S_H = X + c_0 + p_0$$



Straddles are also a race against time. Consider theta.

Collars. Long stock + long put below price + short call above price. Buy downside protection by financing with short call. Often do this with floating rate bonds. Reduces position delta = 1 - long put delta - short call delta. Position gamma will be close to 0, as stock gamma is 0 and long put delta almost offsets short call delta. Outside of the 'range' of the collar, delta will become 0, e.g. above the collar, the stock delta is +1 but the short call delta will approach -1. As the collar widens, more like equity. As the collar narrows, more like FI.

Collars are for risk management, when the investor ALREADY OWNS the

stock. They want to modify its risk properties. Bull and bear spreads are for taking a position.

Calendar spread. Buy and sell same option at same X across time. Short call now and long call later = long calendar call spread. Long near call and short far call = short calendar call spread. Short near put and long far put = long calendar put spread. Long near put and short far put = short calendar put spread. Note a portion of the value, particularly of the NEARER option, is time value - make use of this by shorting near-term option. So, the theta will be POSITIVE generally because the short near call theta is higher in absolute value than the long far call. The position delta will be low due to offsetting deltas.

Note: time value is different from theta. The nearer option will have higher time decay and therefore theta in absolute value than the farther option. However, the farther option will have a greater time value.

Short calendar spreads = buy earlier option and sell later option. Later option is usually priced higher (has more time value). So, results in cash inflow. Opposite for long calendar spreads.

Calendar spread profit. A big fall or rise in the underlying price will allow the investor to crystallize the spread profit on a short calendar spread. Short calendar call spread = a big fall leads to near and far calls having 0 value, so the profit is leftover. A big rise in underlying price leads to big near call gain which should be bigger than the far call gain because of latter's increased uncertainty. Short calendar put spread = a big fall in underlying price leads to a bigger near put gain than far put gain. A big rise in the underlying price leads to both near and far puts being 0 value, so the leftover is the net premium.

Long calendar spread profit. Less near volatility is better, in contrast to short calendar spreads.

Position gammas. Gammas on near term options are generally higher than for farther term options. So, short calendar spreads (puts and calls) are generally position net long gamma. A big price move gets the options further from ATM (which is where gamma is highest), leading to low gammas.

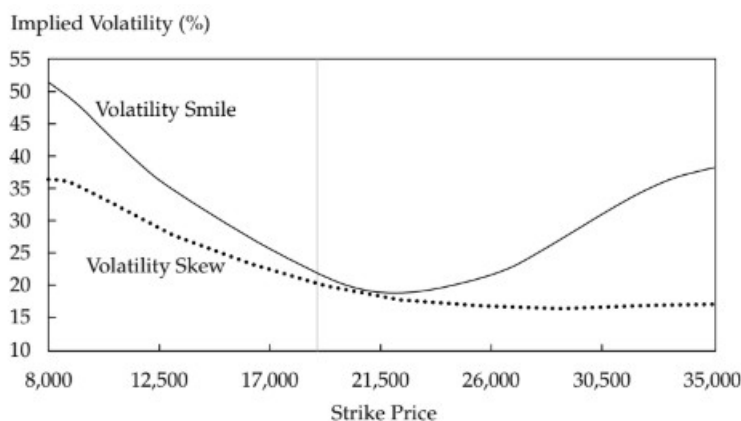
Position vegas. **Further term options have higher vegas and implied volatilities.** Short calendar spreads therefore generally have negative vegas. So, reduced volatility is good for short calendar spreads.

A big move or decrease in implied volatility helps a short calendar spread, whereas a stable market or an increase in implied volatility helps a long calendar spread.

Calendar spread is usually shorting near option and long far option, and maximum profit when near option expires worthless and volatility rises, making long option worth more.

Capturing time value. To "capture" time value in a SHORT calendar strategy, you take advantage of the fact that Theta is not linear. Time value decays much faster as an option gets closer to expiration. So, the earlier-expiring option which is short, loses time value faster (higher theta), and the long calendar option loses time value slower (lower theta).

Implied volatility and volatility skew. Volatilities = SDs, and if have SD for 1 month, then scale up by multiplying by $\sqrt{12}$. Implied volatilities are about future, not current, volatilities nor historical (realized) volatilities. Note **OTM puts** are usually higher value than ATM or OTM calls because of GFC experiences. Implied volatilities depend on their strike prices, leading to a volatility skew and smile. The volatility skew and smile depend on the current underlying price (in a vertical line below):



Volatility skews and smiles are a failure of the BSM's assumption of log-normal returns.

Each volatility smile or skew is referenced on a fixed S_0 or underlying asset price. For European options, due to BSM, the implied volatility of OTM puts and ITM calls at the same X must be the same. So the volatility smile and skew are for both puts and calls. Volatility skew = OTM puts (= ITM calls, if European options) are higher in implied volatility than ATM calls and OTM calls (=ITM puts) because of crashophobia. Volatility smile means there's a market anticipation of a big price jump.

For non-European options, IVs of OTM puts and ITM calls at same X can be different.

Volatility skew = e.g. $X = 0.9 \times S_0$ put implied volatility MINUS $X = 1.1 \times S_0$ call implied volatility (if this is positive, there is volatility skew), and chart this across time - see Exhibit 30.

Term structure of volatility. Plot IV across time (of option with same X). Generally in contango. Negative values = volatility inversion = acute market concerns. Implied volatility surface = 3D of term structure of volatility and price (= volatility smile or skew across time). Here is an example of volatility skew:

Exercise Price	Market Call Price	Market Put Price	Implied Call Volatility	Implied Put Volatility
11,200	526.00	61.90	5.87	17.72
11,400	365.45	102.60	10.80	17.01
11,600	240.00	165.80	12.26	16.44
11,800	135.00	213.00	12.14	16.39
12,000	65.80	370.00	11.98	16.56

The intuition for why Implied Volatility (IV) is assumed to be constant across strike prices in the Black-Scholes-Merton (BSM) model is based on the model's core assumption that the risk of the underlying asset is uniform, regardless of the specific options contract being valued.

Do Q22, p77. When the implied volatility decreases for OTM (out-of-the-money) calls relative to ATM (at-the-money) calls and increases for OTM puts relative to ATM puts, a volatility skew exists. The left side of the IV curve is OTM puts and the right side is OTM calls.

Risk reversal + delta-hedge. There are trading strategies that attempt to profit from the existence of an implied volatility skew and from changes in its shape over time. A common strategy is to take a long or short position in a risk reversal, which is then delta hedged. Using OTM options, a combination of long (short) calls and short (long) puts on the same underlying with the same expiration is a long (short) risk reversal. In particular, when a trader thinks that the put implied volatility is too high relative to the call implied volatility, she creates a long risk reversal, by selling the OTM put and buying the same expiration OTM call. The options position is then delta-hedged by selling the underlying asset. The trader is not aiming to profit from the movement in the overall level in implied volatility. In fact, depending on the strikes of the put and the call, the trade could be vega-neutral. For the trade to be profitable, the trader expects that the call will rise more (or decrease less) in implied volatility terms relative to the put.

Risk reversals have unlimited exposures, whereas bear and bull spreads are limited risk. Risk reversals use COMBINATIONS of puts and calls, whereas bear and bull spreads are put+put or call+call.

Take advantage of volatility skew, by anticipating changes to the skew. Example: trader thinks put IV is too high relative to call IV, create risk reversal by selling OTM put and buying same expiration OTM call, which replicates or almost-replicates the long underlying position, and then delta-hedge by shorting the underlying. Trader is anticipating call will rise less or decrease less in IV relative to the put IV, i.e. IVs converge.

If volatility skew, then bear put spread might be a good deal, as the OTM put is sold at a high IV and ATM put at a lower IV is purchased.

Index	Implied Volatility by Moneyness			90%–110%
	ATM	Put: 90%	Call: 110%	Volatility Skew
Nikkei 225	12.9	18.9	12.4	6.5
S&P 500	10.3	17.7	9.4	8.3
Euro Stoxx 50	12.3	17.8	9.3	8.5
DAX	14.5	20.0	11.0	9.0

Option strategy examples.

- Covered call. Have a stock, don't think it'll rise much, need money now. Write a call.
- Short puts. Want to buy a stock but current price is too high. Write a put at the price you want. If expire worthless, then you get profit. If price falls below strike price, then you buy a bit above the spot price but get a put premium.
- Long straddle. If there's volatility rising.

Note: a high gamma is beneficial on a put. If the underlying price rises, then long put delta goes from -1 to 0, so the long put in an accelerating fashion loses exposure to the adversely-moving stock price. If underlying price falls, long put delta goes from 0 to -1, so long put becomes increasingly exposed but at a decelerating rate to the beneficially-falling stock price. This is convexity.

5.2 Swaps, forwards, and future strategies

Managing interest rate risk.

Interest rate swaps. **Synthetic fixed security.** A firm issues FRNs. Then

changes its view on MRRs. So it enters an IR pay-fixed receive-floating swap, to fix its interest rate payments.

Formula for changing MDur. Assuming yield curve is flat and parallel yield curve shifts, hedges are perfect (no basis risk or spread risk),

$$(MV_p)(MDUR_p) + (N_s)(MDUR_s) = (MV_p)(MDUR_T)$$

which is actually

$$(MV_p)(MDUR_p) + (N_s)(MDUR_s) = (MV_T)(MDUR_T)$$

but in most cases $MV_p = MV_T$, but if different, then it's not just swaps that are involved; it involves a change in the 'effective' value of the PF. If however $MDUR_T = MDUR_P$ then can rearrange to:

$$(N_s)(MDUR_s) = (MV_T - MV_p)(MDUR_T)$$

Anyways, solving the first equation,

$$N_s = \left(\frac{MDUR_T - MDUR_p}{MDUR_s} \right) (MV_p)$$

where P is the portfolio, T is the target.

Intuition I. The exposure to durations increases. Example: if we have a 1% increase in interest rates leading to a 2% value increase before, and we add a swap with a 1% increase in interest rates leading to a 2% value increase, then we are INCREASING our exposure to the interest rate, regardless of whether durations are like weighted averages. It's like doubling or increasing the number of the same-duration bonds - the portfolio duration will rise.

Intuition II. The equation above is like the very average of durations - which is intuitive, e.g.

$$(MV_p)(MDUR_p) + (MV_x)(MDUR_x) = (MV_{p+x})(MDUR_{p+x})$$

$MDUR_s$ is net of modified durations. Position received minus position paid. Example, in a pay-fixed receive-floating, MDur of floating rate bond minus MDur of fixed-rate bond. MDur of FRNs close to 0 since they keep resetting. So, MDur of pay-fixed receive-floating is negative, and benefits from rising interest rates.

MV_p is PF value. $MDUR_p$ is MDur of PF. N_s is notional value of hedge. $MDUR_T$ is the overall MDur of PF and hedge together.

FRAs and IR futures (for a single bond, not a basket of bonds in contrast to FI futures). See Example 3 on p88 and q10 on p126. Want to hedge an interest rate.

- If RECEIVING interest and think rate will FALL: BUY an IR future at $100 - r_h$ where r_h is the hedged interest rate, e.g. 97 for a 3% interest rate, and then later unwind when it rises, e.g. to 98 corresponding to a 2% rate. So, receive 2% on a loan but get a 1% addition from the hedge.
- If PAYING interest and think rate will RISE, SELL an IR future at $100 - r_h$, e.g. 98 corresponding to a 2% rate, and then later unwind when it falls by buying it back at e.g. 97 corresponding to 3%. So, pay 3% on the loan, but get 1% discount from the hedge.

Fixed-income futures. Standardized forward contracts with underlying a basket of deliverable bonds. Within the underlying basket of bonds, the seller will deliver the cheapest-to-deliver (CTD) bond, which is greatest profit. The CF conversion factor converts to e.g. 6% and 20 years to maturity bond. What is delivered to the buyer is the principal invoice amount = (Futures settlement price divided by 100) x CF x contract size. The cost to the seller of buying this bond is the cash dirty price x contract size = bond purchase value.

Hedge ratios = number of FI contracts to sell or buy to hedge a portfolio. Change in price = hedge ratio x change in value of FI futures:

$$\Delta P = (HR)(\Delta F)$$

where

$$\Delta F = \frac{\Delta CTD}{CF}$$

where CTD is the price of the CTD bond. So,

$$HR = \frac{\Delta P}{\Delta F} = \frac{\Delta P}{\Delta CTD} CF$$

Generally,

$$HR \approx CF$$

because

$$\Delta P \approx \Delta CTD$$

Duration-based hedge ratios. This is base-point value BPV. The target BPV is:

$$BPV_T = MDUR_T \times 0.01\% \times MV_p$$

The PF BPV is:

$$BPV_p = MDUR_p \times 0.01\% \times MV_p$$

The CTD BPV:

$$BPV_{CTD} = MDUR_{CTD} \times 0.01\% \times MV_{CTD}$$

$$MV_{CTD} = \frac{CTD}{100} \times N_S$$

The futures FI contract BPV is

$$BPV_F = \frac{BPV_{CTD}}{CF}$$

Combining,

$$BPV_p + BPV_{HR} \times BPV_F = BPV_T$$

which is substantively the same as

$$(MV_p)(MDUR_p) + (N_s)(MDUR_s) = (MV_T)(MDUR_T)$$

Sometimes, it's easier to use the BPV form and other times it's easier to use the MDUR form - it depends on the raw data available. Anyways, re-arranging the BPV form,

$$BPV_{HR} = \left(\frac{BPV_T - BPV_p}{BPV_F} \right) = \left(\frac{BPV_T - BPV_p}{BPV_{CTD}} \right) \times CF$$

For full or complete hedging, $BPV_T = 0$,

$$BPV_p + BPV_{HR} \times BPV_F = 0$$

(Note: $BPV_p = 0$ for cash, so if buying something from scratch or modelling as if buying from scratch, use 0).

$$BPV_{HR} = -\frac{BPV_p}{BPV_F}$$

$$BPV_{HR} = -\frac{BPV_p}{BPV_{CTD}} \times CF$$

In reality, hedging results are imperfect because (1) hedge is done with CTD and CTD bond can change, so $MDUR_{CTD}$ will also change, (2) convexity means relationship between interest rates and bond prices is non-linear, (3) non-parallel interest rate changes.

Fixed income basis. The pricing discrepancy between the price of the cash security and that of the fixed-income futures is the basis.

$$Basis = Spot - Futures \times CF$$

If the basis is negative, a trader would make a profit by “buying the basis”—that is, purchasing the bond and shorting the futures. In contrast, the trader would make a profit by “selling the basis” when the basis is positive; in this case, she would sell the bond and buy the futures.

Short-term. For money market instruments or short durations, the modified duration is just

$$MDUR \approx \frac{Days}{365}$$

So, for Example 3 p89, where the hedge is unwound at $t = 2$, the duration is simply 3 months, or 0.25:

$$\frac{30M \times 0.25}{10,000bps} + 25N = 0$$

This is also why the Eurodollar futures BPV are defined linearly - and this facilitates the futures pricing of $100 - r$.

Currencies. Currency swaps, and currency forwards and futures.

Currency swaps. May or may not exchange notionals at beginning and end; depends on type of currency swap.

Cross-currency basis swap: exchange notionals; get loan or funds in own currency and then swap floating-for-floating payments in different currencies. Net effect is to use own-currency loan to take out a loan in a foreign currency while having a fixed FX rate. There is **no extra on top of MR (i.e. the basis) is paid** in a swap.

Example 1: CAD-USD currency swap example. Canadian company needs US, and USD borrowing is $MRR(USD) + 100$ bp, whereas CAD borrowing is $MRR(CAD) + 65$ bp. If Canadian company borrows CAD and gets CAD-USD swap for -15 bp, then it pays total 80 bp over MRRs. Canadian company 'lends' CAD and 'borrows' USD, and gets paid $MRR(CAD) + 50$ bp, where $65 - 15 = 50$ bp, and pays $MRR(USD) + 0$ bp to the swap dealer, and pays $MRR(CAD) + 65$ bp to the bank, ultimately paying $65 + 15 = 80$ bp on net. This compares with $MRR(USD) + 100$ bp if it borrowed USD. The Canadian company gets 15 bps less than MRR and has to pay the Canadian bank MRR + 65 bps, so effectively pays + 80 bps, i.e. a lower basis, with no FX change risk.

Synthetic domestic yield. Earn extra yield by investing in a foreign bond market and swapping the proceeds into domestic currency. Example: during periods when demand for US dollars is strong relative to demand for JPY, the US-Japan interest rate differential implied by the currency markets may be significantly wider than the actual interest rate differential. Use a cross-currency swap to lend USD and buy JPY, buy Japanese treasuries (which generate JPY to pay the JPY interest), and each period pay yen and receive US dollars on the swap; at beginning and end, swap notionals. When the basis is largely negative, due to relatively weak JPY relative to USD, JPY borrowing costs are low, making the return from lending USD via a cross-currency swap more attractive than buying US treasuries. The rates on the cross-currency basis swaps will depend on the demand for US dollar funding, because when the US dollar reference floating rate is elevated, the counterparty receiving US dollars at initiation of the swap will be willing to receive a lower interest rate on the non-dollar currency periodic payments.

Equities. Equity swaps and equity forwards and futures.

Equity swaps. Usually OTC - so, customizable but counterparty risk. Usually 3 types:

- receive-equity return, pay-fixed
- receive-equity return, pay-floating
- receive-equity return, pay-another equity return

Usually cash-settled, but can be physically settled if only one type of stock is the notional.

There are 'total' instead of 'equity' return-swaps, which includes dividends. Example: if share price is 14, and dividend of 0.1, and share price rises 1%, then total return is

$$\frac{14(1.01) + 0.1 - 14}{14}$$

and then determine whether this exceeds the annual interest rate given. Total returns are OTC and do NOT cede equity voting rights.

Advantages and disadvantages. Advantages: can be used when market access is limited, equity is taxed but swaps are not, equity custodian fees are high, monitoring stocks (e.g. because of corporate actions) is high. Disadvantages: must usually put up collateral (for counterparty risk), illiquid, no voting rights.

Example. Investor has equity PF linked to SP500 and believes in some chance it will decline. Enter pay-equity return, receive-fixed. Then if equity returns fall, receive BOTH equity return AND fixed payments. If equity returns rise, pay equity return and receive fixed payments.

Equity forwards and futures. Let β_s be the PF beta, β_T the target PF beta. Because value of futures contract begins each day at 0, the dollar beta of the combination of stocks and futures is $\beta_T S$ where S is the MV of the PF. Number of futures contract = N_f , beta of futures β_f and F is futures contract value = $Index \times multiplier$.

$$\beta_T S = \beta_S S + N_f \beta_f F$$

which is actually

$$\beta_T S_T = \beta_S S_P + N_f \beta_f F$$

but usually $S_T = S_P$ but if not, then there is a change in the effective PF equities. If however $\beta_T = \beta_S$ then can rearrange to

$$\beta_P (S_T - S_P) = N_f \beta_f F$$

Anyways, rearranging the first equation to:

$$N_F = \left(\frac{S}{F} \right) \left(\frac{\beta_T - \beta_S}{\beta_f} \right)$$

If want to eliminate market risk completely, $\beta_T = 0$, so

$$N_F = - \left(\frac{S}{F} \right) \left(\frac{\beta_S}{\beta_f} \right)$$

and if want to increase $\beta_T > \beta_s$, then $N_f > 0$, i.e. to increase market exposure have to buy futures.

Beta calculation

$$\beta = \frac{\% \text{ total gain}}{\% \text{ market gain}}$$

Cash equitization. Take cash and equitize it by buying futures (or buying calls and shorting puts), all the while the cash earns interest. Note for cash holdings, $\beta_s = 0$,

$$\beta_T S = \beta_S S + N_f \beta_f F$$

so

$$N_F = - \left(\frac{S}{F} \right) \left(\frac{\beta_T}{\beta_f} \right)$$

where $\beta_f = \beta_T = 1$ if just want to track an index and buying futures on that index.

It is not necessary to have any cash to just buy equity futures - but that's highly leveraged speculation. If have cash, it is not leveraged.

Variance swaps. No exchange of cash, but variance notionals are used - variance notionals are dollar amount paid per unit of difference in VARIANCE. Long variance position will gain if realized variance is greater than swap variance strike (e.g. if $X = 20\%$, then $X = 20$). X is a volatility, not variance. Trade size is expressed as vega notionals - vega notionals are dollar change in value for a 1% change in VOLATILITY.

$$\sigma_r^2 = \text{Realized variance} = 252 \times \sum_1^{N-1} \frac{R_i^2}{N-1}$$

$$R_i = \ln \left(\frac{P_{i+1}}{P_i} \right)$$

where 252 is the number of trading days per year. The log quotient form is used instead of $R_{t+1} = \frac{P_{t+1} - P_t}{P_t}$ because you can't just add these returns; this is what you want to do: $(1 + R_t)(1 + R_{t+1})(\dots) - 1$, which is actually done by adding log quotients.

$$R_i = \ln \left(\frac{P_{i+1}}{P_i} \right) \approx \frac{P_{i+1} - P_i}{P_i}$$

because, defining

$$x = \frac{P_{i+1} - P_i}{P_i}$$

$$\frac{P_{i+1}}{P_i} = 1 + x$$

$$R_i = \ln \left(\frac{P_{i+1}}{P_i} \right) = \ln(1 + x)$$

By a log approximation or Maclaurin series,

$$R_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$

$$\text{Variance notional} = \frac{\text{Vega notional}}{2 \times X}$$

where Vega notional = average profit or loss of variance swap for a 1% change in volatility from X. The vega notional is like $\Delta\sigma^2$, so when differentiated wrt to volatility,

$$\Delta\sigma^2 = \frac{d(\sigma^2)}{d\sigma} \times \Delta\sigma + \dots$$

and since

$$\frac{d(\sigma^2)}{d\sigma} = 2\sigma$$

substituting,

$$\Delta\sigma^2 \approx 2\sigma \times \Delta\sigma$$

and since $X = \sigma$,

$$\Delta\sigma^2 \approx 2X \times \Delta\sigma$$

since by definition,

$$\text{Change in value} \approx N_{Var} \Delta\sigma$$

$$\text{Change in value} \approx N_{Vega} \Delta\sigma^2$$

Then combining with the Taylor approximation,

$$\text{Change in value} \approx N_{Var} (2X \times \Delta\sigma)$$

Equating,

$$N_{Vega} \times \Delta\sigma \approx N_{Var} (2X \times \Delta\sigma)$$

$$N_{Vega} \approx N_{Var} (2X)$$

$$\text{Variance notional} = N_{Var} \approx \frac{\text{Vega notional}}{2X} = \frac{N_{Vega}}{2X}$$

which is effectively scaling the vega notional. The settlement amount is variance notional times (realized variance - strike variance).

$$\text{SA} = N_{var} \times (\sigma_r^2 - X^2) = N_{Vega} \left(\frac{\sigma_r^2 - X^2}{2X} \right)$$

Note: X is a **volatility, not variance**. It is % but expressed as a $\% \times 100$. It is generally calculated by implied volatility of 90% moneyness of put.

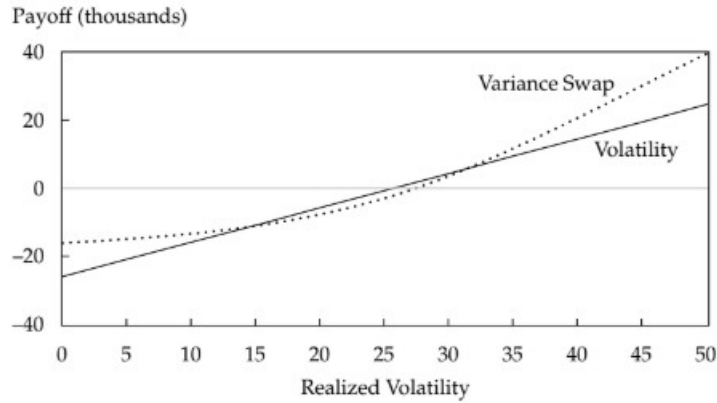
Let variance notional = N_{var} , and IV = implied volatility. The mark-to-market valuation of the variance swap is, where t = current time = time elapsed since start of variance swap, is

$$VarSwap_t = \frac{N_{var}}{(1 + r_{T-t})^{T-t}} \times \left(\frac{t}{T} \times \sigma_r^2 + \frac{T-t}{T} \times (IV)^2 - X^2 \right)$$

Long variance swaps are convex. It is equivalent to long basket of options (which all have positive gamma, which is good for tail risk because it is a squared term) and short underlying (to delta hedge). This isolates the gamma. So, volatility sellers can sell variance swaps at higher price than ATM options. ATM options require delta hedging and have basis risk.

Variance swaps have an advantage over volatility options: variance swaps isolate gamma (whereas options have a delta which needs to be hedged away) and options have directional risk (options are only one-way).

Volatility. Example: variance swaps useful for long-equity positions as insurance, as volatility rises when stocks fall. Example: VIX futures and options. Like commodities, contango term structure will lead to roll yield losses and backwardation will lead to gains.



Application.

- For equity, find N_f if looking for number of futures for hedging. For bonds, use $BPVHR$. For equities, PF market exposure adjustment is the priority, so look at betas. For bonds, PF interest rate exposure adjustment is the priority, so look at durations.
- For $BPVHR$, to calculate BPV_T , use $MDur_T$ of the current relevant portion of the PF if there's no target specified, i.e. status quo modified duration. For $BPV_p = 0$ if just buying some bonds with cash.

- To calculate a bps change on bond futures, $BPV_F \times \Delta \text{yield} \times N_{\text{contracts}}$, where $BPV_F = BPV_{CTD}/CF$.
- Hedge ratio = 1, means a change in PF value = change in hedging instrument value.
- Cannot directly invest in the VIX, can only transact derivatives on it.

Inferring market expectations.

Effective federal funds (FFE) rate - the actual rate transacted between depository institutions, which is different from the target federal funds rate. On the CBOE, Federal funds futures contract price = 100 - expected FFE.

$$\text{Prob of change in FFE} = \frac{\text{FFE implied by futures} - \text{current FFE}}{\text{FFE assuming rate hike} - \text{current FFE}}$$

5.3 Currency management

Foreign exchange concepts.

- Spot markets. Generally, P/B or price over base currency notation. Base currency is EUR, GBP, AUD, NZD, then USD for all others. Generally quote to 4 decimal places, but exceptions like JPY/USD only quoted to 2 decimal places.. If $1.5123EUR/USD$, then 1.15 is the 'handle' and 23 pips. Spot is settled $T + 2$. Bid price = how much of price currency to buy one unit of base currency. Offer price = how much of price currency to sell one unit of base currency.
- Two-sided dealer quote of USD/EUR of 1.1648/1.1652 means the dealer will buy for 1.1648USD and sell for 1.1652USD. A USD/EUR two-sided quote can be seen as trades on the base currency EUR: you can sell EUR and get 1.1648USD or buy EUR at 1.1652USD.
- Forward markets. Settlement after $T + 2$. 'All-in' rate is forward rate = spot rate + forward points (scaled ofc by dividing by 10,000). 'Buying X for delivery against Y' is long a forward contract on X. With mark-to-market, get a net value for a currency to value the forward (which can be converted to a different currency at the FORWARD BID rate), and remember to discount back to the present.
- Swap markets. Currency swaps are different to FX swaps. FX swaps: buy or sell spot and sell or buy forward. Swap legs can be equal size (matched swap) or different (mismatched swap). Involves exchanges of different principal amounts at beginning and end (reversed at maturity) - different because the spot is different from the forward. However, FX swaps have NO interim interest payments. FX swaps can be used to roll forward contracts.

- Currency options. Mostly vanilla but some exotic options.

Currency risk.

Foreign-currency return. Return of foreign asset in foreign currency terms.
Domestic currency return of a foreign asset $R_{DC} = \text{foreign-currency return} + \text{FX}$.

$$R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1$$

where the domestic currency is the price currency in R_{FX} .

$$R_{DC} = R_{FC} + R_{FX} + R_{FC}R_{FX} \approx R_{FC} + R_{FX}$$

and based on this approximation, volatility squared is

$$\sigma^2(R_{DC}) \approx \sigma^2(R_{FC}) + \sigma^2(R_{FX}) + 2\sigma(R_{FC})(\sigma(R_{FX})\rho(R_{FC}, R_{FX}))$$

So, if $\rho > 0$, $\sigma^2(R_{DC}) > \sigma^2(R_{FC})$ but if $\rho < 0$ then it's unclear whether $\sigma^2(R_{DC})$ or $\sigma^2(R_{FC})$ is larger.

For a portfolio, measured using the common currency = domestic currency,

$$R_{DC} = \left(\sum \omega_i (1 + R_{FC,i})(1 + R_{FX,i}) \right) - 1$$

where $\omega < 0$ means short-selling. For

$$R_{DC} = \omega_1 R_{DC,1} + \omega_2 R_{DC,2}$$

the volatility squared is

$$\sigma^2(\omega_1 R_1 + \omega_2 R_2) \approx \omega_1^2 \sigma^2(R_1) + \omega_2^2 \sigma^2(R_2) + 2\omega_1 \omega_2 \sigma(R_1) \sigma(R_2) \rho(R_1, R_2)$$

which means the complexity increases geometrically, and PF optimization gets very complicated. But since $\sigma(R_{DC})$ depends on $\sigma(R_{DC}), \sigma(R_{FX})$ and how each of these interacts with the others ρ . Generally speaking, negative correlations among these variables will help reduce the overall portfolio's risk through diversification effects.

Currency hedging strategy.

- Some believe in long-run, currency effects cancel out to 0, and since depreciation in one currency = appreciation in another, if diversified then no major net effect. Others note pricing inefficiencies and limited speculative capital for arbitrage in market.
- Look at IPS for time horizons, constraints, requirements.
- Above, it was noted PF optimization gets complicated. So, decompose into two steps: (1) PF optimization assuming perfect currency hedges (i.e. $R_{FX} = 0$, and (2) decide how much to relax currency hedges.

- PF optimization also requires EXPECTED values for volatilities and correlations, which is complicated as they interact.
- Currency hedging: diversification and cost considerations.
- Diversification considerations I. In the long-run, there is mean-reversion to international parity conditions. Saves on hedging costs. But might have to wait decades = long-run, and need benchmark index that is also unhedged again currency risk.
- Diversification considerations II. Look at asset composition, i.e. whether R_{FC} , R_{FX} are negatively correlated, which reduces currency exposures.
- Diversification considerations III. Bonds more exposed to currency than equities. Equities depend more on earnings. Bonds are exposed to interest rates, which also affect currency in positively-correlated direction, e.g. if interest rate falls, bond prices rise and foreign currency depreciates.
- Diversification considerations IV. Currency hedging may depend on market conditions. USD, JPY, CHF are safe-haven currencies, and if bonds globally fall in price, these currencies rise. For non-US investors, this would mean under-hedging the currency exposure to the USD (i.e., a hedge ratio less than 100%), whereas for US investors it would mean over-hedging their foreign-currency exposures back into the USD.
- Cost considerations I: trading and opportunity costs. Trading costs: bid-offer spread, currency option premia, forward contracts need to be rolled - which even if are positive cash inflows leads to higher cash volatilities, administrative infrastructure. Opportunity costs: regret.

Specific currency risk management strategies. Note: HR = hedge ratio = (value of currency exposure hedge) divided by (total foreign currency exposure).

- Passive hedging. Rules-based, no discretion for PM. Often, benchmark is a foreign currency benchmark. HRs can drift so need rebalancing. Can use this to avoid all currency risk in a PF.
- Discretionary hedging, e.g. HRs to be kept within a range.
- Active management. Take currency risks and manage them for profit.
- Currency overlay. External PM or in-house. Can separate beta (the currency hedge) from alpha-seeking. Can use this to generate alpha, and generally **alpha-seeking strategies should have minimum correlation to major asset classes and other alpha sources in the PF**. Can use multiple currency overlay managers.
- Foreign exchange as an asset. Free to take FX exposures where value to be harvested, regardless of underlying portfolio.

Currency management program. Fully-hedge or close to it, if short-term investment objectives in IPS, risk-averse and don't mind regret over missed opportunities, need more income and liquidity immediately, more FI assets as foreign assets, more volatile financial markets, PF owner views.

Currency tactical, as opposed to strategic, decisions. Need to begin with market view or forecast.

- Economics. Real exchange rate = ratio of real purchasing power, $RER = NomFX \times \frac{PriceLevel_P}{PriceLevel_F}$. In the LR, real exchange rate will converge to fair value, e.g. international parity conditions. Over ST and MT: $r = i - \pi^e$, risk premiums (country risk premiums). So, ST or MT real exchange rate (base currency) should appreciate if (1) LR real exchange rate rises (e.g. affected by fiscal and monetary policies), (2) home r or i rises, (3) EXPECTED foreign inflation rises, (4) foreign risk premium rises.
- Technical analysis. Market price already reflects fundamentals, and use historical patterns because they repeat. Market psychology persists, so people react similarly to similar situations that repeat, e.g. irrational exuberance. Try to identify where overbought or oversold by looking at support or resistance levels where dealer order boards are clustered - once these are breached, price movement accelerates as get to stop-losses, which further accelerate market change.
- Technical analysis II. 200-day MA is resistance or support level. If spot price is below 200-day MA, it is resistance. If it above, then the 200-day MA is support. If the spot or 50-day MA crosses 200-day MA, this is 'break out' point (moves from below 200-day MA to above it).
- Technical analysis III. Market technicians (1) use historical price data to help forecast future movements, (2) assume historical price patterns repeat to identify trade opportunities, and (3) attempt to determine not where prices SHOULD be (fundamentals) but rather where they WILL be (technical).
- Technical analysis IV. Use technical skills to (1) time entry and exit points, (2) identify historical price trends, when it was overbought or oversold, and when trended too far in one direction and vulnerable to a trend reversal.
- Carry trading. Borrow low-interest rate currencies and invest in high-interest rate currencies. Highly leveraged. Exploits violation of uncovered interest rate parity, which says that over the long run, $\% \Delta S \approx i_H - i_L$, i.e. high-yield currencies should see their currencies depreciate. However, usually over ST and MT, high-yield currencies appreciate, which is reflected in the **forward rate bias**. With the forward rate premium or discount defined as $F_{P/B} - S_{P/B}$ (F = forward, S = spot) the bias in the forward rate bias is that the premium typically overstates the amount of appreciation of the base currency, and the discount overstates the amount of depreciation.

In other words, UIP predicts $E(S_{t+1}) = F_t$ but in fact there are deviations from it. Trading the forward rate bias = carry trade because when base currency has lower i than price currency, base currency will trade a forward premium. Carry trade = short the low i currency (short the premium) and long the high i currency (long the discount). Carry trades are negative skew because funding currencies are usually safe-haven currencies while investing currencies are usually EM currencies. So, low volatility is better for carry trading.

Trading with and against the forward rate bias.

Type of position	What it means	Typical example
Trading <i>with</i> the forward rate bias	You assume the historical bias will persist. You exploit the bias.	Carry trade: Borrow in low-yield (forward premium) currency and invest in high-yield (forward discount) currency.
Trading <i>against</i> the forward rate bias	You assume the bias will correct itself. You bet on UIP holding true .	Take the opposite side of the carry trade: borrow in high-yield currency and invest in low-yield currency.
Investor Type	Likely Position	Explanation
Risk-seeking or tactical investor	<i>Trade with</i> the bias	Exploit the carry trade's historically positive return.
Risk-averse investor	<i>Trade against</i> the bias	Prefers to avoid the risk that high-yield currencies might crash — follows UIP expectations.
1. Implement the carry trade.		Buy (invest in) the high-yield currency and sell (borrow) the low-yield currency.
2. Trade the forward rate bias.		Buy (invest in) the forward discount currency and sell (borrow) the forward premium currency.

The low- i currency trades at a forward premium and the high- i currency trades at a forward discount. Positive roll yield requires shorting the forward premium.

Trading with the forward rate bias. Carry trade. Short low- i currency and long high- i currency. If UIP completely holds, FX rates will change as in CIP so that the high- i currency depreciation completely offsets the interest rate differential. However, UIP doesn't hold like CIP, so the forward rate bias persists, allowing a profit.

Trading against the forward rate bias means **SHORTING** the high- i currency **FORWARD** and investing in the **FORWARD CONTRACT** of the low- i currency in the hope that the high- i currency depreciates enough (UIP holds) and low- i currency appreciates. As the low- i currency will appreciate (long the forward premium currency), the forward will lead to a positive roll yield.

Volatility trading. Delta = price risk = option premium change to underlying price change. Vega = volatility risk = option premium change to change in implied volatility. Note: straddles are automatically delta-hedged because European ATM call option has delta of around +0.5, and European ATM put

is -0.5. **Strangles** are like straddles except OTM call and put.

Deltas for European-style put options range from -1 (deep-in-the-money put) to 0 (deep-out-of-the-money put), and from 0 to $+1$ for calls. Deltas of 0.5 and $+0.5$ occur when the strikes are ATM on a forward basis.

Can have volatility overlay, analogous to currency overlay.

Additional points

- When looking at a carry trade, consider also the implied volatility - want IV to be low.
- If NZ inflation rate will rise relative to JP, then NZ real interest rate will fall $r = i - \pi^e$ relative to JP, so the JPY/NZD will depreciate, i.e. NZD will depreciate.
- If selling MXN at $2MXN/GBP$ is the rate for BUYING GBP. That is not the actual rate of selling MXN, which is $0.5GBP/MXN!!!$
- If the spot rate is $2MXN/GBP$ for an ATM put, if the value falls (MXN appreciates and GBP depreciates), then ITM put. If the value rises, then MXN depreciates and GBP appreciates, so just take the profit.

Investors prefer forward to futures contracts: (1) futures are limited in contracts, (2) futures combinations don't always exist, (3) futures require margins and monitoring these margins. OTC forwards volumes is much bigger than for futures.

Hedge ratio drift. Currency hedges, e.g. with forwards or futures, drift with time. A static hedge (matching current MV of FC exposure with offsetting forward contract) is cheaper but there's drift (as MV of foreign currency assets change). A dynamic hedge means rebalancing (and rolling) but more expensive. The more risk-averse, the more dynamic the hedge.

Carry trades and roll yields are equivalent. A negative roll yield typically indicates that the hedger was trading against the forward rate bias by buying a currency at a forward premium or selling a currency at a forward discount. Given the equivalence between the forward rate bias and the carry trade, by trading against the forward rate bias the hedger with a negative roll yield is also essentially entering into negative carry trade, in effect borrowing at high rates and investing at low rates.

	Buy/Invest	Sell/Borrow	
Implementing the carry trade	High-yield currency	Low-yield currency	
Trading the forward rate bias	Forward discount currency	Forward premium currency	Earning a positive roll yield

Whether to currency hedge. If the expected roll yield is -3% but the market view is the currency will depreciate 4% , then hedge makes sense: better to pay a 3% loss than a 4% loss. If the expected roll yield is -5% , the hedge might still make sense depending on risk-aversion.

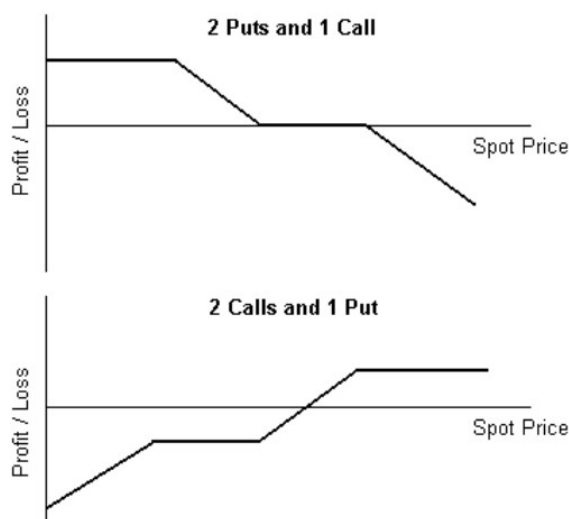
How much to currency hedge. As the forward pips rises in absolute value, the hedge becomes more valuable and more people will hedge.

Currency management strategies. Note: there is passive hedging, discretionary hedging (primary purpose is to hedge the PF), and active currency management (seek alpha). Moreover, there is currency overlay which is tactical for hedging and-or alpha generation, and being part of the policy portfolio which is strategic where positions are taken on currencies for alpha generation.

- Forward contracts. Can use this to completely hedge, e.g. if 'neutral' (i.e. policy) benchmark hedge ratio is 100% for the base currency. A benchmark HR of 100% means (i) the benchmark is in the investor's domestic currency, e.g. S&P500 for a US investor, or (ii) the benchmark has foreign currency exposure (but is denominated in the domestic currency) but is completely hedged.
- Forward contracts II. Can over- or under-hedge. This is a proactive strategy. If PM has market view that base currency will depreciate, then over-hedge the base currency. If view is that it will appreciate, then under-hedge the base currency.
- Forward contracts III. Can follow currency movements in a dynamic but reactive strategy, where there is no market view. Increase HR if base currency depreciated, and reduce HR if base currency appreciated. Increasingly participate in base currency appreciation, increasingly hedge depreciation. It is like adding a put option: appreciations leave the currency position increasingly unhedged, while depreciations mean the currency position is increasingly hedged. It is also like delta-hedging because the downside position is increasingly delta-neutral (because a long put is added to a long base currency exposure). Finally, much like bonds, it adds convexity: convexity increases the bond price more quickly with falling yields and falls more slowly with rising yields.
- Protective puts with OTM options. Dynamic hedging with forward contracts or using ATM options is expensive. Instead, use OTM options,

though with some obvious downsides.

- Risk reversal or collar. A long collar is long position + long OTM put + short OTM call. Short risk reversal = long OTM put + short OTM call. A long risk reversal is long OTM call and short OTM put. Long collar = short risk reversal + long position.
- Put spread. Usually buy OTM put and short even-deeper OTM put. The latter helps partially pay for the former. If want completely zero-cost, then use a 1×2 put spread, where 2x the notional is used for the deep OTM put; this is also a leveraged-position, with more shorting than long.
- Seagull spread. Short seagull spreads are where the 'wings' are short positions, and the 'body' is made from a long ATM put (as in the above diagram below) or a long ATM call or protective put (as in the lower diagram below). Since long puts, long calls and protective puts are expensive, using TWO short premia in the wings defrays the cost.



Long seagull spreads are inverses of the above diagram, where the body is made from e.g. a short ATM call and the wings are made from long positions.

- Exotic options. Vanilla = European options. Exotic = not vanilla, does what vanilla options can do with fine-tuning risk and cost, but better. Most common currency exotics are **knock-in** and **knock-out** features and digital options.
- Exotics: Knock-in and knock-out options. Knock-in options are created when the FX spot touches a barrier. Knock-out options cease to exist when the FX spot touches a barrier. Cheaper than vanilla but more limited protection.

- Exotics: Digital (binary or all-or-nothing) options. Pay fixed amount if ITM. ITM depends on definition, e.g. American digital options = if touch X before expiry. Depending on how ITM, can be very leveraged.

Buying base currency = buy call or long forward contract. Selling base currency = buy put or short forward contract.

Language. If ZAR/GBP and are exposed to foreign currency ZAR and want to hedge it (i.e. protect from ZAR depreciation), then want to LONG CALL ZAR/GBP because appreciating GBP = depreciating ZAR. 'Upside' is when ZAR appreciates, i.e. ZAR/GBP FALLS.

Proxy, cross, macro and other hedges. Proxy hedges are for domestic-foreign currency hedges (hedge foreign currency back to domestic); cross-hedges are foreign-foreign currency hedges (introduces a **basis risk**). Having long and short positions in positively correlated currencies can lead to lower PF risk - see p197. Macro hedges view PF as not as individual assets but macro-risks like term risk, credit risk and liquidity risk. Other hedges for PFs: using gold, volatility overlays, derivatives based on baskets of currencies also work.

Example 7, p196.

$$R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1$$

$$R_{DC} = R_{FX} + R_{FC} + R_{FC}R_{FX}$$

Taking the standard deviation (or variance if squared):

$$\sigma(R_{DC}) = \sigma(R_{FX} + R_{FC} + R_{FC}R_{FX})$$

If $\sigma(R_{FC}) = 0$,

$$\sigma(R_{DC}) = \sigma(R_{FX} + R_{FC}R_{FX})$$

$$\sigma(R_{DC}) = \sigma(R_{FX}(1 + R_{FC}))$$

$$\sigma(R_{DC}) = (1 + R_{FC})\sigma(R_{FX})$$

Minimum-variance hedge ratio. Finds the optimum cross-hedging ratio. Let y_t be the percentage changes in the value of the asset to be hedged and x_t be the percentage change in the value of the hedging instrument. Both y_t and x_t are in the investor's domestic currency. Then minimize ϵ_t :

$$y_t = \alpha + \beta x_t + \epsilon_t$$

So, β is the hedging ratio. This means,

$$\beta = \frac{Cov(x, y)}{\sigma^2(x)} = \rho(x, y) \frac{\sigma(y)}{\sigma(x)}$$

Read example on p200. When looking at the hedging instrument x_t being e.g. $\% \Delta S_{GBP/CHF}$ and the $\beta = +1.35$ = hedging ratio, then long 1CHF assets

corresponding to y_t means short with 1.35CHF. This in turn means short the *GBP/CHF* forward, but long the *CHF/GBP* forward.

A 'direct' hedge between spot y_t changes and forward contract x_t changes, will usually result in $\beta \approx 1$, though this depends on how liquid the forward contract is, interest rates, maturity time, etc. However, can regress y_t as changes in $R_{DC} = (1 + R_{FC})(1 + R_{FX}) - 1$ and the forward contract x_t being R_{FX} (which effectively minimizes $\sigma^2(R_{DC})$, which will give a β usually different from 1 - this joint optimization means $\rho(R_{FC}, R_{FX})$ is minimized.

Indirect hedges. Cross and macro hedges are 'indirect' hedges which will result in more 'useful' β results. However, indirect hedges will result in basis risk from e.g. market risk (price correlations changing), credit risk, liquidity risk.

Minimum-variance hedge ratio example, p200. Remember the domestic currency = price currency here.

$$(1 + R_{FC})(1 + R_{FX}) - 1 = R_{GBP} = \alpha + \beta(\% \Delta S_{GBP/CHF}) + \epsilon$$

If the minimum variance hedge ratio $\beta > 0$, then foreign currency strengthening correlates positively with domestic currency earnings. So want to REVERSE this in a hedge to stabilize return variance. So, SHORT the GBP-CHF, which means LONG the CHF-GBP.

The cost of the hedge is focussed on the 'core'. For a manager with a long exposure to a currency, the cost of this "core" hedge will be the implicit costs of a short position in a forward contract (no upside potential, possible negative roll yield) or the upfront premium on a long position in a put option. Either of these two forms of insurance can be expensive, but can be defrayed by writing options, varying X, varying notional amounts, using exotic features like knock-ins or knock-outs or digital features.

Additional points

- If an investor wants to make a sudden withdrawal from the fund, and the fund is completely currency-hedged, then just leave it until the withdrawal.
- Using a deep OTM put can be better on a relative price-change basis than a slightly OTM put or ATM put, as the relative price changes are larger.

Emerging market currency exposures are thinly-traded with fewer derivatives, particularly between EM-EM currencies, so any small change can lead to major fluctuations. Moreover, governments often intervene. This leads to: (1) higher trading costs, (2) increased probability of severe liquidity issues. There are often currency crises.

EM II. The occurrence of currency crises can also affect hedging strategies based

on forward contracts. Recall that hedging a long exposure to a foreign currency typically involves selling the foreign currency forward. However, when currencies are under severe downward pressure, central banks often react by hiking the policy rate to support the domestic currency. But recall that the higher interest rates go in a country, then, all else equal, the deeper the forward discount for its currency (enforced by the arbitrage conditions of covered interest rate parity). Having to sell the currency forward at increasingly deep discounts will cause losses through negative roll yield and undermine the cost effectiveness of the hedging program.

EM III. Return distributions. Fatter tails in probability distributions, and a negative skew compared to normal distribution.

Non-deliverable forwards NDFs. Often used when there are capital controls in one country. So long or short a forward to a non-controlled currency, usually USD. Basically forward contract, but no physical cash exchanged; instead, there's just netting off and then conversion to the USD. Credit risk of NDF is typically lower because of netting off and using the USD. Priced with tail risk and not consistent with covered interest parity because of capital controls.

Additional points

- Fixed income PFs typically associated with hedge ratios close to 100%, and more strict compared to equity PFs.
- EM currency exposure = use active currency management. EM usually have higher carry trade yields and volatile FX rates.

Speculative traders versus hedgers.

- Speculative volatility traders want to be net short volatility because most options expire out of the money. They would want to flip their position and be long volatility ahead of the volatility spikes, but these episodes can be notoriously difficult to time. Most hedgers, in contrast, typically run option positions that are net long volatility because they are buying protection from the unanticipated price volatility.
- Speculative traders can take a position. Hedgers are just protecting their existing positions.
- Speculative traders can take either side of trades, while hedgers are just protecting their positions.

Additional points

- Rebalancing and rolling over mismatched hedges is a dynamic hedge - it can be comprised of static hedges each month however.

- When rolling over a forward or future, there is payment to settle the previous hedge by reversing it and netting it. However, there is no cash inflow or outflow THAT DAY for the NEW HEDGE because it will happen later.

6 Portfolio management pathway

6.1 Index-based strategies

Three requirements for index to be basis for equity investment strategy: (1) rule-based, (2) transparent, (3) investable. Index investing. Seek market return (CAPM beta) but not outperformance (alpha). Requires constant trading to match an index.

Factor investing. Can be active or passive, just as long as using factors. Passive factor investing = smart beta (cheap like passive but some active-like features). Factors are more than just beta, e.g. Fama-French factors like size, BTM. Can overweight or underweight them (factor tilts). Factor rotation - changing exposures to different factors.

Index-based factor strategies. Index investing where the index is factor-tilt index. Active management up-front but not too much continuously. Transparent. But susceptible to replication, and riskier than broad market equity indices.

Types of index strategies:

- Return-oriented strategies. Dividend-yield strategies, momentum, fundamentally-weighted.
- Risk-oriented strategies. Reduce downside volatility and risk. Strategies: volatility weighting (yes this is a strategy) where all assets are weighted by inverse of relative price volatility. Minimum variance investing - use MVO to minimize volatility of returns based on historical price returns. Risk-weighting: based only on historical data.
- Diversification-oriented strategies. Equal-weighted indices. Maximum-diversification strategies.

Contrast **market cap-weighted strategies** - most common passive strategies, to **fundamental weighting strategies**. Market cap-weighting leads to large-cap companies moving the most - even if their price moves slightly, since they're so big, their weighting changes a lot. Using fundamental value - book value, cashflow, revenue sales, dividends, employee count - and taking the percentage of a company's statistics of these as a proportion of the entire index sum of companies, and then taking a simple average of the proportions, gives fundamental weighting. This is less susceptible to price movements and investing in it outperforms capitalization-weighted indices.

Other weightings and strategies. Weight by size of dividend yield. Momentum, which can be determined in various ways. Inverse of volatility. Diversification and equal-weighting (low single-stock risk).

Pooled investments. Mutual funds - buy from advisor, funds marketplace, from fund.

- ETFs - unlike with traditional open-end mutual funds, ETF shares can be bought by investors using margin borrowing.
- Investors can take short positions in an ETF and closed-end MFs.
- ETF has greater tax efficiency generally: while the ETF can buy and sell securities to adjust the fund constituents (e.g. to track a recently-reconstituted index), this will trigger capital gains or losses; instead, the ETF can wait for APs to redeem ETF units (then the ETF can GET RID OF in-kind low-basis holding baskets to the APs) or ETFs can wait for APs to ask the ETF to create new ETF units (then the ETF can ADD the new securities it wants to buy to the creation basket) - so the in-kind redemptions and creations mean no tax consequences. MFs must buy and sell securities, trigger capital gains which are passed onto everyone in the fund.
- The ETF-AP transaction is not taxable, and if the AP redeems ETF units and receives low-basis securities, then the AP's 'basis reference' resets to the FMV, so when the AP sells those securities, they do not pay the 'embedded basis' that the ETF was holding. So, it is **tax avoidance**.
- Even though it is tax avoidance, in the end the investor pays capital gains when they sell the ETF shares. The ETF units, because of the tax avoidance, has 'embedded capital gains', and the investor ends up paying for this (unless they die and the units are inherited - in which case the capital gains are reset to 0). So, this is a system of **tax deferral**.
- ETF disadvantage if market is illiquid, but this is also true for closed MFs.
- There are factor-based ETFs (15-20%) while most of rest are index-based ETFs.

Buffering is a rules-based adjustment mechanism used in index replication strategies. When a stock moves just outside the criteria for inclusion in an index (e.g. market cap, sector classification), buffering allows the portfolio manager to wait before making a trade — instead of immediately buying or selling the security. Reduces tracking error.

Derivatives-based approaches. For ST, often faster and more liquid than cash-based purchases to get exposure. But for LT, better to use cash-based (derivatives have time limit).

Derivative overlays. Use derivatives to adjust PF. A **completion overlay** is to adjust the beta. A **rebalancing overlay** is to adjust policy weights. A **currency overlay** is to hedge currencies.

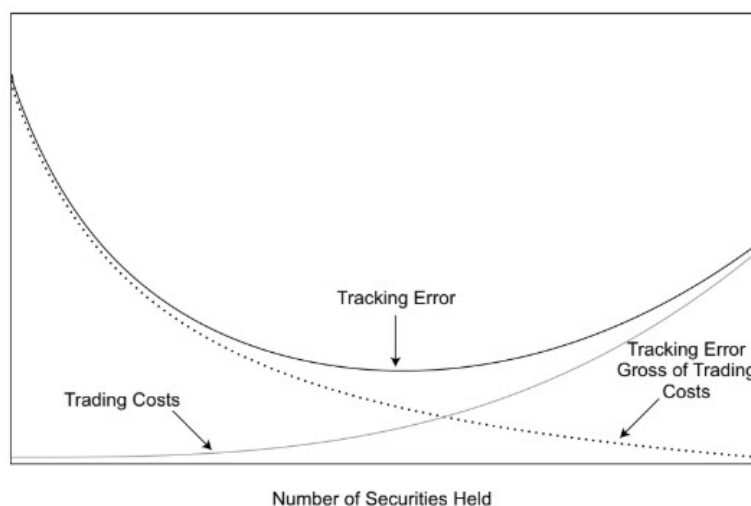
Trading notes. (1) Program trading is strategy of buying and selling many stocks simultaneously. OMS - order management system. (2) MOC (Market on close) orders usually for index PFs, as the PFs are measured against indices at the end of the day.

Additional points about derivatives.

- Even a small change in underlying can lead to big changes in margin requirements for futures.
- Equity swaps. Often used to reduce taxes on equity returns. Dividends and capital gains for equity, but equity swap returns are often treated as interest (lower tax rate). But very illiquid, though can OTC almost anything. More frequent the settlements, the lower the counterparty risk.

Building equity index PFs approaches: full replication, stratified sampling, optimization. Also: synthetic replication and-or OTC derivatives. Can blend approaches.

- Full replication. Matches index. But asset size has to be sufficient, and sufficient liquidity. Generally, having more of the equities of the index reduces tracking error, but adding more stocks that are smaller and less liquid will increase transaction costs and investor can begin to affect the smaller stock prices upon trading. Rising transaction costs can lead to increased tracking errors. So there is an optimum.



- Stratified sampling. Along country, industry, etc. lines often using standard classifications. Mutually exclusive and exhaustive stratifications, and can stratify along multiple dimensions. Often start with largest securities (which are the least-costly) and then adjust to weight by stratifications.
- Optimization. Mean-variance optimization using constraints, e.g. number of securities, transaction costs, covariances, to minimize e.g. lowest tracking error, or lowest volatility (or make total volatility = benchmark volatility). Post-optimization, can delete lowest-weighted stocks (often having high transaction costs that exceed diversification benefits), and delete positively correlated stock from pairs of such stocks. Optimization only lasts for so long, so have to re-optimize as conditions change - this is costly.

Tracking errors. Excess return = $R_p - R_b$ and tracking error is the SD of excess return. An index with a large number of equities will have a larger tracking error because harder to fully replicate. Other tracking error sources: transaction and commission costs (note: commission costs higher for active managers, suspected of having informational advantage), and cash drag (equitize using, e.g. futures contracts).

$$R_p - R_b = \text{Active factor risk \%} + \text{Active security selection \%}$$

Answer Feedback: Correct. The percentage of excess return arising from active factor weighting is 18.18%. In comparing the weights between the fund and the benchmark, the factors with different weights are Growth and Quality. The total contribution to the return caused by active factor weighting is:
 (Underweighting of the Growth factor + Overweighting of the Quality factor) = $\frac{(-0.24\% + 0.20\%)}{-0.22\%} = \frac{-0.04\%}{-0.22\%} = 18.18\%$. The fund's holding of Momentum securities was less than the benchmark's (24 versus 30), and thus, the fund incurred active security selection risk. But it did not incur active factor risk, since the factor weight is the same as that of the benchmark.

	River Valley Fund		Benchmark		Return Contribution		Attribution Analysis	Difference Due to:	
Factor	Weight (A)	Return (B)	Weight (C)	Return (D)	River Valley (A) x (B) = (E)	Benchmark (C) x (D) = (F)	Total Effect (E) - (F) = (G)	Factor Weight	Security Selection
Growth	0.22	7.9%	0.25	7.9%	1.74%	1.98%	-0.24%	-0.24%	
Value	0.19	5.2%	0.19	5.2%	0.99%	0.99%	0.00%		
Quality	0.29	6.7%	0.26	6.7%	1.94%	1.74%	0.20%	0.20%	
Momentum	0.30	3.9%	0.30	4.5%	1.17%	1.35%	-0.18%		-0.18%
Total	1.00		1.00		5.84%	6.06%	-0.22%	-0.04%	-0.18%

Securities lending. Can help offset tracking errors by offsetting transaction costs. Has risks however, e.g. counterparty risk.

Investor activism. Although activist funds are well known for trying to effect change through voting, index-based managers could also - they are shareholders and improving corporate governance can improve their client's returns. However, has to be balanced against costs of getting information to vote.

HHI index. Since

$$HHI = \sum w_i^2$$

then the effective number of stocks, i.e. equi-number, is

$$HHI = \sum \left(\frac{1}{N} \right)^2 = N \left(\frac{1}{N} \right)^2 = \frac{1}{N}$$

so, the number of effective stocks is the reciprocal of the HHI.

6.2 Active equity investing: strategies

Fundamental (discretionary) versus quantitative (systematic).

- Fundamental. Overweight or underweight v benchmark. Top-down and/or bottom-up. Concentrate in-depth on a smaller number of individual stocks. Monitored and adjusted at any time when too high or low.
- Quantitative. Optimization of e.g. alpha or information ratio. Focus on factors, spread bet across many stocks. Adjusted or rebalanced at programmed intervals.

Factors in investing. Rewarded factors are empirically tested, unrewarded are not.

Bottom-up strategies. Often focus on: (1) business model and branding, (2) competitive advantages, and (3) company management and corporate governance. Bottom-up strategies often characterized as: (1) value or (2) growth oriented, based on relative value (e.g. P-B or P-E ratio) to peers.

Value investing.

- Relative value. Default value investing.
- Contrarian investors. While value investors usually look at fundamentals of relative value, contrarian investors look at behavioral aspects like sentiment and sharp price movements.
- 'High-quality value' = Warren Buffet, which is value but with good fundamentals. Value below intrinsic value (which is NOT the same as relative value, which is value compared to peers).
- Income investing = good dividends, which often correlates with good stock price.
- Deep-value investing = extremely low values. Often financially distressed companies, where deep-value investors have informational advantage (e.g. specialize in restructuring), which is similar to 'restructuring and distressed investing'.
- Special situations investing = investing in companies with corporate events such as spinoffs and M&As.

Growth investing. Growing faster than industry or market by revenues, earnings, cash flow. High-quality companies usually. Look at y-y growth of revenue, income. Less emphasis on P-B, P-E, D ratios.

GARP (growth at a reasonable price) is a sub-discipline within BOTTOM-UP growth investing. This approach is used by investors who seek out companies with above-average growth that trade at reasonable valuation multiples, and is often referred to as a hybrid of growth and value investing. Many investors who use GARP rely on the P/E-to-growth (PEG) ratio—calculated as the stock's P/E divided by the expected earnings growth rate (in percentage terms)—while also paying attention to variations in risk and duration of growth.

Note: value metrics (e.g. P-E ratio) are very different from growth metrics (e.g. income growth).

Note2: even if a fund uses value and relative value, its P-B or P-E ratios might be high compared to the benchmark if it is leaning heavily to high-P sectors, for instance.

Top-down strategies. Geographical, sector and industry, volatility, investment themes (e.g. technologies). Often use futures, ETFs, swaps. If bottom-up strategies lead to inadvertent portfolio drift from policy weights, can use portfolio overlays (e.g. derivatives) to adjust back.

Factor-based strategies. Some rewarded, many unrewarded factors. Backtest new factor candidates - even if it works, must make sense; otherwise, it's data mining. Most widely used factors are **hedged portfolio approach** like Fama-French, using quantiles, where stocks are equally or market weighted within each quantile, and then short lowest quantile (e.g. increased debt the most that quarter) and long highest quantile (e.g. reduced debt the most). So, they're long-short portfolios. Note disadvantages: middle quantiles not used, linear relationships assumed, stock positions are concentrated, must be able to short stocks, and hedged portfolios are not 'pure' factor exposures.

Factor tilts. Long-only with exposure to a factor. Tracks benchmark index but also factor exposure.

Factor-mimicking portfolio. Theoretically exposed to one factor and no other factors. But expensive to construct.

Other factors. Value, momentum (note: complex empirical results, and better to neutralize sector exposures), growth, and quality (e.g. accruals).

Example 5. For bond yields, normalize them by subtracting the moving average (e.g. over last 12 months). And then regress factor returns against the normalized bond yields.

Activist strategies. Target companies that could be improved - less about restructuring companies (which is more for other funds). Many tactics, but also countertactics like founder's shares, poison pills to issue shares at deep discount to dilute votes, staggered boards, etc. Market usually reacts favorably to targeted companies.

Other active strategies: statistical arbitrage. Use technical analysis, time-series analysis and econometrics, ML. Take advantage of mean reversion, e.g. pairs trading.

Other active strategies: event-driven. Take advantage of corporate events like M&A, restructuring, share buybacks, special dividends and spinoffs.

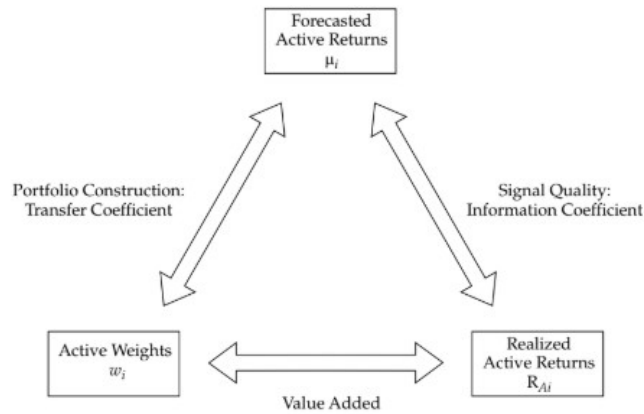
Fundamental active investment process. Define investment thesis, prescreen, analyze, forecast, construct forecast portfolio, rebalance, add buy and sell disciplines. Sell disciplines are when the stock in a portfolio is sold because its price exceeds its estimated intrinsic value.

Pitfalls in active investment. **Cognitive errors:** confirmation bias, illusion of control, availability bias, overconfidence bias. **Emotional bias:** loss aversion (holding onto losers, selling winners too soon) and regret aversion bias (inaction and not taking action, herding behavior). **Value trap:** low price stock but could fall lower; more common in fundamental investing but also in quantitative strategies where there's no future profitability considered. **Growth trap:** good company results but might be overpriced already or will perform poorly - growth trap refers to growing EARNINGS not PRICE.

Quantitative active investment strategy.

- Quant process: define market opportunity, acquire and process data, find factors and combine factors in multifactor model (some factors will be colinear so should be removed), back-test, evaluate. Mind risk modeling (Cov matrices) and trading costs, liquidity, how long the model advantage will last.
- Want to find relationship between PF returns next period and factor returns this period. This is also what is backtested.
- Information coefficient IC. Correlation of factor exposures and holding period returns. Better than hedging portfolios like Fama-French because it uses all the data (even the middle quartiles). Pearson IC = ρ . Spearman rank IC = uses ordinal rank = less susceptible to outliers.
- Quant crowding. When many quant traders adopt same approach, then less gains, e.g. 'crowding coefficient' (regress shorts on price momentum). So, use different factors!

- Compared to fundamental investment. Quant strategy returns generally not correlated with fundamental approaches, so perhaps some diversification benefits. However, quant requires more data, infrastructure, model development. Be careful of quant crowding and look-ahead and survivorship biases.



Quantitative investment pitfalls.

- Survivorship bias: When back-tests use only those companies that are currently in business today, they ignore the stocks that have left the investment universe due to bankruptcy, delisting, or acquisition. This leads to overly optimistic results and can cause investors to arrive at incorrect conclusions.
- Look-ahead bias: This comes from using information that was unknown or unavailable at the time an investment decision was made.
- Data mining and overfitting: Data mining refers to automated computational processes for discovering patterns in large datasets. It can introduce a bias that results in model overfitting. Model overfitting is an excessive search and analysis of past financial data to uncover patterns and to conform to a pre-determined model for potential use in investing.
- Transaction costs: Trading is not free. Transaction costs can easily erode returns significantly.
- Constraints on turnover: Managers may face constraints on turnover, which may impact their ability to capture a model's expected excess return in a live trading process.
- Availability of positions: If a position is not readily available to purchase or sell short, it can also limit the ability to capture a model's expected excess return.

Style classification. Identify investment styles of active managers. PF is better by including employing managers with different investment styles. Two style analyses: **holdings-based** and **returns-based**. Use manager's self-identified styles as a starting point. Holdings-based more accurate as it's bottom-up. Note: limited availability on derivatives markets means hard to determine how much derivative overlay is used in portfolios.

Holdings-based. Bottom-up, evaluate style of each holding and add them up. Evaluate style using style indices from BBG, etc. but can make own - often based on splitting total market cap into value, core, growth; and by size.

Returns-based. Many PMs don't disclose details of PFs, so use returns-based.

$$r_t = a + \sum \beta R_i + e_t$$

where r is the return on the fund, β is the fund's style exposure, and R is the return of the style index, where the style index series is from an information provider.

6.3 Active equity investing: portfolio construction

Using active return,

$$R_A = \sum \Delta w_i R_i$$

PMs generate active returns by: (1) strategically adjusting rewarded risks (betas and factor exposures) different from benchmark, (2) tactically adjusting active weights to generate alpha, (3) idiosyncratic risk.

$$R_A = \sum (\beta_{p,i} - \beta_{b,i}) F_i + (\alpha + e)$$

where betas are factor sensitivities for the PF and benchmark, and F is return of factor, and α is the alpha (skill) and e is idiosyncratic return (luck). Both quant and fundamental investing will use this, though quant will use explicitly and fundamentals will use implicitly. Fundamentals, because their PFs will be fewer stocks and individual-stock-research based, will have more alpha attribution.

Alphas. Top-down managers have alphas usually from factor TIMING. Bottom-up managers have alphas intentionally or as a by-product of stock picking. If the R^2 of the return driver equation is low, it means even if there are factor returns and factor exposures, that most of the returns are 'unknown' = $\alpha + e$.

Portfolio construction depends on: (1) rewarded factor weightings, (2) alpha, (3) position sizing, (4) breadth of expertise. The market PF has $\beta = 1$ and 0 exposure net to all factors.

1. is about changing the betas over the long term. More used by systematic managers.

2. is about (a) timing changes in rewarded factors, (b) exposure to unrewarded (thematic) factors like regional, sector, commodity prices. Factor timing is more likely with top-down (macro) factors versus bottom-up (security) factors, and factor timing is more likely with discretionary managers versus systematic managers.
3. is about how concentrated the PF should be, e.g. can get same average beta exposure whether using 20 (stock picker) or 200 (factor-oriented manager) securities, but the former is more concentrated and has a greater impact on $\alpha + e$ and therefore idiosyncratic volatility (active risk). Active risk is

$$\sigma_A = \sqrt{\frac{\sum (R_{A,t})^2}{T-1}}$$

Systematic managers have less concentrated positions than discretionary managers.

4. is about number of independent decisions from fundamental law of active management:

$$E(R_A) = IC\sqrt{BR}\sigma_A(TC)$$

A PM considering many different metrics has a greater BR than one that doesn't. Systematic managers have lower idiosyncratic risk than discretionary managers.

Active share

$$AS = \frac{1}{2} \sum |Weight_{p,i} - Weight_{b,i}|$$

If the PF has a security that are not in the benchmark at all, then

$$|1 - 0| + |0 - 1| = 2$$

and so counts as 2. So, the Active Share is 0 for a portfolio that matches the benchmark and 100% for a portfolio that shares no investments with those of the benchmark. If active share is 80%, then 20% was invested in the same way as the benchmark. If two portfolios are managed against the same benchmark (and if they invest only in securities that are part of the benchmark), the portfolio with fewer securities will have a higher level of Active Share than the highly diversified portfolio.

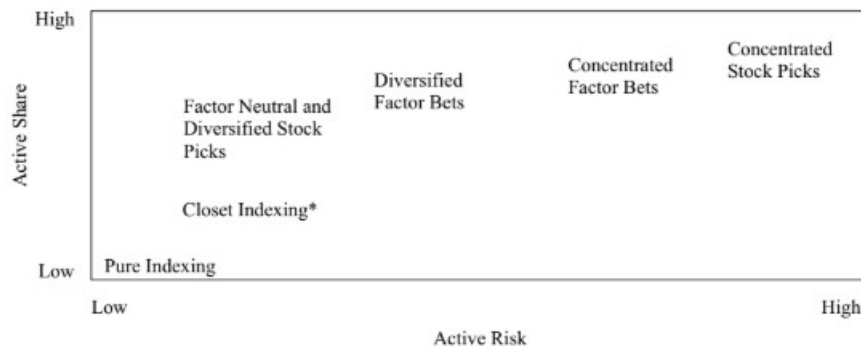
Active risk (relative to a benchmark). Tracking error or tracking risk.

$$\sigma_{R_A} = \sigma_A = \sqrt{\sigma^2(\sum (\beta_{p,i} - \beta_{b,i})F_i) + \sigma_e^2}$$

Active risk increases when a PF becomes more uncorrelated with its benchmark. Note: cash is uncorrelated with most equity benchmarks, so adding cash to a PF will INCREASE active risk.

Active share and active risk relationship. Higher active share generally means

higher active risk. However, this also depends on the cross-correlations encoded in the variance-covariance matrix. If a bank stock A is overweighted and bank stock B is underweighted, this has almost no effect on active risk because bank stocks are highly correlated. However, if a bank stock A is overweighted and a auto producer stock is underweighted, then this will affect active risk more because they are not highly correlated. Characterize PMs along active share versus active risk:



Active share and active risk are positively correlated, but this doesn't say anything about active RETURNS. However, lower active share = investors have expectation of lower fees.

Niche equity strategies. Hard to calculate active share versus a benchmark. Statistical arbitrage, event-driven, activist investing. Have high active risk from high idiosyncratic risk though.

Objectives and constraints. Objective functions are explicit (including making explicit forecasts) or implicit. Constraints are relative (wrt to a benchmark) or absolute (without).

Objective functions. Example: maximize risk-adjusted returns either using predicted active risk (information ratio) or predicted volatility (Sharpe ratio). Can optimize with respect to risk, e.g. portfolio volatility, downside risk, max diversification, drawdowns - usually risk-optimized portfolios have low beta and value factor orientation. Can optimize with respect to anything, e.g. weighted average of size, value, momentum. Can make up own optimization functions.

Constraints. Limits on geography, sector, industry, transaction costs, specific factors.

PF styles and strategies examples. Note: risk factors = beta factors.

- Concentrated stock-picker versus factor-better.
- Rewarded factors, no factor timing, LT. Diversified to minimize idiosyncratic risk and active risk is low.

- Sector shifter, no individual security selection, lots of sector exposure deviations. Tactical exposures to sectors, which are unrewarded risks, success will be mostly alpha. Active risk from sector weights will be larger than active risk from security weights.
- Systematic and based on individual securities = quant and bottom-up.
- If timing, and using technical and econometric forecasts. This is not systematic; it's discretionary despite the technical and econometric forecasts.
- Some PFs can be security-diversified, but factor-concentrated. Example: timing sector exposures.

Absolute versus relative risk. Determine whether absolute or relative risk is better depending on the strategy (e.g. if benchmark is cash + 5% risk measure should be total return volatility - absolute risk, if benchmark is cap-weighted index then active risk - relative risk).

Absolute risk (not relative to a benchmark). Rises or falls depending on what asset is added to what portfolio. Total variance is (note: the weights w can be factor coefficients!):

$$V_p = \sum_i \sum_j w_i w_j \text{Cov}(r_i, r_j) = \sigma_r^2$$

Do Example 5, n2 on p158. Contribution of asset to portfolio variance is:

$$CV_i = \sum_j w_i w_j \text{Cov}(r_i, r_j) = w_i \text{Cov}(r_i, r_p)$$

Regarding the last equation, note since

$$x \text{Cov}(y, z) = \text{Cov}(xy, z) = \text{Cov}(y, xz)$$

that

$$\begin{aligned} & w_1 w_1 \text{Cov}(r_1, r_1) + w_1 w_2 \text{Cov}(r_1, r_2) + w_1 w_3 \text{Cov}(r_1, r_3) \\ &= w_1 (w_1 \text{Cov}(r_1, r_1) + w_2 \text{Cov}(r_1, r_2) + w_3 \text{Cov}(r_1, r_3)) \\ &= w_1 (\text{Cov}(r_1, w_1 r_1) + \text{Cov}(r_1, w_2 r_2) + \text{Cov}(r_1, w_3 r_3)) \\ &= w_1 \text{Cov}(r_1, \sum_j w_j r_j) = w_1 \text{Cov}(r_1, r_p) \end{aligned}$$

Proportion of total portfolio variance contributed by an asset is:

$$CV_i / V_p$$

Note the returns (r) can be sectors, industries, portfolios with a factor exposure. It can be negative (which means that element, factor, etc. is **diversifying** because it is negatively correlated with other elements, factors, etc.). An element,

factor, etc. risk contribution of $> 100\%$ is like that because of the existence of negative CV/V , and it must be that $\sum CV/V = 1$ - so it's basically normalization. If w 's are factor exposures (β), then CV/V is **factor risk contribution**.

Minimize portfolio risk not explained by PM's specialty, e.g. sector exposure or factor exposure by minimizing the second last term:

$$V_p = Var(\sum(\beta_{i,p}F_i)) + Var(e_p)$$

The first additive term is R^2 . If the manager's portfolio were the market portfolio, all the variance of the portfolio returns would be explained by a beta of 1 to the Market factor. Idiosyncratic risks would be fully diversified. However, as we move away from the market portfolio, total portfolio variance will be influenced by other factor exposures and other risks unexplained by factors.

Relative risk. Variance of the portfolio's active return:

$$AV_p = \sum_i \sum_j (w_i - w_{bi})(w_j - w_{bj})Cov(r_i - r_{bi}, r_j - r_{bj})$$

where r_{bi} is the return of the benchmark component corresponding to asset i . The contribution of each asset to the portfolio active variance is:

$$CAV_i = (w_i - w_{bi})Cov(r_i - r_{bi}, r_p - r_b)$$

If you are assessing risk using a relative risk construct, you can no longer assume that a lower-risk asset reduces active risk or that a higher-risk asset increases it. In fact, depending on the composition of the benchmark, a lower-risk asset could increase active risk whereas a higher-risk asset might reduce it.

Note: if there are only two assets in equal weights in an index, then correlations of active returns between the benchmark and a and between the exact same benchmark and b are both -1. Outperformance of one means underperformance of the other.

$$r_B = 0.5r_a + 0.5r_b$$

$$r_{a,active} = r_a - r_B = r_a - 0.5r_a - 0.5r_b = 0.5(r_a - r_b)$$

$$r_{b,active} = r_b - r_B = r_b - 0.5r_a - 0.5r_b = 0.5(r_b - r_a) = -r_{a,active}$$

So,

$$\rho(r_{a,active}, r_{b,active}) = -1$$

Moreover, cash is positively correlated to a (b) and negatively correlated with b (a). Cash tends to move opposite to whichever index is outperforming the benchmark. Because the benchmark is half-half a and b, when a is above average, b is below average. Hence, cash's active return will be positively correlated with the underperforming index (the one "down") as they are both 'pushed down' from the benchmark, and negatively correlated with the outperforming

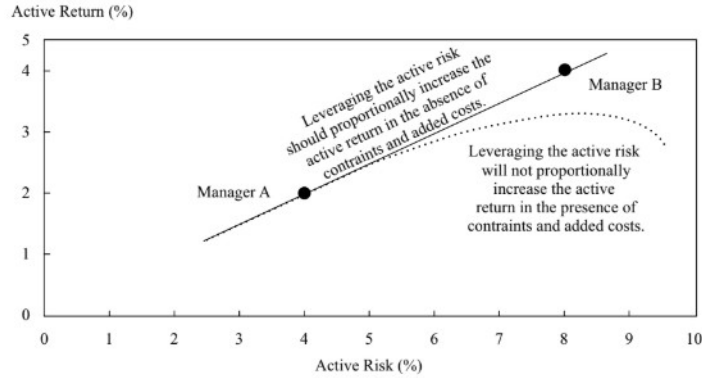
one.

Information ratio. Active return to active risk.

$$IR = \frac{R_A}{\sigma_A}$$

Determining risk level. Sometimes, use absolute risk target, other times use absolute risk target. However, sometimes the risk target limits the PF or the risk target is high such that PF can't fully use it:

- Portfolios may face implementation constraints that degrade the information ratio if active risk increases beyond a specific level. For instance, not enough liquidity, can't leverage (to increase overweighting) or short (to increase underweighting beyond 0 of the stock).



- Portfolios with high absolute risk targets face limited diversification opportunities, which may lead to a decrease in the Sharpe ratio. Twice absolute risk will not lead to twice the return from the Markowitz frontier - run out of diversification opportunities.
- There is a level of leverage beyond which volatility reduces expected compounded returns. There is a linear relationship between risk and return in a one-period setting if can borrow or lend at risk-free rate.

Geometric and arithmetic return relationship.

$$R_g = R_a - \frac{\sigma^2}{2}$$

Derivation. First derive:

$$R_g = E(\ln(1 + r))$$

from

$$(1 + R_{total}) = (1 + r_1)(1 + r_2) \dots (1 + r_T)$$

$$1 + R_g = [(1 + r_1)(1 + r_2) \dots (1 + r_T)]^{1/T}$$

$$\ln(1 + R_g) = \frac{1}{T} \sum \ln(1 + r_t)$$

Since $\ln(1 + R_g) \approx R_g$ if R_g is small and the RHS can be re-expressed as an expectation:

$$R_g \approx E[\ln(1 + r)]$$

Returning to the derivation. If r is small (as in monthly or annual returns), we can approximate the natural log:

$$\ln(1 + r) \approx r - \frac{1}{2}r^2$$

which are the first two terms from the Maclaurin series about $x = 0$. Then,

$$E[\ln(1 + r)] \approx E[r] - \frac{1}{2}E[r^2]$$

and recall

$$\begin{aligned} \text{Var}(r) &= E[r^2] - (E[r])^2 \\ E[r^2] &= \text{Var}(r) + (E[r])^2 = \sigma^2 + r^2 \end{aligned}$$

and for small r^2 ,

$$E[r^2] = \text{Var}(r) + (E[r])^2 \approx \sigma^2$$

Substituting,

$$\begin{aligned} E[\ln(1 + r)] &\approx E[R_a] - \frac{1}{2}\sigma^2 \\ R_g &\approx R_a - \frac{1}{2}\sigma^2 \end{aligned}$$

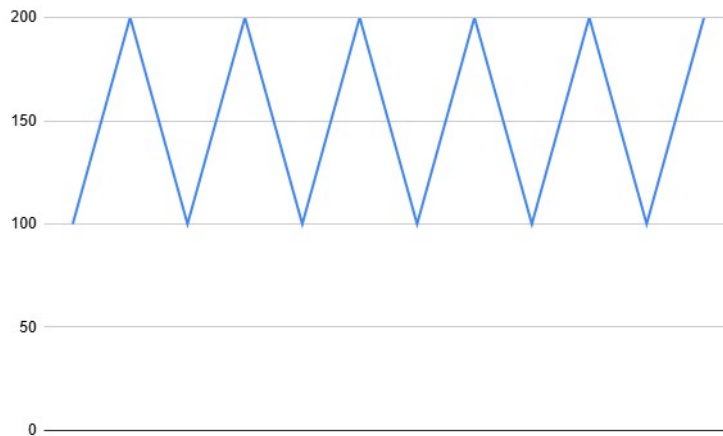
With leverage by x times (so, x is a constant), xR_a and

$$\begin{aligned} \text{Var}(xR_a) &= x^2\text{Var}(R_a) \\ \sigma(xR_a) &= x\sigma(R_a) \end{aligned}$$

With leverage, the R_a rises but because of the squared term, the variance rises faster. So, leverage leads to stagnating geometric returns, even if the arithmetic returns are high. This is due to compounding drag: e.g. starting with 100, an increase of 50% ($\times 3/2$) is 150, and a subsequent fall of 50% ($\times 1/2$) is 75, and if this process is repeated, there is a gradual decline:



The intuition is that plus-minus percentages are non-linear (because in the above example, $3/2$ is not a reciprocal of $1/2$). A 50% decrease ($\times 1/2$) requires a 100% recovery ($\times 2/1$), and a 50% gain ($\times 2/3$) requires a 33.3% loss ($\times 2/3$). There is a re-basing effect. If starting from 100, there is instead a 100% increase and a 50% fall, then the trajectory is:



So, greater volatility (σ) leads to a drag on the geometric return. It gets worse if leverage transaction costs are incorporated.

With increasing leverage, the Sharpe ratio also falls quickly. The return rises to a point, and then volatility (the denominator) keeps rising.

Additional risk measures. Heuristic (intuitive rules, e.g. cap on liquidity, sector exposure, transaction costs) versus formal (volatility, VaR, skew, drawdowns) which are more statistical. Risk measures often not in IPS because hard to forecast.

Likely risk measures.

- Long-short equity likely to target a volatility within a range.
- Benchmark-agnostic targeting absolute return will likely target active risk and-or drawdowns.
- Hard to find risk measures for PFs with low number of securities, as the outcomes are wide.

Implicit costs. Price movement and market impact from trades can erode alpha.

Capitalization and trading volume. Market capitalization distribution is skewed, dropping quickly. A smaller absolute volume of small-cap stocks are traded each day, but the proportion of small-cap market-cap is much bigger than for large-cap stocks.

- Small-AUM fund will have few real position and trading limits. However, large-AUM funds will likely reach some of these limits - so they may need to change the limits, trade less, rebalance less, or hold a different capitalization distribution of stocks.
- Small-cap (not to be confused with small-AUM funds) funds. Hard to concentrate positions with trading limits because their trading volumes are small. May need to reduce AUM, diversify more, limit turnover, etc. to reduce trading costs.
- A fund limited to a small number of securities, i.e. a concentrated position, would quickly run into trading limits on small-cap stocks, so will probably need to trade large-cap stocks or increase holding period, or increase the number of holdings.
- Increase in AUM usually leads to difficulties in executing small-cap equities. Also, difficulties in momentum of small-cap stocks which requires a lot of turnover.

Trading limits

- Max position size = lesser of 10x index weight or index weight + 150 bps. Crossover point: $10w = w + 1.5\%$.
- If the max position is 1.5 million, and the minimum index weight is 0.015%, then the max AUM is 1.5 million divided by 0.015% to get 1 billion AUM.

Trading costs

- Cost of slippage. Slippage = difference between execution price and mid-point of bid-ask at time the trade was first entered. Includes effect of both

volatility-trend costs (costs with buying in a rising market and selling in a falling market) and market impact from trading. When a larger trade is executed in increments over multiple days, the estimate of market impact costs for later trades does not account for the impact of earlier trades on subsequent execution prices.

- Slippage costs are generally higher for small-cap stocks because of lower ADVs and thinner order books.
- Small-cap stocks have higher effective costs than large-cap stocks. Illiquidity is cyclical. Larger trading volumes have larger market impact.
- Opportunity costs.

Well-constructed PF. If two products have comparable factor exposures, the product with a lower absolute volatility and lower active risk will likely be preferred (assuming similar costs). If two products have similar active and absolute risks, the portfolios have similar costs, and the alpha skills of the managers are similar, the product having a higher Active Share is preferable, because it leverages the alpha skills of the manager and will have higher expected returns.

Note: low volatility PFs usually have large-cap, value, and quality orientations.

Notes on advanced factors.

- BAB. Betting against beta. Factor. Empirically find that SML line, which relates high beta to high returns, is very flat. Happens because investors overweight high beta stocks and there's not enough leveraging in the market. So, factor is about shorting high-beta stocks (deleverage to $\beta = 1$) and long low-beta stocks (leverage to $\beta = 1$). Long minus short = BAB factor.
- Low-beta stocks can be leveraged up to $\beta = 1$ because

$$\beta_p = w_1\beta_1 + w_2\beta_2$$

and leveraging the low-beta stock 1 means $w_1 > 1$ and borrowed funds is 2, i.e. $w_2 < 0$ since $w_1 + w_2 = 0$ in a closed system. Further, $\beta_1 < 1$ and $\beta_2 = 0$ since borrowed funds are just the riskless asset.

- The above linear relationship of betas is intuitive, but can be proved by

$$r_p = w_1r_1 + w_2r_2$$

$$\beta_p = \frac{Cov(r_p, r_m)}{\sigma_m^2} = \frac{Cov((w_1r_1 + w_2r_2), r_m)}{\sigma_m^2}$$

The denominator is

$$Cov((w_1r_1 + w_2r_2), r_m) = w_1Cov(r_1, r_m) + w_2Cov(r_2, r_m)$$

Substituting,

$$\beta_p = \frac{w_1 \text{Cov}(r_1, r_m) + w_2 \text{Cov}(r_2, r_m)}{\sigma_m^2}$$

$$\beta_p = w_1 \left(\frac{\text{Cov}(r_1, r_m)}{\sigma_m^2} \right) + w_2 \left(\frac{\text{Cov}(r_2, r_m)}{\sigma_m^2} \right)$$

$$\beta_p = w_1 \beta_1 + w_2 \beta_2$$

- Quality factor. Good financial health and business model. Long basket of high-quality stocks and shorting low-quality stocks. Quality minus junk. Look at ROE, ROA, margins, earnings, accruals, dividend stability, leverage, coverage ratios, liquidity ratios. Investors looking ST and overlooking safe stocks.

Alphas. Positive alpha = higher returns than risks taken. There can be negative alpha.

Long, long-short, market-neutral strategies. Long-short equity strategies. Can be market-neutral, which is 0 net market exposure, e.g. 100% long and 100% short. Long extension is like 130% long and 30% short - so it's long enhanced by some long-shorting.

- Long strategies. Less complex, safer, etc.
- Long-short. Shorting can reduce exposures to sectors, regions, factors if have a 'set' PF that is bought. This can free managers to work on their specialties. Need shorts to extract full value from market. Greater alpha.
- When shorting is involved, there is net exposure and gross exposure.
- Long extension. 'Enhanced active equity' strategies. More efficient exposure to factors. Greater alpha.
- Market-neutral. Removes exposure to certain factors, e.g. market but also possibly other factors. If however all risks and exposures are eliminated, it will generate no money (or negative with the transaction costs). Correlation of market-neutrality with other strategies is low. Examples of market-neutral strategies: pairs trading, statistical arbitrage.

Adjusting market exposure. Increase market exposure by buying higher-beta stocks, leveraging. Decrease market exposure by buying lower-beta stocks (and cash) and shorting.

Notes.

- Average of cross-correlations between different rewarded factors is close to 0 or negative, allocating across factors could bring significant diversification benefits.

- Adding ability to leverage and short = improve transfer coefficient.
- The market factor usually dominates risk, and being able to short means reducing exposure to market factor, which should lead to more balanced diversification of risk. This in turn should reduce tracking error (active risk).
- With market-neutrality, it's not that volatile so need much more leverage than usual to get the same results of absolute risk targets.
- There are 3 building blocks for portfolio construction: (1) alpha skills (including timing), (2) position sizing, and (3) factor weightings.
- Long-only strategies generally aren't limited by capacity (except if a large fund wants to long short-caps), but long-short limited by securities to borrow.
- Gross exposures. Long-short can be 100%, e.g. 50% long and 50% short.

6.4 Liability-driven index-based strategies

Immunization = structuring and managing a FI PF to minimize variance in realized rate of return and lock in CF yield (IRR) on PF.

Can immunize a single debt liability with a zero-coupon bond or futures, swap, or other derivative. A zero-coupon bond, while having greater initial capital outlay, doesn't have an initial margin requirement, no MTM settlements, no rollovers (e.g. because futures are only available for near-term contracts), no counterparty risk. Other derivatives, like an interest rate swap, a receive-fixed swap, still has counterparty risk and periodic settlements. Swaptions have a premium associated with them.

Managing the interest rate risk of a single liability.

Immunization. Assuming default probability of 0, minimize FI portfolio variance, where the variance is from future interest rate fluctuations. A zero-coupon bond held to maturity = perfect immunization, but often these aren't available.

Macaulay duration. Weighted average of cashflow PVs = when dividend reinvestment changes offset price changes when interest rates change. So, investor having horizon = MD is immunized.

Immunization example (p205).

Time	Date	Cash Flow	PV of Cash Flow	Weight	Time x Weight	Dispersion	Convexity
0	15-Feb-21	-200,052,250					
1	15-Aug-21	3,323,625	3,262,282	0.0163	0.0163	1.9735	0.0326
2	15-Feb-22	3,323,625	3,202,071	0.0160	0.0320	1.6009	0.0960
3	15-Aug-22	3,323,625	3,142,971	0.0157	0.0471	1.2728	0.1885
4	15-Feb-23	3,323,625	3,084,962	0.0154	0.0617	0.9871	0.3084
5	15-Aug-23	50,323,625	45,847,871	0.2292	1.1459	11.2324	6.8754
6	15-Feb-24	2,971,125	2,656,915	0.0133	0.0797	0.4782	0.5578
7	15-Aug-24	2,971,125	2,607,877	0.0130	0.0913	0.3260	0.7300
8	15-Feb-25	2,971,125	2,559,744	0.0128	0.1024	0.2048	0.9213
9	15-Aug-25	2,971,125	2,512,500	0.0126	0.1130	0.1131	1.1303
10	15-Feb-26	2,971,125	2,466,127	0.0123	0.1233	0.0493	1.3560
11	15-Aug-26	2,971,125	2,420,610	0.0121	0.1331	0.0121	1.5972
12	15-Feb-27	2,971,125	2,375,934	0.0119	0.1425	0.0000	1.8527
13	15-Aug-27	2,971,125	2,332,082	0.0117	0.1515	0.0116	2.1216
14	15-Feb-28	100,271,125	77,251,729	0.3862	5.4062	1.5434	81.0931
15	15-Aug-28	1,390,000	1,051,130	0.0053	0.0788	0.0473	1.2610
16	15-Feb-29	1,390,000	1,031,730	0.0052	0.0825	0.0825	1.4028
17	15-Aug-29	1,390,000	1,012,688	0.0051	0.0861	0.1265	1.5490
18	15-Feb-30	1,390,000	993,997	0.0050	0.0894	0.1788	1.6993
19	15-Aug-30	1,390,000	975,651	0.0049	0.0927	0.2389	1.8533
20	15-Feb-31	56,990,000	39,263,380	0.1963	3.9253	12.5585	82.4316
			200,052,250	1.0000	12.0008	33.0378	189.0580

- Get bond portfolio. Manually calculate total PF cashflows, then find IRR that will make PV cashflows = 0, i.e. $200052250 = 200052250$. The IRR x 2 (for semi-annual payments) will be higher than the weighted average (by market value, not par value) of bonds x YTM's (at time = 0) if the yield curve is upwardly-rising (instead of being flat).
- Note: if the yield curve is sloping upwards (because the coupon rates - and therefore the YTM's - of the longer-dated bonds are higher than the coupon rates - and therefore the YTM's - of the shorter-dated bonds), then the long-dated bonds will 'raise' the IRRs above the market-value-weighted average YTM of the bond portfolio, because the long-dated bonds have the longest maturity obviously, and this means their coupons are the most influential (the long-dated bond is distributing coupons from $t = 0 \dots T$, i.e. even while short-dated bonds are also distributing coupons). So the weighted average of YTM's is like averaging different lengths of a string, but the actual IRR should be more like averaging different lengths of SQUARES or RECTANGLES or other higher-dimensional objects. This 'pulls' the IRR towards the coupon rate or YTM of the long-dated bond.
- Weight = PV_t / PV_T . Sums to 1.
- Sum of (time x weight) divided by 2 (for semi-annual payments) = MD. This will be slightly HIGHER than the quick rule of weighted average MD's of the individual bonds in the bond PF, if the yield curve is upwardly-rising - the higher IRR (like coupons) are associated with cashflows occurring later, so MD's are higher.

- Portfolio dispersion. Like the variance of MD = $(t - MD)^2 \times weight$. Total dispersion = sum = 33.03784/4 (divide by $2^2 = 4$ due to periodicity squared = semi-annual squared).

- Convexity.

$$\frac{\sum t(t+1)(weight)}{(1+IRR)^2}$$

The calculated, actual value when ANNUALIZED is slightly higher than the weighted average (by market value) of the convexities of the bond PF because of yield curve sloping up. More of the heavy cashflows and weights are near maturity, and the IRRs (like coupons) are associated with cashflows occurring later.

- Portfolio convexity statistic = convexity

$$\frac{MD^2 + MD + Dispersion}{(1+IRR)^2} = Convexity$$

where the squared in the denominator is for semi-annual bond payments.

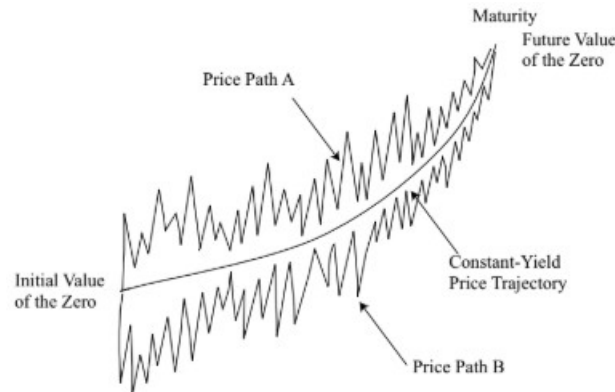
- **Structural risks** of immunization strategies. From potential for shifts and twists in yield curve. Use portfolio dispersion and convexity to assess structural risks.
- Demonstrating that interest rate changes = Using the IRR, find the NPV of each of the cashflows to the MD date, Feb 2027. Summing all of the modified cashflows in this way will = sum of unmodified PV cashflows. This equation says that **minimizing PF dispersion is the same as minimizing PF convexity for a given MD and cash flow yield**.
- **Holding period rate of return (ROR), also called the horizon yield.** Like the YTM of a portfolio of bonds:

$$\sum (MV)_{bonds, t=0} = \frac{\sum PV(CF)_t}{(1 + \frac{ROR}{2})^{12}}$$

where the PV(CF) can be the modified (using scenario IRRs, i.e. 'cashflow yield'; coupon rates don't change though coupon reinvestment rates are at the scenario cash flow yields) or unmodified (using IRRs) cashflows PV sum, they're the same. The 12 is the time from that cash payment time to the Macaulay duration (even if approximate; usually equal to or close to the investment horizon; positive if before MD and negative if after MD date), not the time to maturity. The ROR = YTM if coupons reinvested at ROR and if bond held to maturity or sold at constant-price trajectory. If the IRR changes, then CFs will change because the dividend values will change in one direction, and the price of the bond will change in the opposite direction, but leading to the SAME value at $t = MD$. At the MD, the ROR will not change under any scenario IRR.

Additional points about immunization.

- As time passes, PF's MD changes but note the biggest change is from the number of years to the MD shrinking.
- Paths for zero-coupon yield and high and low interest rates - all pulled to par near maturity.



- Perfect immunization is zero-coupon bond. Any change in cash flow yield on bond PF is = change in YTM on zero-coupon bond.
- A sufficient but not necessary condition of immunization is a flat or parallel change in rates. But can immunization is possible even with non-parallel rate changes, as long as change in cash flow yield on bond PF is = change in YTM on zero-coupon bond. Generally, change in cash flow yield on bond PF is $\neq$ change in YTM on zero-coupon bond, because of non-parallel yield curve changes: pivots, 'butterflies' in the yield curve.
- Higher convexities = greater susceptibility to non-parallel shifts. Low convexities = less susceptibility, more like zero-coupon bonds and $\Delta CF = \Delta YTM$.
- **Structural risks.** Structural risk arises from portfolio design, particularly the choice of the portfolio allocations. The risk is that yield curve twists and non-parallel shifts lead to changes in the cash flow yield that do not match the yield to maturity of the zero-coupon bond that provides for perfect immunization. Structural risk is reduced by minimizing the dispersion of the bond positions, going from a barbell design to more of a bullet portfolio that concentrates the component bonds' durations around the investment horizon. At the limit, a zero-coupon bond that matches the date of the single obligation has, by design, no structural risk because $\Delta CF = \Delta YTM$.

- Portfolio convexity statistic = convexity.

$$\frac{MD^2 + MD + Dispersion}{(1 + IRR)^2} = Convexity$$

minimizing portfolio dispersion is the same as minimizing the portfolio convexity for a given Macaulay duration and cash flow yield. An advantage to using convexity to measure the extent of structural risk is that the portfolio statistic can be approximated by the market value-weighted average of the individual bond convexities. A problem with estimating portfolio dispersion using the weighted average of dispersion statistics for individual bonds is that it can be misleading. Consider a portfolio of all zero-coupon bonds of varying maturities. Each individual bond has zero dispersion (because it has only one payment), so the market value-weighted average is also zero. Clearly, the portfolio overall can have significant (non-zero) dispersion. This **minimizes risk from changes in the yield curve**. Moreover, a more convex bond gains more if yield falls and loses less if yield rises.

Convexity and dispersion. Compared to a low-dispersion bond cashflow, a high-dispersion bond cashflow has higher convexity. Intuitively, a high-dispersion bond is more sensitive to rate declines because there're more long-term cashflows; it's less sensitive to rate increases because more short-term cashflows buffer the drop.

Immunizing assets should have higher convexities than liabilities to be immunized (to minimize effect of non-linear rate changes) - if rates decline, the assets will be overfunded, and if rates rise, they will also be overfunded. However, want the immunizing assets to have **ONLY SLIGHTLY HIGHER** convexities - large ones will lead to less immunization. By the convexity statistic, this means dispersion for the immunizing assets should also be slightly higher.

To immunize a **SINGLE liability** with bond PF (whether single or multiple asset PF): (1) bond PF initial MV equals or exceeds PV of liability (if there's underfunding, i.e. cash inflow less than cash outflow, then become pressured financially and may have to sell bonds and ruin immunization program), (2) portfolio MD matches liabilities due date, (3) minimizes convexity (but convexity is still higher than that of the liability). Dispersion of immunizing assets should be higher than single liability to 'mitigate the effect of non-parallel shifts in the yield curve'.

Money duration.

- MoneyDur = ModDur x MV. $ModDur = MD/(1 + r)$, where MV = PV(debt), and r is per-period. NOTE: Even if MD is annualized, if the payments are semi-annual, then r is the rate per 6-months.
- BPV. MoneyDur x 0.0001. Similar to PVBP.

- MoneyDur better for multiple liabilities to immunize. Match money durations because MVs and CF yields of assets and liabilities are not equal. A single liability has an MD, and if the immunizing bond PF has that same MD, and we assume asset and liability are sensitive to the same interest rate factor (even if CFs are different) so their rates move parallel (not twist), then the MVs must be the same for immunization, so the MoneyDur will be equal. For multiple liabilities, even if the MDs are the same for the immunizing bond PF and liabilities, the CF yields and chosen MVs are generally different, so MoneyDur will be different. And MoneyDur is what needs to actually be the same. Since the market value and cash flow yields of the assets and liabilities are almost certainly not equal in a multiple-liability scenario, matching the Money Duration ensures that the total change in the dollar value of the assets perfectly offsets the total change in the dollar value of the liabilities for a small, parallel shift in the yield curve.
- Using MoneyDur, if CF yield of immunizing PF is lower than the liability PF, then this could lead to the MV_0 of the immunizing bond PF being higher at the beginning than the MV_0 liabilities to be immunized.

Use Macaulay durations for immunizing single liabilities, money durations for immunizing multiple liabilities. Immunization is about matching the **timing** of assets and liabilities, so we use a time-based duration, which is Macaulay duration matching of assets to the single liability. However, for multiple liabilities, there is no 'single' timing. So we generalize immunization to match dollar sensitivities and therefore use money duration. Money duration is **not about timing**, but rather about **dollar sensitivities to interest rates**. MDs are not about dollar sensitivities to interest rates (this is sort of dealt with in immunization by having the initial MVs of the assets and liability match).

However, since if MVs of assets and liabilities are matched, and MDs are matched, then even single liability immunization can just simply combine conditions (1) + (2) above into matching MoneyDur.

With MULTIPLE liabilities, there tend to be more duration gaps. **Duration gap** is the money duration gap (asset MoneyDur - liability MoneyDur) arising from changing rates and other conditions. The duration gap can be filled by

- Buy or sell different bonds and government securities of different durations.
- **Derivatives overlay.** Buy or sell interest rate futures contracts. Make sure that MV of assets is larger than MV of liabilities to begin with, however.

$$\text{Asset portfolio BPV} + N_f \times \text{futures BPV} = \text{Liability portfolio BPV}$$

Note futures BPV is complicated but simplify it with

$$\text{Futures BPV} \approx \frac{BPV_{CTD}}{CF_{CTD}}$$

- **Contingent immunization.** Surplus = MV immunizing assets - MV liabilities. If there's no surplus, then there's trouble: not enough assets to meet the liabilities in the first place (this is also why can't just use derivative overlays, e.g. futures or swaptions, for immunization - need some assets to actually meet the liabilities). Big surplus means can use contingent immunization because have a big buffer in case liabilities grow and/or surplus shrinks, which is a hybrid passive-active strategy. Basically, use surplus to invest to reduce debt obligation costs (so that can pay down debt earlier or with less expense - can buy risky assets to do so; if it evaporates, back to duration-matching). Initially, can be that the BPV of assets will be much smaller than BPV liabilities, even though the MV of assets is much larger than MV of liabilities if the assets are shorter-duration. Example: under-immunize initially, thinking that interest rates would rise; but then later decide that rates will fall, so fill negative duration gap + overhedge with long futures.

Example: DB pension plan

- Timeline: Person has so far worked G years, will work another T years, and then retire for Z years. There's a longevity risk ofc - that the person will outlive Z years, but ignore risk. Variables: m is wage multiplier, W is wage.
- Accumulated benefit obligation (ABO) liabilities. Current legal liability of plan. More important for M&A acquirers.

$$\begin{aligned} ABO &= \frac{1}{(1+r)^T} \left(\frac{mGW_0}{1+r} + \frac{mGW_0}{(1+r)^2} + \dots + \frac{mGW_0}{(1+r)^Z} \right) \\ &= \frac{mGW_0}{(1+r)^T} \left(\frac{1}{r} - \frac{1}{r(1+r)^Z} \right) \end{aligned}$$

- Projected benefit obligation (PBO) liabilities. Financial status and accounts. Funded status. More important for pension designers.

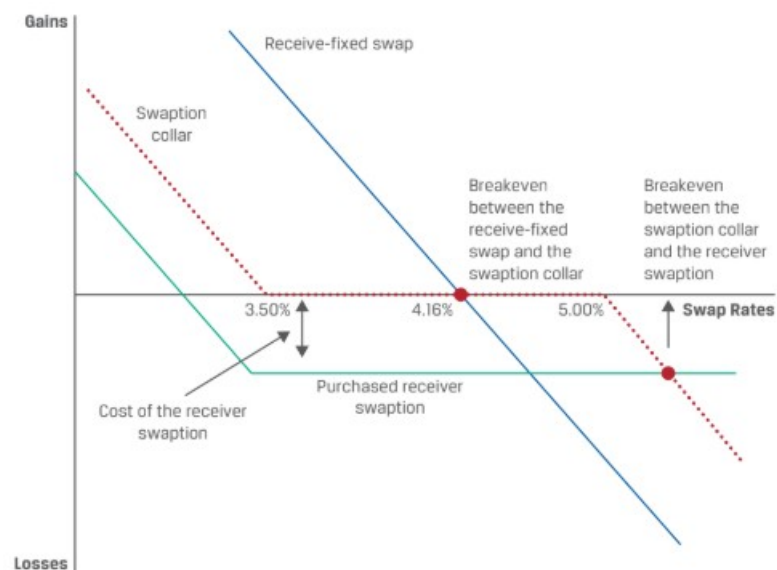
$$\begin{aligned} PBO &= \frac{1}{(1+r)^T} \left(\frac{mGW_T}{1+r} + \frac{mGW_T}{(1+r)^2} + \dots + \frac{mGW_T}{(1+r)^Z} \right) \\ &= \frac{mGW_T}{(1+r)^T} \left(\frac{1}{r} - \frac{1}{r(1+r)^Z} \right) \\ &= \frac{mGW_0(1+w)^T}{(1+r)^T} \left(\frac{1}{r} - \frac{1}{r(1+r)^Z} \right) \end{aligned}$$

- There will be two different durations. The plan sponsor must decide which liability measure to use for risk management and asset allocation. For example, if the corporation anticipates that it might be a target for an acquisition and that the acquirer likely would want to convert the retirement plan from defined benefit to defined contribution, the ABO measure matters more than the PBO.
- Calculate effective durations of liabilities with

$$EffDur = \frac{PV_- - PV_+}{2\Delta r(PV_0)}$$

and then convert to BPV by x0.0001. Take plan surplus (MV assets - MV liabilities) and divide by BPV to get **interest rate sensitivity of surplus**, i.e. a 5 bps fall will lead to X increase (liabilities rise if discount factor is lower) in liabilities, which might = surplus (in which case, a 5 bps increase wipes out the surplus).

- Calculate BPV of assets = weighted average of BPVs. Often assume equity and alternative asset durations = 0. However, they do actually change, but it's just not as predictable or stable as in fixed income. Get BPV of bonds by looking it up (someone else calculates this).
- Duration gap = BPV assets - BPV liabilities. If negative and r falls, then problem.
- Address duration gap with (1) futures, (2) interest rate swaps, (3) receiver swaptions, (4) swaption collars.



Annuities. Ordinary simple, ordinary general, simple annuity due, general annuity due.

Simple perpetuity, where the first payment starts at the end of this year:

$$PV_p = \frac{P}{r}$$

Simple ordinary annuity, where the first payment starts at the end of this year:

$$PV_a = \frac{P}{1+r} + \frac{P}{(1+r)^2} + \frac{P}{(1+r)^3} + \dots + \frac{P}{(1+r)^n}$$

$$PV_a = PV_p - \left(\frac{P}{(1+r)^{n+1}} + \frac{P}{(1+r)^{n+2}} + \dots \right)$$

$$PV_a = PV_p - \left(\frac{P/r}{(1+r)^n} \right)$$

$$PV_a = \frac{P}{r} \left(1 - \left(\frac{1}{1+r} \right)^n \right)$$

Simple ordinary annuity with growth:

$$PV_a = \frac{P}{1+r} + \frac{P(1+g)}{(1+r)^2} + \frac{P(1+g)^2}{(1+r)^3} + \dots + \frac{P(1+g)^{n-1}}{(1+r)^n}$$

$$PV_a = PV_p - \left(\frac{P(1+g)^n}{(1+r)^{n+1}} + \frac{P(1+g)^{n+1}}{(1+r)^{n+2}} + \dots \right)$$

$$PV_a = PV_p - \left(\frac{P(1+g)^n/r}{(1+r)^n} \right)$$

$$PV_a = \frac{P}{r-g} \left(1 - \left(\frac{1+g}{1+r} \right)^n \right)$$

Other points

- Problem with futures margins. Because of MTM, have to settle daily - so can have large realized losses, even if there are unrealized gains elsewhere (e.g. reduction in liabilities). This is 'collateral exhaustion risk'.
- Durations of equity and alternative assets. Often assume equity and alternative asset durations = 0. However, they do actually change, but it's just not as predictable or stable as in fixed income.
- Swap durations. Example. Receive-fixed, pay-floating. Fixed leg duration is e.g. 5 and pay-floating leg duration is 0. So, positive. For Pay-fixed, receive-floating, net swap duration is negative.
- Receiver swaptions. Option to enter receive-fixed swap. If rate falls below the strike price, then gain. Otherwise, expires.

- Callable bond = issuer buys a call option. The strike price is like a bond price (it's not the rate, because then it's not long a call option).
- Issuer's callable bond can be immunized by writing a receiver swaption. If rates rise, call option expires worthless but issuer still gets premium from receiver swaption. If rates fall, issuer gains on the callable bond but loses on the receiver swaption which is now in the money. Note: buying a payer swaption isn't the same thing - have to PAY a premium in the first place.
- Spread risk. The spread risk is between high-quality corporate bond yields and swap rates. Typically, there is less volatility in the corporate-swap spread than in the corporate-Treasury spread because both the MRR and corporate bond yields contain credit risk vis-à-vis Treasuries. Therefore, one of the usual advantages to hedging corporate bond risk with interest rate swaps is that those derivatives pose less spread risk than Treasury futures contracts.

Interest rate hedging. Full hedge:

$$BPV_{assets}\Delta Assets + BPV_{hedge}\Delta Yields_{hedge} \approx BPV_{liabilities}\Delta Yield_{liability}$$

Approximation because doesn't include higher-order terms like convexity. Risks: (1) model risk - parameter assumptions, measurement error, changes to interest rate structure; arises whenever assumptions made about future events and approximations are used to measure key parameters, (2) spread risk - if using an r depending on IG bonds and the credit spread to treasuries moves; note IG bond yields often less volatile than treasuries, (3) counterparty credit risk, (4) asset liquidity risk.

Fixed-income markets. Use duration (bonds) instead of beta (equity) for market exposure and sensitivity measure. Low duration = low risk. Much bigger than equity, much more diverse, have more characteristics, and bond indexes have to keep updating as bonds expire.

Bond indexes. There is pure indexing (full replication), and there's enhanced indexing (like stratified sampling or cell approach). Have to rebalance more than equity indices - this means higher transaction costs. Alternatives to directly investing in bonds or bond PFs: ETFs, mutual funds, TRSs.

Bond indexes II. To reduce tracking error, match duration (including option-adjusted duration, to account for embedded calls in bonds) and convexity (for potentially large changes), key rate durations (for individual rate changes), weight in sector and bond quality, sector and quality spread duration (spread duration = change in bond's price given widening or narrowing of spread compared with benchmark), sector-coupon-maturity call weights, and issuer exposure (match duration from each issuer to the duration weight in the index). Can also **PV of distribution of CF methodology**, which is dividing timeline into

cash flow vertices, aggregating CF in each vertex (including CF from embedded calls based on probability of call), then multiplying the aggregated CFs by the time of the cash vertex to get something like MoneyDur.

Bond indexes III. Enhanced strategies mean reducing costs (e.g. trading costs), better models, individual bond selection, call exposure enhancements (e.g. consider callable bonds - if r falls, price rises but hits a ceiling because issuer can call - so they are not sensitive to duration, so PM should use something other than duration).

Bond indexes IV. Use total return swap for quick, liquid, cheap exposure. Receiver receives both appreciation and dividend cash flows.

Bond indexes V. Indices require daily valuations, not weekly ones.

Bond portfolios and liquidity management. A bullet portfolio of 3, 5, 8 years duration versus barbell portfolio with durations 1 and 12 years. Bullet PF will have a steady stream of staggered bonds near maturity due to its bullet structure, so it's better for liquidity, even if the barbell has bonds with 1 year duration because the constant stream of additional bonds is probably very uneven.

6.5 Yield curve strategies

Formulas.

- $E(R) \approx \text{coupons} + \text{rolldown returns} + \Delta P$ investor's view of benchmark yields + ΔP investor's view of yield spreads + ΔP investor's view of FX value changes. Note: returns = coupon income + price appreciation = rolldown return + ΔP due to investor's view of benchmark yield.
- The following is for **parallel yield or rate changes only**. It doesn't work for yield slope changes:

$$\% \Delta PV \approx -(ModDur \times \Delta yield) + \frac{1}{2} Conv \times (\Delta yield)^2$$

and take the MV weighted average ModDur and convexities.

- Futures BPV = BPV_{CTD} / CF_{CTD}
- Swaps (see below about receiver swaps). Swap return = swap carry (the swap initial 'offset') + swap MTM gain (= difference between the fixed and floating legs $\approx$ swap BPV $\times \Delta bps$). Swap BPV = swap duration $\times$ notional. See p290 Example 3.

Note: bonds with higher durations usually have higher convexities. Higher dispersion means higher convexities. Barbell portfolios usually have higher convexities as they have higher dispersions.

Static yield curve strategies. If yield curve is **upwardly sloping**, want to **increase duration**:

- Buy-and-hold. Buy bonds with durations above the portfolio duration. Like rolling down the yield curve, but no active trading of the purchased long-duration bonds. Coupon income.
- Rolling down yield curve. Coupon income + rolldown return (by active trading).
- Repo carry trade. Buy LT bond and short ST (repo sale) to finance the LT purchase.
- Derivatives, i.e. synthetic strategies. Long futures contract - similar to buy-and-hold and rolling down yield curve. Receive-fixed swaps - the remaining fixed rate payments in value as the future ST rates will be lower than current LT rates (i.e. like buy-and-hold and rolling down yield curve), i.e. **swap carry**.

Strategy	Description	Targeted Return	Goal
Long futures position	Purchase contract for future bond delivery	$(\Delta \text{Price} / \Delta \text{Bond yield}) - \text{Margin cost}$	Synthetically increase duration (up-front margin and daily mark-to-market valuation)
Receive-fixed swap	Fixed-rate receiver on an interest rate swap	$(\text{Swap rate} - \text{MRR}) + (\Delta \text{Swap mark-to-market} / \Delta \text{Swap yield})$	Synthetically increase portfolio duration (up-front / mark-to-market collateral) + / - Swap carry

Note on Payer-swaps. Negative duration. Can reduce PF duration, and reduce costs from an increase in interest rates.

Dynamic yield curve. Assume instantaneous YTM changes affecting ΔP due to investor's view of benchmark yields.

- Yield level and slope changes. If anticipate rates will overall rise (whether or not slope changes), then reduce duration by shorting bonds and/or buying short-dated bonds, pay-fixed interest rate swap, short futures. If anticipate overall rates will fall (whether or not slope changes), increase duration with cash bond purchases (bullets), receive-fixed swaps, long interest rate futures. Calculate by MV average of ModDur and convexities and applying the $\% \Delta P$ formula above.
- Yield level and slope changes II. If anticipate yield slope changes but no overall change in level, go with net zero duration, but short or long the ST and LT positions depending on whether the slope steepens or flattens.
- Techniques: bond purchases or sales, receive-fixed or receive-floating swaps, long or short futures.

- Anticipating yield slope changes. Regardless of yield level changes, if anticipating yield slope steepening, then ST rates anticipated to fall and LT rates to rise, so that means LONG ST bonds and short LT bonds. Reverse if anticipating yield curve flattening. Calculate manually each individual position, e.g. calculate $\% \Delta P$ for each bond in a barbell 2-bond portfolio and sum them. This sum won't be reflected in the overall portfolio $\% \Delta P$ formula above.
- Note that the formula below doesn't work for yield slope changes.

$$\% \Delta PV \approx -(ModDur \times \Delta yield) + \frac{1}{2} Conv \times (\Delta yield)^2$$

- Bond value accretion. Example with 1 5y zero-coupon bond with $ParValue = Notional = FV = 100$. At $t = 0$, $PV = \frac{100}{(1+r)^5}$. At $t = 2$, $PV = \frac{100}{(1+r)^3}$. At $t = 5$, $PV = 100 = FV$. If going on long on the bond, then $FV > 0$ and the bond premium accretes positively (it rises in value). If shorting bond, bond premium accretes negatively. If bond $YTM > 0$ then long bond premium accretes positively. If bond $YTM < 0$ then long bond premium accretes negatively.
- Rolldown return in above formula is final bond accretion minus initial bond accretion **with initial rates**. Example with 10y zero-coupon long bond $Notional = 20$ and initial yield (rate) r_1 and at $t = 0.5$, rolldown return is

$$20 \times \left(\frac{1}{(1+r_1)^{9.5}} - \frac{1}{(1+r_1)^{10}} \right)$$

and for a 2y zero-coupon short bond of $Notional = -100$ and initial rate r_2 ,

$$-100 \times \left(\frac{1}{(1+r_2)^{1.5}} - \frac{1}{(1+r_2)^2} \right)$$

- ΔP due to benchmark yield changes. This is bond value differences at $t = t$ **using both initial and final rates**. Example with 10y long bond $Notional = 100$ with initial rate r_1 and final rate r_{1t} at $t = 0.5$

$$20 \times \left(\frac{1}{(1+r_{1t})^{9.5}} - \frac{1}{(1+r_1)^{9.5}} \right)$$

and for a 2y zero-coupon short bond of $Notional = -100$ and initial rate r_2 and final rate r_{2t} ,

$$-100 \times \left(\frac{1}{(1+r_{2t})^{1.5}} - \frac{1}{(1+r_2)^{1.5}} \right)$$

- Add rolldown return + ΔP due to benchmark yield changes.

Yield curve shapes. Butterfly strategy. Butterfly is about CHANGING spreads.

- Butterfly spread = - (ST yield) + (2 x MT yield) - (LT yield).
- Positive butterfly: go from positive spread to negative spread. Negative butterfly: go from negative spread to positive spread.

Issuer owning a callable bond. This is the same as 'selling volatility' to the issuer.

Effective duration and convexity (review).

$$EffDur = \frac{PV_- - PV_+}{2\Delta r(PV_0)}$$

$$EffCon = \frac{PV_- + PV_+ - 2PV_0}{(\Delta r)^2(PV_0)}$$

Options. Interest rate put and call options are based on bond PRICE not interest rate. **Long call options ADD convexity = as rate falls, bond price rises, and call option value rises. Long put option adds convexity.** Short put and call options only add some premium and REDUCE convexity.

Swaptions. Right to enter into IR swap. Payer-swaptions. Option to enter into a pay-fixed receive-floating swap based on a strike interest rate K. Long payer-swaption wants rate to rise - then, long payer-swaption owner can exercise by paying K (doesn't change) while receiving the HIGHER rate (= MRR). Long payer-swaption decreases duration. Long receiver-swaption increases duration.

Options on bond futures. Right to enter into a futures position.

Pay-fixed swap. Pay-fixed = short fixed-rate bond. Receive-floating = long FRN. Negative duration.

Strategy	Description	Targeted Return	Portfolio Duration Impact
Long bond call option	Purchase right to take forward bond delivery	Max (Bond price at lower yield - Strike price, 0) - Call premium	Increase portfolio duration
Long bond put option	Purchase right to deliver bond in the future	Max (Strike price - Bond price at higher yield, 0) - Put premium	Decrease portfolio duration
Long payer swaption	Own the right to pay-fixed on an interest rate swap at a strike rate	Max (Strike rate - Swap rate, 0) - Swaption premium	Decrease in portfolio duration
Long receiver swaption	Own the right to receive-fixed on an interest rate swap at a strike rate	Max (Swap rate - Strike rate, 0) - Swaption premium	Increase in portfolio duration
Long call option on bond future	Own the right to take forward bond delivery at a strike price	Max (Bond futures price at lower yield - Strike price, 0) - Call premium	Increase in portfolio duration
Long put option on bond future	Own the right to deliver bond in the future at a strike price	Max (Strike price - Bond futures price at higher yield, 0) - Put premium	Decrease in portfolio duration

Key rate duration. Each key rate duration is determined within the context of the **entire portfolio**. The ENTIRE portfolio.

$$KeyRateDur_k = \frac{-1}{PV} \times \frac{\Delta PV}{\Delta r_k}$$

$$\sum_k KeyRateDur_k = EffDur$$

Key rate duration and ModDur are same for zero-coupon treasuries:

$$KeyRateDur_k = ModDur \times PFweight_k$$

(Does this only apply to zero-coupon bonds?????)

Active FI management across currencies. Want to know domestic currency returns:

$$R_{DC} = \sum w_i(1 + r_{FC,i})(1 + r_{FX,i}) - 1$$

Covered interest rate parity versus uncovered. UIP suggests that over time, returns on unhedged foreign currency exposure will be same as on domestic currency investments. Domestic currency investments = long domestic bonds. However, UIP doesn't hold as there is a forward rate bias (forward premium puzzle) due to e.g. higher-interest rate currencies being more volatile (they tend to be in developing countries) - this allows carry trades (if UIP held, better to just invest domestically - get same returns at lower transaction costs). The puzzle is not addressed by the Mundell-Fleming model, which is only for ST or interest rate and currency changes NOW, i.e. they are already part of the UIP before the domestic investor begins investing in the foreign currency.

Synthetic interest rates. Example with JPY investor and USD. The JPY investor borrows JPY at 1%, the local rate TONA. Suppose the spot JPY-USD rate is 150 but the CIP-predicted forward is 145.7. If CIP and UIP hold, the implied USD rate is 4% = synthetic USD interest rate. If the actual interest rate is 5% (as in next paragraph example), then the basis is 1%. Under CIP and UIP, this should be 0. If positive, the market expects the USD to be stronger than expected and the JPY weaker than expected under UIP. Non-0 values can arise from economic policies, financial intermediation cost changes, largely supply of savings in low rate currencies, etc.

Carry trades are very volatile - see Example 12 on p316. They work in stable environments.

Cross-country basis swap. From JPY investor perspective who wants to buy USD assets. JPY investor could borrow JPY from a local bank, exchange it for USD, and buy USD assets. But the JPY investor is then exposed to FX risk. Or JPY investor can borrow directly from US bank, but this is logistically complicated, and the profit in USD is still subject to FX risk. Instead, to reduce

foreign currency exposure, JPY investor can enter into a swap to exchange borrowed JPY for USD from a swap counterparty e.g. dealer initially (and reversed at end), and buys USD assets. In the swap, the JPY investor receives interest from USD assets and then pays USD interest to the swap dealer e.g. floating SOFR, 5%, etc. and the dealer pays JPY interest (which can be used to pay the local JPY bank) to the JPY investor. The JPY investor in sum

1. pays JPY of principal + receives USD principal + buys USD bond = pay JPY principal
2. receive USD asset payments (fixed; but can be floating) + pay USD interest (floating, e.g. SOFR; but can be fixed) + receive JPY interest (floating) = residual (from fixed-floating difference) + receive JPY interest (floating)
3. sell USD assets and receive USD + pay USD principal + receive JPY principal = receive JPY principal

So, JPY investor in sum pays JPY principal, gets JPY interest, and receives back JPY principal - only exposed to JPY, **replicates domestic currency return**. If CIP and UIP holds, JPY investor only earns domestic return. JPY investor gets advantages of FX hedging and no logical complications in the swap.

Carry trades are exposed to currency changes. In the above cross-currency basis swap, the USD payments received are fixed but have to pay floating; address by USD pay-fixed, receive-floating. Also, if JPY investor has to pay JPY bank in fixed, then add a further swap: JPY receive-fixed pay-floating rate swap. Adding the cross-currency basis swap and USD fixed-floating and JPY floating-fixed swaps = **Fixed-fixed cross-currency swap**.

Bear flattening example. If foreign rates will experience bear flattening and domestic rates will be stable and upward-sloping, then best carry trade is receive-floating, pay-floating cross-currency swap. Best to receive floating because the foreign rates will rise; best to pay floating because the future domestic rates will fall if term structure is constant. Note: carry trade is about RATES, they are NOT bond prices. Note2: if domestic rates will be stable, then ST rates will always remain low (as the yield curve is 'stable' which means 'moving stably').

Summary. To solve for expected return over a 1y investment horizon using the equation $E(R) \approx \text{coupons} + \text{rolldown returns} + \Delta P$ investor's view of benchmark yields + ΔP investor's view of yield spreads + ΔP investor's view of FX value changes.

- Rolling yield. $(PV_1 - PV_0)/PV_0$.
- ΔP investor's view of benchmark yields. $\% \Delta PV \approx -(\text{ModDur} \times \Delta \text{yield}) + \frac{1}{2} \text{Conv} \times (\Delta \text{yield})^2$
- $R_{DC} = \sum w_i(1 + r_{FC,i})(1 + r_{FX,i}) - 1$

Note: if a bond is trading at a premium, it is priced above par (e.g. $P > 100$). So, instead of bond accretion where the bond rises in value as it is 'pulled to par' (which happens for bonds trading at a discount $P < 100$), there is **bond amortization** and it LOSES value as it is 'pulled to par'.

6.6 Fixed-income active management: credit strategies

Spreads. Examining the fourth component of the equation: $E(R) \approx$ coupons + rolldown returns + ΔP investor's view of benchmark yields + ΔP investor's view of yield spreads + ΔP investor's view of FX value changes.

Yield spreads (benchmark spreads). YTM minus the benchmark yield (= YTM of an on-the-run government bond of similar maturity). Includes bid-offer spreads, risk.

CVA measures the credit risk portion of the yield spread, as a PV. Default risk = POD. Loss severity = $LGD = 1 - RR$, where LGD is expressed as percentage of par value. Then, $CVA = POD \times LGD$.

Credit loss rate = realized percentage of defaults. Rises and falls with economic conditions.

Credit migration. Usually fall in credit rating.

Credit spread curves. Spread between curves over same time horizon, for different dates. See Exhibit 7, p341, and the below graph example on OAS spreads.

Credit cycle.

	Early Expansion (Recovery)	Late Expansion	Peak	Contraction (Recession)
Economic Activity	Stable	Accelerating	Decelerating	Declining
Corporate Profitability	Rising	Peak	Stable	Falling
Corporate Leverage	Falling	Stable	Rising	Peak
Corporate Defaults	Peak	Falling	Stable	Rising
Credit Spread Level	Stable	Falling	Rising	Peak
Credit Spread Slope	Stable for high grade, inverted for low ratings	Steeper for both higher and lower ratings	Steeper for both higher and lower ratings	Flatter for high grade, inverted for low ratings

Durations. Use with YTM changes to estimate bond price changes.

Analytical duration is often wrong. For example, issuer financial distress will cause a bond's price to diverge from what a model using benchmark rates would suggest. As an issuer nears default, the price of its bond approaches the estimated recovery rate ($POD \approx 1$, so $CVA = 1 \times (1 - RR)$, so the price percentage of the bond drops by $1 - RR$, and therefore becomes RR) regardless of the current benchmark YTM, because investors no longer expect to receive risky future coupon payments. Under a 'flight to quality' market stress scenario, investors sell high-risk, low-rated bonds, which fall in price, and purchase government bonds, which experience price appreciation.

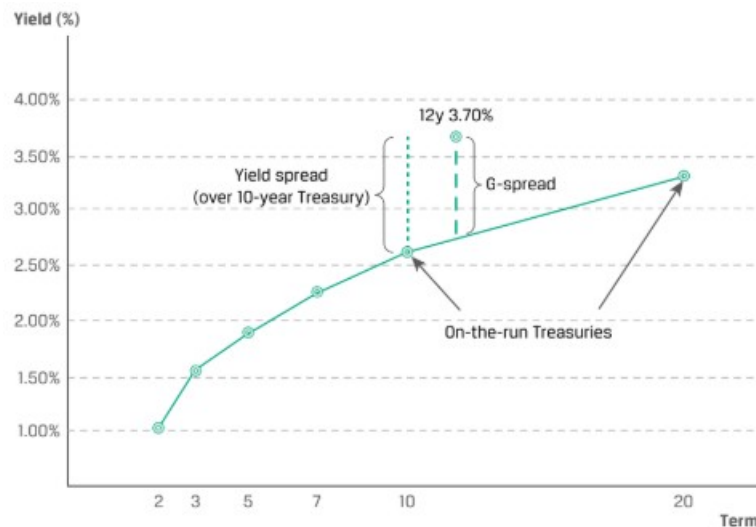
Empirical duration. The above observed negative correlation between HY credit spreads and government benchmark yields to maturity often leads fixed-income practitioners to use statistical models and historical bond market data to estimate the empirical duration. Empirical duration often differs between higher-quality and lower-quality bonds:

- When there is a flight to quality. HY credit spreads widen and IG falls, resulting in difference between analytical and empirical duration. High risk bonds fall in price, government bonds rise in price.
- When a bond nears default. regardless of benchmark YTM, bond approaches estimated recovery rate.

Empirical duration of risky bonds will almost always be lower than its analytical duration because the benchmark yield moves in the opposite direction to the credit spread. The analytical duration calculation is based on the YTM (which includes the credit spread). If there is a benchmark yield shift of 50bps in a worsening economy, then this should increase the price of the bond, but this is silently accompanied by an increase in the credit spread, which should push the price down. So the actual price is less sensitive to the benchmark yield change - so if the change in yield = benchmark yield shift, then the empirical duration will be smaller than the analytical duration.

Credit spread measures: yield, G-, I-, asset swap, Z-, CDS, OAS.

- Yield spreads. Relative to nearest actual benchmark bond, even if the maturities are different, even if the coupons are different. The difference in maturities leads to **measurement bias**.

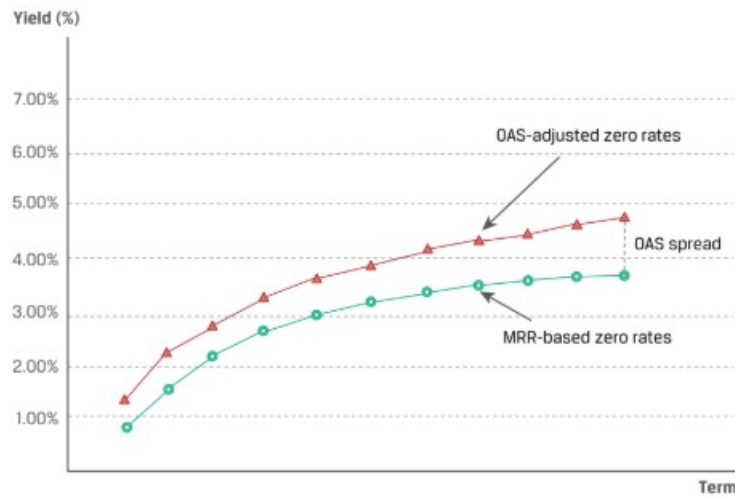


- G-spread. Linearly interpolate benchmark yield between the nearest benchmark bonds. Coupon rates don't matter. Less measurement bias because it considers more of the term structure.
- Swap spread = **swap rate** - government bond YTM. Use linear interpolation to calculate the swap rate.
- I-spread. I-spread = bond current YTM - swap rate. Same tenors. Single point spread. Doesn't consider options, e.g. callable bonds.
- Asset swap spread (ASW) = bond coupon rate - MRR. Matching coupon dates for remaining life of bond.
- Z-spread. Spread over government spot curve or interest rate swap spot curve. More accurate than G- or I-spreads, but more complex.

$$PV_{bond} = \frac{PMT}{(1 + z_1 + Z)} + \frac{PMT}{(1 + z_2 + Z)^2} + \dots + \frac{PMT}{(1 + z_T + Z)^T}$$

Not for use with embedded options (i.e. no volatility trees).

- CDS basis. CDS spread minus Z-spread of same tenor. Unlike ASW, terminates after credit event.
- OAS - option-adjusted spread. Generalizes Z-spread to include options under assumed volatility. Most comprehensive and accurate but complex and depends on volatility and other assumptions.



Spread	Description	Advantages	Disadvantages
Yield spread	Difference between bond YTM and government benchmark of similar tenor	Simple to calculate and observe	Maturity mismatch, curve slope bias, and inconsistent over time
G-spread (Government spread)	Spread over interpolated government bond	Transparent and maturity matching default risk-free bond	Subject to changes in government bond demand
I-spread (Interpolated spread)	Yield spread over swap rate of same tenor	Spread versus market based (MRR) measure often used as hedge or for carry trade	Point estimate of term structure and limited to option-free bonds
ASW (Asset swap)	Spread over MRR of fixed bond coupon	Traded spread to convert current bond coupon to MRR plus a spread	Tradable spread rather than spread measure corresponding to cashflows and limited to option-free bonds
Z-spread (Zero volatility spread)	Yield spread over a government (or swap) spot curve	Accurately captures term structure of government or swap zero rates	More complex calculation limited to option-free bonds
CDS Basis	Yield spread versus CDS spread of same tenor	Interpolated CDS spread versus Z-spread	Traded spread rather than spread measure corresponding to cashflows and limited to option-free bonds
OAS	Yield spread using Z-spread including bond option volatility	Provides generalized comparison for valuing risky option-free bonds with bonds with embedded options	Complex calculation based on volatility and prepayment assumptions; bonds with embedded options are unlikely to earn OAS over time

Example 6, p350. Portfolio of bonds. Given OASs of each of the bonds of a

bond PF, get the bond PF OAS by weighted averages, weighted by current bond market values.

- In a portfolio of bonds, the $Price = P + AI$.
- Don't need the tenors - just looking for spreads.
- 2 ways to solve: (1) use current bond MV-weighted average (see also calculating weights for interpolation in Example 4), (2) calculate PF combined yield, and calculate PF combined yield with OAS's, and then find difference. In each case of (2), take the individual spreads to calculate the individual bond prices, then combine the bond prices with FVs to get the combined PF values, and from there, calculate the combined PF yields.
- OAS is **already** in the bond $P + AI$. To get the 'pure' price, **add (by e.g. 1.0125%)** the OAS's to each of the bonds and calculate the combined yield. Add it because a positive spread = discount, which means the 'pure' price is actually higher.

1. A European portfolio manager is presented with the following information on a portfolio of two bonds. Calculate the OAS of the portfolio.

Issuer	Face Value	Price*	Accrued Interest*	OAS (bps p.a.)
Company A	€ 100,000,000	95	1.5	125
Company B	€ 200,000,000	97	2.0	150

*Per 100 of par value

Solution:

- Solve for bond and portfolio values:
 - Company A bond: €96,500,000 = €100,000,000 × (0.95 + 0.015)
 - Company B bond: €198,000,000 = €200,000,000 × (0.97 + 0.02)
 - Portfolio: €294,500,000 = €96,500,000 + €198,000,000
- Solve for portfolio weights w_A and w_B :
 - Company A (w_A): 32.8% = €96,500,000/€294,500,000
 - Company B (w_B): 67.2% = €198,000,000/€294,500,000
- Solve for portfolio OAS using $(OAS_A \times w_A) + (OAS_B \times w_B)$.
 OAS = 142 bps p.a. = (0.328 × 125 bps) + (0.672 × 150 bps)

Floating rate spread measures. Use current MRR below (so, MRR is static):

$$PV = \frac{\frac{(MRR+QM)FV}{m}}{\left(1 + \frac{(MRR+DM)}{m}\right)^1} + \frac{\frac{(MRR+QM)FV}{m}}{\left(1 + \frac{(MRR+DM)}{m}\right)^2} + \dots + \frac{\frac{(MRR+QM)FV}{m} + FV}{\left(1 + \frac{(MRR+DM)}{m}\right)^T}$$

Quoted margin **doesn't capture changes in credit risk over time** because it's static. Discount margin = required margin = yield spread over MRR in

order to price the FRN at par. Note that MRR is in every numerator, if the MRR changes, all future numerators change too. DM assumes flat MRR zero curve.

Zero-discount margin. Note the z 's in the second and subsequent terms, instead of MRR, which are forward MRRs.

$$PV = \frac{\frac{(MRR+QM)FV}{m}}{\left(1 + \frac{(MRR+(Z-DM))}{m}\right)^1} + \frac{\frac{(z_2+QM)FV}{m}}{\left(1 + \frac{(z_2+(Z-DM))}{m}\right)^2} + \dots + \frac{\frac{(z_T+QM)FV}{m} + FV}{\left(1 + \frac{(z_T+(Z-DM))}{m}\right)^T}$$

Z-DM makes the FRN priced at par. Will differ from DM depending on the yield curve shape.

The **annualized rolling difference** = calculate by comparing the value of two bonds at a given point in time. It's the same as returns on bonds with credit spreads, which will be higher than a government bond without a credit spread: (1) rolldown returns will be higher, as initially there will be a credit spread but as it is pulled to par in addition to rolling down, yields will be higher, and (2) if a corporate bond has a credit spread, then its coupon rate will be higher if it is priced at par. Coupon rate $\approx$ benchmark rate + credit spread.

Using a longer-dated maturity government bond or interpolated government bond benchmark, where the yield curve is sloping up, will lead to LOWER annualized rolling differences. See Example 10, p355.

$E\Delta P$ due to investor's view of yield spreads. Below equation is limited to yield spread changes:

$$\frac{\Delta PV}{\Delta Spread} = \% \Delta PV_{spread} = - \underbrace{(EffSpreadDur \times \Delta Spread)}_{\text{Spread (OAS) duration}} + \frac{1}{2}(EffSpreadConv \times (\Delta Spread)^2)$$

where ΔP Spreads are usually OAS.

$$EffSpreadDur = \frac{PV_- - PV_+}{2(\Delta Spread)PV_0}$$

$$EffSpreadConv = \frac{PV_- + PV_+ - 2PV_0}{2(\Delta Spread)PV_0}$$

Duration times spread DTS. Since spread CHANGES are proportional to the spreads themselves, it is better to use

$$\frac{\Delta Spread}{Spread}$$

Since

$$EffSpreadDur \approx \frac{\Delta P}{\Delta Spread}$$

$$EffSpreadDur \times Spread \approx \frac{\Delta P}{\Delta Spread} \times Spread = \frac{\Delta P}{\Delta Spread / Spread} = DTS$$

So, DTS is like effective 'proportional spread' duration. So, take the percentage change and multiply by DTS, e.g. if the OAS changes by 10 bps when it is 125 bps, then $DTS \times (10/125)$.

$$DTS \times \frac{\Delta Spread}{Spread} = \Delta P$$

Excess spread.

$$ExcessSpread = \underbrace{Spread_0 / \text{periods per year}}_{\text{Initial spread income}} - \underbrace{(EffSpreadDur \times \Delta Spread)}_{\text{Capital gain or loss}}$$

It is negative because if the spread rises, it is a loss. If the spread decreases, it's a gain.

Excess spread return. As default becomes more likely, bond value approaches PV of expected recovery, so the loss is then $(POD \times LGD)$, so remove this:

$$E(r_{excessspread}) = \underbrace{Spread_0 / \text{periods per year}}_{\text{Initial spread income}} - \underbrace{(EffSpreadDur \times \Delta Spread)}_{\text{Capital gain or loss}} - (POD \times LGD) / \text{periods per year}$$

This is basically a RETURN from the excess credit spread above default risk or expected loss ($EL = POD \times LGD$) and changing credit spread, so depending on what is put into the original $Spread_0$, it will also capture liquidity and other risks beyond credit risks. But it only addresses changing CREDIT risks - changing liquidity and other risks are not addressed. Spread = default risk ($POD \times LGD$) + term premium + liquidity premium + tax premium + others.

We need 'periods per year' for initial spread and $POD \times LGD$ but NOT for capital gains or losses because the former two are RATES. Capital gains or losses are STOCK.

If the excess spread return is positive, then there is sufficient compensation for risk.

The $E(R) = coupon + rolldown + \dots$ equation's ΔP investor's view of credit spreads $= -EffSpreadDur \times \Delta Spread + \frac{1}{2}EffSpreadConv \times (\Delta Spread)^2$.

Duration measures for FRNs.

$$EffRateDur_{FRN} = \frac{PV_- - PV_+}{2(\Delta MRR)PV_0}$$

$$EffSpreadDur_{FRN} = \frac{PV_- - PV_+}{2(\Delta DM)PV_0}$$

Credit strategies.

Bottom-up strategies.

- Key financial ratios. But hard to compare across industries and sectors and geographies. Problems with using historical financial statements.

Ratio	Description	Advantages	Disadvantages
EBITDA/ Total Assets	Profitability Cash flow as a percentage of assets	Combines operating income with non-cash expense	Ignores capital expenditures and working capital changes
Debt/ Capital	Leverage Fraction of company's capital financed with debt	Direct measure of relative reliance on debt financing	More relevant for investment-grade than high-yield issuers
EBITDA/ Interest Expense	Coverage Cash flow available to service debt	Measures relative issuer ability to meet debt payments	Volatile measure for firms with high cash flow variability

- Models I. Reduced form credit models. Solve for default intensity or the POD over a specific time. Use observable company and macro variables. Altman Z-score model is an example.
- Models II. Structural credit models. Probability of assets falling below liabilities. Use market-based asset value and other variables.
- Relative value analysis. Compare excess spread returns, for instance. Use ratios like profitability, leverage, coverage ratios.
- May need to interpolate multiple times, see Example 17, p 365.

Notes: (1) May need to interpolate multiple times, see Example 17, p 365, (2) 'new premium of X bps' means to ADD X bps to the spread.

Top-down strategies.

- Assess credit quality, e.g. ordinal or some kind of 'weighted' scale ranking. Public credit ratings - but note they can be different, e.g. S&P focus on POD while Moody's on EL. Moreover, public credit ratings are for specific time periods. So, many active credit managers use OAS to measure quality (for FI PF quality, sum OAS weighted by individual credit market values).
- Spreads and YTM relationship. Normally assume in equations that they're uncorrelated - the MRR moves around independently of the spread. But they can be correlated, particularly during market change periods. The DTS (which reflects bonds with wide spreads experience proportionally larger changes) is an example.
- Sector allocation.

- Factor-based credit strategies.

Factor	Rationale	Measures Used
Carry	Expected return measure if POD or aggregate risk premium is unchanged	OAS
Defensive	Empirical research suggests safer low-risk assets deliver higher risk-adjusted returns	Market-based leverage, gross profitability, and low duration
Momentum	Bonds with higher recent returns outperform those with lower recent returns	Trailing six-month excess bond and equity returns
Value	Low market value versus fundamental value indicates greater than expected return	Bond spread less default probability measure, which includes rating, duration, and excess return volatility

- ESG. ESG and credit ratings often correlated. Green bonds are pari passu with issuer's outstanding senior unsecured bonds.

More on duration. There is **(credit) spread duration** and **benchmark duration**. Latter is the normal duration - price sensitivity to interest rates.

Bond prices. Use matrix pricing. Trading cost = bond + half the bid-offer spread.

Liquidity risk. Use CDSs, bond ETFs, swaps, etc. to address PFs that have illiquid bonds. Bond ETFs have relatively constant duration and pay variable interest - don't mature and don't have duration drift.

Tail risk. Look at tail risk using parametric methods, historical simulations, Monte Carlo, hypotheticals, stress tests. Use VaR measures below. If PF exceeds tail risk, then use position limits, risk budgeting, derivatives (swaptions, credit default swaptions).

Example 20, p376. VaR for bond prices at X% confidence interval in N distribution. Use: $-ModDur \times \Delta yield$. Calculate ModDur with Excel, $\Delta yield$ as the CI from volatility (volatility = SD).

Other VaRs (especially during market stress): CVaR, incremental (partial) VaR, relative VaR. Relative VaR = measures expected tracking error versus a benchmark by calculating VaR or CVaR on active PF minus benchmark PF.

Synthetic credit strategies. Use CDSs.

Credit default swap (CDS). Manages credit risk SEPARATELY from interest rate risk. As a percentage of notional:

$$\text{CDS price} \approx 1 + (\text{Fixed coupon} - \text{CDS spread}) \times \text{EffSpreadDur}_{CDS}$$

The fixed-coupon is usually 1% for IG and 5% for HY issuers. The CDS spread = PV of difference between swap-zero curve (term structure of interest rates, basically) and swap-zero curve multiplied by hazard rate (probability that issuer credit event will occur given not occurred in prior period). The MTM gain or loss is difference between CDS prices, or use

$$\Delta(\text{CDS price}) \approx -\Delta(\text{CDS spread}) \times \text{EffSpreadDur}_{\text{CDS}}$$

If the CDS spread < fixed-coupon, then protection buyer receives upfront premium. If the CDS spread > fixed-coupon, then protection buyer pays upfront premium.

Instrument	Description	Targeted Return	Portfolio Impact
Single-Name CDS	Protection buyer pays premium to seller in exchange for payment if credit event occurs	Buyer gains and seller loses if single-name credit spread widens or credit event occurs	Short (buyer) or long (seller) single-name credit spread exposure
Index-Based CDS	Protection buyer pays premium in exchange for partial payment if credit event occurs for index member	Buyer gains and seller loses if index member spreads widen or if credit event occurs	Short (buyer) or long (seller) index-based credit spread exposure
Payer Option on CDS Index	Option buyer pays premium for right to buy protection ("pay" coupons) on CDS index contract at a future date	Max (Credit spread at expiration – CDS Credit Spread Strike, 0) – Option Premium	Short CDS index-based credit spread exposure
Receiver Option on CDS Index	Option buyer pays premium for right to sell protection ("receive" coupons) on CDS index contract at a future date	Max (CDS Credit Spread Strike – CDS Credit Spread at expiration, 0) – Option Premium	Long CDS index-based credit spread exposure

CDS curve. CDS spreads across maturities for single issuer or index. This is purely about credit risk. It is a pure credit curve, whereas others (e.g. yield spread curve) have other risks - including interest rate or benchmark yield risk, liquidity risk, included. Use CDS curves for (1) overweighting or underweighting bonds in PF instead of using bonds themselves: instead, overweight by selling CDS and underweight by buying CDS, and (2) other active credit management tools (other than CDSs). e.g fixed-income ETFs and derivatives using them as the underlying.

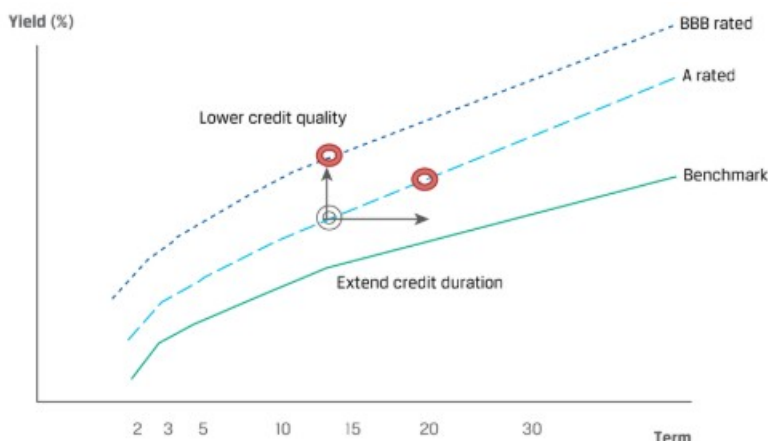
CDS long-short strategies. Top-down or bottom-up.

- Long CDS (long protection, short risk) on wherever credit spread is expected to rise, and short CDS wherever credit spread expected to fall using

CDSs of same maturity and tenor on the underlying bonds. In the cash bond market, would instead have to do this by shorting bonds whose credit spreads will widen and then buying duration-matched amounts (MoneyDur-matched to reduce interest rate risk, which CDSs do not have assuming credit risk is independent of interest rate risk) of bonds whose credit spreads will narrow.

- CDS curve trades. Benefit from relative changes in CDS curve. MoneyDur-matched long and short positions because otherwise, the strategy is exposed to absolute CDS spread change in addition to relative CDS spread changes across maturities. See Example 25 p384.

Static credit strategies. If yield curve is expected to be stable, can (1) reduce credit quality or (2) extend credit duration.



1. Going from the A-rated bond to the BBB-rated bond in the diagram. $E(ExcessSpread) \approx Spread_0 - (EffSpreadDur \times \Delta Spread) - (POD \times LGD)$, if spread, POD and LGD remain static OVER THE HOLDING PERIOD (not between A and BBB bonds, for instance). In this case, $\Delta Spread = 0$ and $\Delta(POD \times LGD) = 0$ ACROSS TIME. Generally, the excess spread return will be higher for BBB bonds than A bonds.
2. 'Buy and hold' or 'Rolling down yield curve'. The first involves buying risky bonds with durations above the benchmark without active trading during a subsequent period. If the relationship between long- and short-term credit spreads remains stable over the investment horizon, the manager is rewarded with greater return from the higher spread duration. "Rolling down" the credit curve not only generates incremental coupon income (adjusted over time for any price difference from par) due to wider spreads but also adds return from the passage of time and the investor's ability to sell the shorter-maturity position in the future at a lower credit spread at the end of the investment horizon.

Example 26. If going long a 15y semiannual coupon bond by X USD over a 10y semiannual coupon bond over 6m, when there are 15y and 10y government semiannual coupon treasuries, want to compare X USD 15y bond returns versus X USD 10y bond returns over the 6m. Divide into (1) incremental coupon returns (15y over 10y over 6m; can be divided into benchmark (government coupon differences) and credit spread (corporate bond coupon difference minus government coupon difference)), and (2) incremental price appreciation (15y over 10y over 6m - here, will need to interpolate YTM's at 6m). Then divide each of (1) and (2) into returns from (a) benchmark yields and (b) credit spreads. For (1) don't need to interpolate years: just compare X USD 15y semiannual coupon versus X USD 15y government semiannual coupon, and do the same for 10y. For (2), **interpolate the YTM's for each of the 15y and 10y bonds by looking at the 15, 10, 5y bond YTM's** (and do the same for government bond YTM's). Then calculate rolldown returns for each of the 15y and 10y bonds separately using the interpolated YTM's, and then roughly split them on the basis of **how much the yield spread changes over the holding period, and how much the benchmark changes over the holding period**. Then combine (1) and (2).

Dynamic credit strategies.

- Tactical credit strategies. Long-short cash bonds. Calculate profit by using excess spread return formula.
- Synthetic credit strategies. Long-short using CDSs. Use CDS price formula to calculate profit.

Global credit strategies. Consider FX rates changing; even with UIP if assume bullish foreign economy then it may violate UIP and so may want to not hedge.

Structured credit. CDOs, CLOs, MBS, ABS, covered bonds.

Instrument	Description	Exposure	Portfolio Applications
Collateralized Debt Obligations (CDOs)	Fixed-income securities backed by a diversified pool of debt obligations	Redistribute portfolio debt cash flows across ratings spectrum	Create tailored portfolio-based debt exposure categories/profiles unavailable in the cash bond market
Collateralized Loan Obligations (CLOs)	Fixed-income securities backed by a diversified pool of floating-rate leveraged loan obligations	Redistribute portfolio loan cash flows across ratings spectrum	Create tailored portfolio-based loan and interest rate exposure profiles unavailable in the cash bond market
Mortgage-Backed Securities (MBS)	Fixed-income securities backed by a pool of commercial or residential mortgage loans	Provide portfolio-based exposure to real estate cash flows	Offer active managers exposure to real estate and to volatility (prepayment/extension risk) unavailable in the cash bond market
Asset-Backed Securities (ABS)	Fixed-income securities backed by a pool of credit card, auto, and other loans	Provide portfolio-based exposure to consumer loan cash flows	Offer active managers direct exposure to consumer loans and to volatility unavailable in the cash bond market
Covered Bonds	Senior debt obligations backed by pool of commercial/residential mortgages or public sector assets	Provide portfolio-based exposure to real estate cash flows with recourse to issuer	Offer active managers direct exposure to consumer loans and to real estate/public sector cash flows unavailable in the cash bond market

- If there is economic recovery, get more exposure to credit spreads.
- MBS has an embedded call option: prepayment if interest rates fall. So, increased volatility means the call option (which mortgage owners own) increases in value, so the MBS value falls. Upward shift in benchmark yields mean call option is out of the money, so the MBS value rises.
- CLOs (bond-only exposure; no interest rate exposure) are a type of CDO. CDOs are broad, covering a wide variety of debt - both interest rates and bonds. CLOs concentrate leveraged loans, syndicated bank loans, and borrowers diversified across industries (unlike CDOs, which have highly-correlated risks like subprime MBS). CLOs loans are usually secured.

Notes

- $\Delta P = -ModDur \times \Delta r - EffSpreadDur \times \Delta Spread$
- Q10, p410. Which spread duration to use when it changes?
- Q12, p410. A decline in yield can automatically mean a decline in the credit yield spread.
- Q17, p411. A 10% rise in the spread is a RELATIVE figure; it means we need to first calculate the ABSOLUTE spread rise first.
- Q18, p411. To reduce liquidity risk of an illiquid bond, allocate to buy-and-hold or enter pay-received, receive-floating swap.
- Q28, p414. Credit curve roll-down strategy is about CREDIT RISK, which must be separated from benchmark yield effects!

- Q31, p415. Fixed exchange rate regimes usually result in greater instability. High domestic currency YTM's in emerging economies are a sign of expected currency depreciation over time.
- **UIP causality.** If UIP holds, currency depreciation expectations lead to higher YTM's. Not the other way around: higher YTM's lead to expected FX depreciation.
- Q32, p415. Remember foreign returns are $(1 + r)(1 + r_{FX}) - 1$.

6.7 Trade strategy and execution

Reasons for trading.

- Profit. Have a view different from market.
- Risk management and hedging.
- Cashflow needs. Usually low trade urgency.
- Corporate actions, index reconstitutions and rebalances, margin calls, dividends, coupon receipts, distributions.

Trading concepts.

- Lit and unlit (dark pool) markets. Unlit markets: can't see the bids and sells as with lit markets, so only know there's a trade when you got a trade. Post-trade transactions and quantities even in unlit venues are recorded.
- Trade urgency.
- Alpha decay. Alpha = trading alpha = trade alpha.
- Execution risk. Price moves from decision time to execution time.

Trade considerations

- Order characteristics (buy or sell), order size, ADV order relative size
- Security characteristics. Security type, whether there's a ST alpha, price volatility, liquidity. Whether market is rising or falling.
- Market conditions.
- Investor, PM and trader risk aversion. Affects for e.g. trade urgency.
- Market impact. Usually, price will revert a bit after a big trade. Use market orders instead of just naively buying and selling, to reduce information leakage to dealers.

Reference prices = price benchmarks. Which benchmark to use as a 'target' for reference calculations.

- Pre-trade benchmarks. Decision price (price at exact time of investment decision; a record - used when PM wants to mark time from alpha decay), previous close (often used by quant traders), opening price (determined by auction from accumulated bids and asks overnight; the first published price) (using the opening price reduces **overnight risk**), arrival price (price upon arrival at market; important if alpha decay is fast).
- Intraday benchmarks. Volume-weighted average price VWAP (need both list of bids and asks) which allows determination of where market tightness is. Time-weighted average price TWAP = Equal-weighted average price throughout the day, excludes potential trade outliers (big one-time trades).
- Post-trade benchmarks. Closing price - used for NAV calculations.
- Price-target benchmarks, e.g. if think intrinsic value is X, then buy at or below X. BUT, if buying far below X and there is ST alpha, then use pre-trade benchmark (decision price, arrival price).

Trading strategies

- Short-term alpha. Don't use dark pools (too slow, too uncertain). Use **open market trading** - don't have luxury of passive trading. If sending to opening price auction, note that opening auction typically makes up 1 – 1.25% of day's trading volume, so don't send too much.
- Long-term alpha. **Passive trading, e.g. VWAP, TWAP**. Don't execute over multiple time periods with decision prices or reference prices - there will be unrelated market moves which will be unpredictable.
- Risk rebalance trade. Think about whether positive and negative betas are in balance. If reducing both long and short positions, try to not disturb price too much and use TWAP to see if generally undisturbed prices after trading. Reducing both long and short positions in ratio will reduce trade execution risk.
- Client redemption (cash outflow) trade. Want to trade at **closing auction price** (which matches NAV and benchmark calculations). Otherwise, trader takes a risk. Closing auction (market on close MOC): by computer at end of day settlement, and then that closing price is guaranteed for all.
- New mandate (cash inflow). If lots of cash and can't efficiently invest quickly, equitize the cash.

Trade execution.

Human involvement.

- Larger blocks of trade. Need to be manually broken up efficiently and traded.

- **Principal trades** or 'broker risk trades' (where the dealer trades, the dealer doesn't just look for other counterparties, which is 'agency trade').
- OTC (quote-driven or off-exchange) market trades.
- **Request for quote RFQ**, which is a custom quote which is nonbinding, provided upon request.

Automated trading. Liquid, high-volume trades, where human involvement is susceptible to front-running. Usually involves **direct market access DMA** or **algorithmic trading**.

Algorithmic trading: (1) (Trade) execution algorithm - customized by PM, and (2) profit-seeking.

- Scheduled algorithms. Based on percentage of volume POV, volume-weighted average price VWAP or time-weighted average price TWAP. POV (participation algorithms) trade more when market trades more (but higher costs if market is rising, and may not complete all trades on time as it depends on market activity). TWAP trade based on time-specified schedule, finish all trades but could be more expensive. VWAP trades based on historical intraday volume patterns (but might not finish all trades on time if volumes are too low),
- Scheduled algorithms II. Appropriate for orders in which portfolio managers or traders do not have expectations for adverse price movement during the trade horizon. These algorithms are also used by portfolio managers and traders who have greater risk tolerance for longer execution time periods and are more concerned with **minimizing market impact**. Scheduled algorithms are often appropriate when the order size is relatively **SMALL** (e.g., no more than 5%–10% of expected volume), the security is relatively **LIQUID**, or the orders are part of a risk-balanced basket and trading all orders at a similar pace will maintain the risk balance.
- Profit-seeking I: Liquidity-seeking (opportunistic) algorithms. Good for large orders to execute quickly without market impact. Avoid information leakage. Often these algorithms 'sweep the book' looking for best deals. Can use dark pools. Good for **LARGE ILLIQUID AND LIQUID** securities. Might trade only small amounts of shares if liquidity is not present at good prices.
- Profit-seeking II: Arrival price. Seek to trade close to market prices at arrival time. Trade more early, then less later. Use when believe price will move adversely against later, so trade earlier. Volume or time-based. Arrival price algorithms trade aggressively to capture alpha ahead of others during the day. Can be time or volume based. For risk-averse PMs (who want to reduce execution risk), fast, where there is alpha decay. Good

for **MEDIUM-SIZED LIQUID** securities. Used to minimize cost of executing an order relative to the arrival price. Reduce IS quickly (ahead of adverse price movement). Used for orders where portfolio manager or trader believes prices are likely to move unfavorably and wishes to trade more aggressively to capture alpha, used when security is relatively liquid or the order is not outsized (size less than 15% of the expected volume).

- Dark strategy or liquidity aggregators. Used when information leakage is a problem. **LARGE ILLIQUID** orders to minimize market impact. Probably won't execute whole order, however.
- Smart order routers (SORs). A type of algorithm which determines best route and deal for order. Good for **SMALL CONTINUOUS** orders. Two types: (1) market orders (if market moves quickly), and (2) limit orders.

Algorithmic selection: clustering ML to decide how similar stocks should be traded; LASSO models for market forecasting.

Trading markets.

- Equities. Mostly exchanges and dark pools.
Dark pools. More than 50 globally, mostly in US and Europe. In US: 'alternative trading systems ATS' and in Europe: 'multilateral trading facilities MTF' and 'systematic internalizers SI'.
- Fixed-income. Much lower market transparency than equities. Usually only exchanges for on-the-run government bonds. Others usually OTC and RFQs.
- Exchange-traded derivatives. Usually large markets, electronic trading.
- OTC derivatives. Dealers. Usually traded by institutions, so large trade sizes.
- Spot FX. All OTC. Huge market, opaque. Interbank (lowest bid-ask spreads) market, banks (medium spread), retail and commercial (largest spreads).

Large urgent trades = broker risk (principal) trades. Large less urgent trades = agency trades. Small less urgent = exchanges, electronic trading, DMA, TWAP, etc.

Trade costs.

Implementation shortfall IS. Trade costs from investment decision to trade completion by trader.

$$IS = \text{Paper return} - \text{Actual return}$$

where the paper return is the full return that the PM thought they could get theoretically (getting 100% of the trade, no execution risk or market impact cost), and the actual return is return on what was actually transacted (less than full trade, execution risk cost and market impact cost included). **FOR SALES, S MUST BE NEGATIVE!!!!**

$$\text{Paper return} = (P_n - P_d)S = SP_n - SP_d$$

where P_n is current price, P_d is decision price, S is shares ORDERED but not necessarily executed ($S > 0$ for long position).

$$\text{Actual return} = (\sum s_j)(P_n) - \sum s_j p_j - Fees$$

$$\text{Actual return} = \sum s_j P_n - \sum s_j p_j - Fees$$

where s is shares actually executed, and p is the transaction price of the share.

$$\begin{aligned} IS &= S(P_n - P_d) - \sum s_j P_n + \sum s_j p_j + Fees \\ &= \sum s_j p_j + S(P_n - P_d) - \sum s_j P_n + Fees \\ &= \sum s_j p_j - \sum s_j P_d + S(P_n - P_d) - \sum s_j P_n + \sum s_j P_d + Fees \end{aligned}$$

Rearranging,

$$IS = \underbrace{\sum s_j p_j - (\sum s_j)P_d}_{\text{Execution cost}} + \underbrace{(S - \sum s_j)(P_n - P_d)}_{\text{Opportunity cost}} + Fees$$

where subscript j = actual transaction. Execution cost is like market impact cost on actual trade (from decision time to transaction time). Opportunity cost is from not being able to sell the entire order, $\sum s < S$, due to liquidity, etc. Opportunity cost is cost of not being able to execute all shares at decision price. Expanded IS.

$$IS = \underbrace{(\sum s_j)p_0 - (\sum s_j)p_d}_{\text{Delay cost}} + \underbrace{\sum s_j p_j - (\sum s_j)p_0}_{\text{Trading cost}} + \underbrace{(S - \sum s_j)(P_n - P_d)}_{\text{Opportunity cost}} + Fees$$

where p_0 is the arrival price, i.e. price at time order was released to market for execution.

IS in bps.

$$IS(bps) = \frac{IS_{\$}}{S \times p_d} \times 10,000bps$$

Trade cost analysis.

Cost in total dollars (\$):

$$\text{Cost (\$)} = \text{Side} \times (\bar{P} - P^*) \times \text{Shares}$$

Cost in dollars per share (\$/share):

$$\text{Cost (\$/share)} = \text{Side} \times (\bar{P} - P^*)$$

Cost in basis points (bps):

$$\text{Cost (bps)} = \text{Side} \times \frac{(\bar{P} - P^*)}{P^*} \times 10,000 \text{ bps}$$

$$\text{Side} = \begin{cases} +1 & \text{Buy order} \\ -1 & \text{Sell order} \end{cases}$$

$\bar{P}$ = Average execution price of order

P^* = Reference price

Shares = Shares executed

Arrival cost. Difference between market price at time of order released to market and actual transaction.

$$\text{Arrival cost (bps)} = \text{Side} \times \frac{\bar{P} - P_0}{P_0} \times 10^4 \text{bps}$$

where $\bar{P}$ is the actual, transacted average price, and P_0 is the arrival price.

VWAP cost. Whether a fair cost was received for the trading day. Reflects market noise, trading trends throughout day.

$$\text{VWAP cost (bps)} = \text{Side} \times \frac{\bar{P} - VWAP}{VWAP} \times 10^4 \text{bps}$$

Usually, orders will underperform because of bid-ask spread and market impact cost.

TWAP cost. Whether a fair cost was received for the trading day. Removes outlier trades.

$$\text{TWAP cost (bps)} = \text{Side} \times \frac{\bar{P} - TWAP}{TWAP} \times 10^4 \text{bps}$$

Market-on-close (MOC) cost. Use for benchmarks, NAV, mutual funds, etc.

$$\text{MOC cost (bps)} = \text{Side} \times \frac{\bar{P} - Close}{Close} \times 10^4 \text{bps}$$

Market-adjusted cost. For stocks. Takes into consideration market trends (whether price indices are rising or falling).

$$\text{Market-adjusted cost (bps)} = \text{Arrival cost (bps)} - \beta \times \text{Index cost (bps)}$$

where

$$\text{Index cost (bps)} = \text{Side} \times \frac{\text{Index VWAP} - \text{Index arrival price}}{\text{Index arrival price}} \times 10^4$$

Index VWAP usually calculated using overall market index or a related ETF. It is the index VWAP over the trading period. **Compare market-adjusted cost (MAC) with the index cost; if small, then trade cost is mostly from market movements.**

Added value. The pre-trade cost below is estimated and given in a question:

$$\text{Added value (bps)} = \text{Arrival cost (bps)} - \text{Est. pre-trade cost (bps)}$$

Trade governance. Investment managers to have a trade policy document, which is mandated in most jurisdictions. Trade policy document: (1) defines best execution (based on price, cost, speed, likelihood, order sizes, nature of trade), (2) lists factors for determining optimal order execution (urgency, liquidity, market size, standard security or custom, lit or unlit venue, investment strategy, trading rationale), (3) listing eligible brokers and execution venues (based on quality, stability, reputation, capabilities, speed, cost, whether they do principal risk trades) - really important for OTC markets, (4) process to monitor execution arrangements. Should also have (a) trade aggregation and allocation policy (b) trade error policy (goal is to reduce adverse impact on client, NOT to completely disclose all errors to them), (c) best execution monitoring committee, and (d) storage of historical trade data.

MIFID II. Requires firms to take 'all sufficient steps' to obtain best execution. MIFID II also prohibits bundling trading commissions with research (soft dollar). Must separately pay for research. Other jurisdictions might follow EU in MIFID II.

Additional points from questions.

- Cash equitization can be through futures and-or ETFs.

6.8 Case study: Institutional

Liquidity management. Policies and practices that ensure that the portfolio complies with investment policy yet can meet cash outflow needs in a timely and orderly manner without incurring excessive costs. Tools for liquidity management:

- Liquidity profiling and time-to-cash tables. Liquidity profiling - see Exhibit 2. It is used in liquidity classification schedule (time-to-cash table), which defines portfolio categories (or "buckets") based on the estimated time needed in the normal course of business to convert assets in that particular category into cash. The liquidity budget assigns portfolio weights considered acceptable to each liquidity classification in the time-to-cash table and establishes a liquidity benchmark for the portfolio construction process. Liquidity = doesn't significantly impact market. More than one year is illiquid, before is liquid. Liquidity budget needed for stress testing.

- Rebalancing, capital commitments (e.g. for PE)
- Stress testing.
- Derivatives and leverage. Frees up cash.

However, illiquid assets earn higher premium. Can use the value of a put option with X at liquid price of illiquid asset, to value the illiquid-liquid spread.

Quadrivium university endowment case study.

- Spending rules. Geometric smoothing = Yale formula. Spending for the upcoming year is a weighted average of previous year + some percentage (e.g. 5%) of the endowment at end of year. Weights are empirical. Then, nominal return target of endowment should then add inflation rate for this year.

Using the Yale rule or smoothing effect (while providing smooth spending for the university) will lead to counter-cyclicality and bad liquidity fund management: when economy and endowment does well, less than 5% but when economy and endowment do badly then more than 5%.

Use cashflow-forecasting and commitment-pacing models to plan spending, as well as stress tests.

- Return projections. Use MVO, Monte Carlo, etc. Use **net of fees** returns.
- Low return expectations in the future on endowment assets. Then need to take greater risk to increase return expectations, but only if risk-adjusted return (e.g. Sharpe ratio) is higher.
- Illiquid assets. Can increase the Markowitz frontier, but need to de-smooth PE, RE returns.
- Allocations to funds. Avoid large allocations to a small number of funds to diversify idiosyncratic risks.
- Stress and scenario tests. If there's another GFC-like event. Then fund could be squeezed due to (1) have to support university with more funds on short notice, (2) MTM shock on fund value, (3) low capital distributions from private markets, and (3a) accelerating capital calls as opportunities arise in distressed markets. Liquidity in a stress situation will deteriorate: (4) faster capital calls, (5) activation of gates, (6) smoothing effect.

Smoothing effect. Investments in private markets tend to incorporate market valuations with a lag that leads to a relative increase in their portfolio weighting during periods of market stress and a relative decrease in the portfolio weighting of more liquid assets. This does not reduce the effective liquidity of the portfolio in dollar terms, but it does impact the percentage of assets in the overall portfolio that could be used to satisfy liquidity needs in periods of market stress.

- Illiquidity. Can make rebalancing difficult or more expensive.
- Vintage years. Diversify PE funds across vintage years to avoid overexposure to certain parts of the business cycle. Avoid large allocations to a few funds, which will reduce big sudden changes in capital calls.
- Cash usage. More efficient to use derivatives than using cash on securities or ETFs.
- Funding costs. Notionals and MRRs for swaps, implied funding costs for futures (have to pay more for future price rather than buying outright). But derivative funding costs are still less than cash securities' funding costs (i.e. borrowing the whole amount or opportunity cost of whole amount).
- Other funding costs. Cash drag, rolling futures, moving cash from one external manager to another.
- External manager fund movement and rebalancing. Moving funds between external PMs is costly and would be used for more permanent changes. For more temporary changes, use derivatives. But for permanent changes, it's costly to keep using derivatives which expire. Also, using multiple derivatives is even more costly.
- ESG strategies: (1) screening and exclusion, (2) ESG factor incorporation into analysis.

Additional points

- Rebalancing bands. Wider if return SD is larger, smaller if SD is smaller.
- Illiquidity premia. Studies show longer term illiquid investments have higher illiquidity premia. Illiquid public equities also have illiquid premia but depends on security (other security properties could be interfering in determining how much of the premia is from illiquidity).
- Q5, p538. Futures incurs leverage (implicit) costs too. 3x leverage means 1 part cash and 2 parts borrowing. Three-month MRR rate = annualized rate.
- If introducing more illiquidity to fund, (1) monitor more and (2) rebalance more.
- Using derivatives instead of cash markets = tracking error or basis error.